

# Function of One Complex Variable I

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# Chapter 1

## The Complex Number System

### 1.1 The real numbers

### 1.2 The field of complex numbers

**6.** Let  $R(z)$  be a rational function of  $z$ . Show that  $\overline{R(z)} = R(\bar{z})$  if all the coefficients in  $R(z)$  are real.

*Solution.* Let:

$$R(z) = \frac{P(z)}{Q(z)},$$

where  $P(z)$  and  $Q(z)$  are polynomials with real coefficients and  $Q(z) \neq 0$ .

$$P(z) = \sum_{k=0}^n a_k z^k, \quad a_k \in \mathbb{R}.$$

Then

$$\overline{P(z)} = \sum_{k=0}^n \overline{a_k z^k} = \sum_{k=0}^n \overline{a_k} \bar{z}^k = \sum_{k=0}^n a_k (\bar{z})^k = P(\bar{z}).$$

Similarly,

$$\overline{Q(z)} = Q(\bar{z}).$$

Therefore:

$$\overline{R(z)} = \overline{P(z)/Q(z)} = \overline{P(z)}/\overline{Q(z)} = P(\bar{z})/Q(\bar{z}) = R(\bar{z}).$$

■

## 1.3 The complex plane

## 1.4 Polar representation and roots of complex numbers

2. Calculate the following:

1. the square roots of  $i$
2. the cube roots of  $i$
3. the square roots of  $\sqrt{3} + 3i$

*Solution.* To solve these questions we need solve  $z^2 = i$ ,  $z^3 = i$  and  $z^2 = \sqrt{3} + 3i$ . In general case we have:

$$z = \sqrt[n]{r} e^{\frac{\theta + 2k\pi}{n}i}, \quad k = 0, 1, \dots, n-1$$

Then:

1.  $\sqrt{i} = e^{\frac{\pi}{4}i}, e^{\frac{5\pi}{4}i}, \quad \theta = \frac{\pi}{2}, k = 0, 1$
2.  $\sqrt[3]{i} = e^{\frac{\pi}{6}i}, e^{\frac{5\pi}{6}i}, e^{\frac{9\pi}{6}i}, \quad \theta = \frac{\pi}{2}, k = 0, 1, 2$

For the third one we need to write it in polar coordinates:

$$r = 2\sqrt{3}, \quad \theta = \tan^{-1}\left(\frac{3}{\sqrt{3}}\right) = \frac{\pi}{3}.$$

So the square roots are

$$z = \sqrt{2\sqrt{3}} e^{i(\pi/6 + k\pi)}, \quad k = 0, 1.$$

■

5. Let  $z = e^{\frac{2\pi}{n}i}$  for an integer  $n \geq 2$ . Show that  $1 + z + \dots + z^{n-1} = 0$ .

*Solution.* The equation  $z^n = 1$  has exactly  $n$  distinct solutions, namely the  $n$ -th roots of unity:

$$z_k = e^{i\frac{2\pi}{n}k}, \quad k = 0, 1, \dots, n-1.$$

On the other hand, by factoring we have

$$z^n - 1 = (z - 1)(1 + z + \dots + z^{n-1}).$$

For  $z = e^{i\frac{2\pi}{n}k}$  with  $k = 1, \dots, n-1$ , it is clear that  $z \neq 1$ . Hence the factor  $z - 1$  is nonzero, which forces

$$1 + z + z^2 + \dots + z^{n-1} = 0.$$

In particular, this holds when  $z = e^{\frac{2\pi i}{n}}$ , which is the desired case. ■

## 1.5 Lines and half planes in the complex plane

1. Let  $C$  be the circle  $\{z : |z - c| = r\}$ ,  $r > 0$ ; let  $a = c + re^{i\alpha}$  and put

$$L_\beta = \left\{ z : \operatorname{Im}\left(\frac{z - a}{b}\right) = 0 \right\}$$

where  $b = e^{i\beta}$ . Find necessary and sufficient conditions in terms of  $\beta$  that  $L_\beta$  be tangent to  $C$  at  $a$ .

*Proof.* The line  $L_\beta$  passes through the point  $a$  with slope  $\tan \beta$ . For  $L_\beta$  to be tangent to the circle  $C$  at  $a$ , it must be perpendicular to the radius drawn from the center  $c$  to the point  $a$ . Since:

$$a - c = re^{i\alpha},$$

the direction of the radius vector is given by angle  $\alpha$ . For perpendicularity, the direction of the tangent line must differ by  $\frac{\pi}{2} \pmod{\pi}$ . Therefore:

$$\beta = \alpha + \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}.$$

■

## 1.6 The extended plane and its spherical representation

4. Let  $\Lambda$  be a circle lying in  $S$ . Then there is a unique plane  $P$  in  $\mathbb{R}^3$  such that  $P \cap S = \Lambda$ . Recall from analytic geometry that

$$P = \{(x_1, x_2, x_3) : x_1\beta_1 + x_2\beta_2 + x_3\beta_3 = \ell\}$$

where  $(\beta_1, \beta_2, \beta_3)$  is a vector orthogonal to  $P$  and  $\ell$  is some real number. It can be assumed that

$$\beta_1^2 + \beta_2^2 + \beta_3^2 = 1.$$

Use this information to show that if  $\Lambda$  contains the point  $N$  then its stereographic projection on  $\mathbb{C}$  is a straight line. Otherwise,  $\Lambda$  projects onto a circle in  $\mathbb{C}$ .

*Proof.* Let  $z = u + vi$  and  $Z = (x_1, x_2, x_3)$  its corresponding point on the circle. Then:

$$x_1 = \frac{2u}{u^2 + v^2 + 1}, \quad x_2 = \frac{2v}{u^2 + v^2 + 1}, \quad x_3 = \frac{u^2 + v^2 - 1}{u^2 + v^2 + 1}.$$

Point  $Z$  is also belong to the  $P$  so:

$$x_1\beta_1 + x_2\beta_2 + x_3\beta_3 = \ell$$

Now substitute into the plane equation and multiply by  $u^2 + v^2 + 1$ :

$$2\beta_1u + 2\beta_2v + \beta_3(u^2 + v^2 - 1) = \ell(u^2 + v^2 + 1).$$

Rearranging gives

$$(\beta_3 - \ell)(u^2 + v^2) + 2\beta_1u + 2\beta_2v - (\beta_3 + \ell) = 0. \quad (1.1)$$

*Case 1:*  $N \in \Lambda$ . Then  $N \in P$ , so  $\ell = \beta_3$ , and (1.1) reduces to

$$2\beta_1u + 2\beta_2v - 2\beta_3 = 0 \quad \Longleftrightarrow \quad \beta_1u + \beta_2v = \beta_3,$$

the equation of a straight line in the  $(u, v)$ -plane. Thus if  $N \in \Lambda$ , the image is a line.

*Case 2:*  $N \notin \Lambda$ . Then  $\ell \neq \beta_3$ . Divide (1.1) by  $\beta_3 - \ell$  and complete the squares:

$$\begin{aligned} u^2 + v^2 + \frac{2\beta_1}{\beta_3 - \ell}u + \frac{2\beta_2}{\beta_3 - \ell}v - \frac{\beta_3 + \ell}{\beta_3 - \ell} &= 0 \\ \Longleftrightarrow \left(u + \frac{\beta_1}{\beta_3 - \ell}\right)^2 + \left(v + \frac{\beta_2}{\beta_3 - \ell}\right)^2 &= \frac{\beta_1^2 + \beta_2^2 + \beta_3^2 - \ell^2}{(\beta_3 - \ell)^2} = \frac{1 - \ell^2}{(\beta_3 - \ell)^2}. \end{aligned}$$

Hence the image is the circle with radius  $\frac{\sqrt{1-\ell^2}}{|\beta_3-\ell|}$  and center

$$\left(-\frac{\beta_1}{\beta_3 - \ell}, -\frac{\beta_2}{\beta_3 - \ell}\right).$$

■

# Chapter 2

## Metric Spaces and the Topology of $\mathbb{C}$

### 2.1 Definition and examples of metric spaces

**3.** If  $(X, d)$  is any metric space show that every open ball is an open set. Also, show that every closed ball is a closed set.

*Proof.* Fix  $a \in X$  and  $r \geq 0$ .

**(1) Open balls are open.** Let  $B(a, r) := \{x \in X : d(x, a) < r\}$ . If  $r = 0$ , then  $B(a, 0) = \emptyset$ , which is open. Assume  $r > 0$  and take any  $y \in B(a, r)$ , so  $d(y, a) < r$ . Set

$$\varepsilon := r - d(y, a) > 0.$$

We claim that  $B(y, \varepsilon) \subset B(a, r)$ . Indeed, if  $z \in B(y, \varepsilon)$ , then by the triangle inequality,

$$d(z, a) \leq d(z, y) + d(y, a) < \varepsilon + d(y, a) = r,$$

so  $z \in B(a, r)$ . Thus every  $y \in B(a, r)$  has a neighborhood contained in  $B(a, r)$ , hence  $B(a, r)$  is open.

**(2) Closed balls are closed.** Let  $\overline{B}(a, r) := \{x \in X : d(x, a) \leq r\}$  and consider its complement

$$X \setminus \overline{B}(a, r) = \{x \in X : d(x, a) > r\}.$$

Take any  $y$  in this complement, so  $d(y, a) > r$ . Define

$$\varepsilon := \frac{d(y, a) - r}{2} > 0.$$

If  $z \in B(y, \varepsilon)$ , then by the reverse triangle inequality,

$$d(z, a) \geq d(y, a) - d(z, y) > d(y, a) - \varepsilon = d(y, a) - \frac{d(y, a) - r}{2} = \frac{d(y, a) + r}{2} > r.$$



Hence  $z$  also lies in the complement. Therefore  $X \setminus \overline{B}(a, r)$  is open, and so  $\overline{B}(a, r)$  is closed. ■

**8.** Let  $(X, d)$  be a metric space and  $Y \subset X$ .

1. If  $G \subset X$  is open (in  $X$ ), show that  $G \cap Y$  is open in the subspace  $(Y, d)$ .
2. Conversely, if  $G_1 \subset Y$  is open in  $(Y, d)$ , show there exists an open  $G \subset X$  such that  $G_1 = G \cap Y$ .

*Proof.* Recall that open balls in the subspace  $(Y, d)$  are of the form

$$B_Y(y, r) = \{z \in Y : d(z, y) < r\} = B_X(y, r) \cap Y.$$

**(1)** Let  $G \subset X$  be open. For any  $y \in G \cap Y$ , since  $G$  is open in  $X$ , there exists  $r > 0$  with  $B_X(y, r) \subset G$ . Intersecting with  $Y$  gives

$$B_Y(y, r) = B_X(y, r) \cap Y \subset G \cap Y.$$

Thus each  $y \in G \cap Y$  has a subspace ball contained in  $G \cap Y$ , so  $G \cap Y$  is open in  $(Y, d)$ .

**(2)** Let  $G_1 \subset Y$  be open in  $(Y, d)$ . For each  $y \in G_1$  choose  $r_y > 0$  such that  $B_Y(y, r_y) \subset G_1$ ; equivalently,  $B_X(y, r_y) \cap Y \subset G_1$ . Define

$$G := \bigcup_{y \in G_1} B_X(y, r_y) \subset X.$$

Each  $B_X(y, r_y)$  is open in  $X$ , hence  $G$  is open in  $X$  (union of open sets). Moreover,

$$G \cap Y = \bigcup_{y \in G_1} (B_X(y, r_y) \cap Y) = \bigcup_{y \in G_1} B_Y(y, r_y) = G_1,$$

because the last union covers  $G_1$  and is contained in it by construction. Therefore there exists an open  $G \subset X$  with  $G_1 = G \cap Y$ . ■

**9.** Do Exercise 8 with "closed" in place of "open".

*Proof.* (1) If  $F$  is closed in  $X$ , then  $X \setminus F$  is open in  $X$ . By Exercise 8,  $(X \setminus F) \cap Y$  is open in  $(Y, d)$ . Since

$$(X \setminus F) \cap Y = Y \setminus (F \cap Y),$$

the complement in  $Y$  of  $F \cap Y$  is open in  $Y$ , hence  $F \cap Y$  is closed in  $(Y, d)$ .

(2) Suppose  $F_1 \subset Y$  is closed in  $(Y, d)$ . Then  $Y \setminus F_1$  is open in  $(Y, d)$ . By

Exercise 8, there exists an open set  $G \subset X$  such that

$$Y \setminus F_1 = G \cap Y.$$

Let  $F := X \setminus G$ . Then  $F$  is closed in  $X$ , and

$$F \cap Y = (X \setminus G) \cap Y = Y \setminus (G \cap Y) = Y \setminus (Y \setminus F_1) = F_1,$$

as desired. ■

## 2.2 Connectedness

4. Prove the following generalization of Lemma 2.6. If  $\{D_j : j \in J\}$  is a collection of connected subsets of  $X$  and if for each  $j, k \in J$  we have  $D_j \cap D_k \neq \emptyset$ , then  $D = \bigcup_{j \in J} D_j$  is connected.

*Proof.* Fix an index  $j_0 \in J$  and, for each  $k \in J$ , set

$$E_k := D_{j_0} \cup D_k.$$

Each  $E_k$  is connected because it is the union of two connected sets with nonempty intersection: indeed  $D_{j_0} \cap D_k \neq \emptyset$  by hypothesis.

Moreover, the family  $\{E_k : k \in J\}$  has a common point (in fact, a common subset):

$$\bigcap_{k \in J} E_k \supset D_{j_0} \neq \emptyset.$$

By Lemma 2.6 (union of connected sets with a point in common is connected), the union  $\bigcup_{k \in J} E_k$  is connected. Finally,

$$\bigcup_{k \in J} E_k = \bigcup_{k \in J} (D_{j_0} \cup D_k) = D_{j_0} \cup \bigcup_{k \in J} D_k = \bigcup_{j \in J} D_j = D.$$

Therefore  $D$  is connected. ■

5. Show that if  $F \subset X$  is closed and connected, then for every pair of points  $a, b \in F$  and each  $\varepsilon > 0$  there exist points  $z_0, z_1, \dots, z_n \in F$  with  $z_0 = a$ ,  $z_n = b$ , and  $d(z_{k-1}, z_k) < \varepsilon$  for  $1 \leq k \leq n$ . Is the hypothesis that  $F$  be closed needed? If a set  $F$  satisfies this property, then  $F$  is not necessarily connected, even if  $F$  is closed. Give an example to illustrate this.

*Proof.* Fix  $\varepsilon > 0$  and for  $x \in F$ , set

$$C_\varepsilon(x) := \{y \in F : \text{there exist } \varepsilon\text{-chain from } y \text{ to } x\}.$$

We claim that  $C_\varepsilon(x)$  is both open and closed in the subspace  $F$ .

*Open in  $F$ :* If  $y \in C_\varepsilon(x)$  and  $y' \in F$  with  $d(y, y') < \varepsilon/2$ , then concatenating an  $\varepsilon$ -chain from  $x$  to  $y$  with the step  $y \rightarrow y'$  yields an  $\varepsilon$ -chain from  $x$  to  $y'$ . Hence  $B(y, \varepsilon/2) \cap F \subset C_\varepsilon(x)$ .

*Closed in  $F$ :* If  $y \notin C_\varepsilon(x)$  and  $y' \in F$  with  $d(y, y') < \varepsilon/2$ , then any  $\varepsilon$ -chain from  $x$  to  $y'$  could be extended by  $y' \rightarrow y$ , a step of length  $< \varepsilon$ . This would imply  $y \in C_\varepsilon(x)$ , a contradiction. Thus  $B(y, \varepsilon/2) \cap F \subset F \setminus C_\varepsilon(x)$ , so  $F \setminus C_\varepsilon(x)$  is open in  $F$ .

Therefore  $C_\varepsilon(x)$  is clopen in  $F$ . Since  $F$  is connected, we must have  $C_\varepsilon(x) = F$ . In particular, for any  $a, b \in F$ ,  $b \in C_\varepsilon(a)$ , which is exactly the desired  $\varepsilon$ -chain. The hypothesis that  $F$  be closed is *not* needed; the proof uses only connectedness. ■

## 2.3 Sequences and completeness

**3.** Show that  $\text{diam } A = \text{diam } \overline{A}$ .

*Proof.* Since  $A \subset \overline{A}$ , we have

$$\text{diam } A \leq \text{diam } \overline{A}.$$

For the reverse inequality, fix  $x, y \in \overline{A}$ . There exist sequences  $(x_n), (y_n) \subset A$  with  $x_n \rightarrow x$  and  $y_n \rightarrow y$ . By continuity of  $d$ ,

$$d(x, y) = \lim_{n \rightarrow \infty} d(x_n, y_n) \leq \sup\{d(u, v) : u, v \in A\} = \text{diam } A.$$

Taking the supremum over all  $x, y \in \overline{A}$  yields  $\text{diam } \overline{A} \leq \text{diam } A$ . Hence  $\text{diam } A = \text{diam } \overline{A}$ . ■

**8.** Suppose  $\{x_n\}$  is a Cauchy sequence and  $\{x_{n_k}\}$  is a convergent subsequence. Show that  $\{x_n\}$  must be convergent.

*Proof.* Fix  $\varepsilon > 0$ . Since  $(x_n)$  is Cauchy, there exists  $N_1$  such that  $d(x_m, x_n) < \varepsilon/2$  for all  $m, n \geq N_1$ . Since  $x_{n_k} \rightarrow x$ , there exists  $N_2$  such that  $d(x_{n_k}, x) < \varepsilon/2$  for all  $k \geq N_2$ . Let  $N = \max\{N_1, n_{N_2}\}$ . For any  $n \geq N$  choose  $k$  with  $n_k \geq n$  (possible

because  $(n_k)$  is strictly increasing and unbounded). Then

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Hence  $x_n \rightarrow x$ . Therefore the Cauchy sequence is convergent, with limit equal to the limit of its convergent subsequence. ■

## 2.4 Compactness

4. Show that the union of a finite number of compact sets is compact.

*Proof.* Let  $K_1, K_2, \dots, K_n$  be compact subsets of a topological space  $X$ , and let

$$K = \bigcup_{i=1}^n K_i.$$

Consider an open cover  $\mathcal{U}$  of  $K$ . Then  $\mathcal{U}$  is also an open cover of each  $K_i$ . Since each  $K_i$  is compact, for every  $i$  there exists a finite subcollection  $\mathcal{U}_i \subseteq \mathcal{U}$  that covers  $K_i$ . Because there are only finitely many sets  $K_1, \dots, K_n$ , the union

$$\mathcal{V} = \bigcup_{i=1}^n \mathcal{U}_i$$

is a finite union of finite collections, hence itself finite. Furthermore,  $\mathcal{V}$  covers all of  $K$ , since each  $K_i$  is covered by  $\mathcal{U}_i$ . Therefore,  $K$  has a finite subcover of  $\mathcal{U}$ , and thus  $K$  is compact. ■

5. Let  $X$  be the set of all bounded sequences of complex numbers. That is,  $\{x_n\} \in X \iff \sup\{|x_n| : n \geq 1\} < \infty$ . If  $x = \{x_n\}$  and  $y = \{y_n\}$ , define

$$d(x, y) = \sup\{|x_n - y_n| : n \geq 1\}.$$

Show that for each  $x \in X$  and  $\epsilon > 0$ ,  $\overline{B}(x; \epsilon)$  is not totally bounded although it is complete. (Hint: you might have an easier time of it if you first show that you can assume  $x = (0, 0, \dots)$ .)

*Proof. Reduction to the center 0.* The metric is translation-invariant:  $d(x + z, y + z) = d(x, y)$  for all  $x, y, z \in X$ . Hence translation by  $-x$  is an isometry sending  $\overline{B}(x; \epsilon)$  to  $\overline{B}(0; \epsilon)$ . Completeness and total boundedness are preserved by isometries, so it suffices to treat  $\overline{B}(0; \epsilon)$ .

**Completeness.** First,  $(X, d)$  is complete (i.e.  $\ell^\infty$  is a Banach space). Let

$(x^{(m)})_{m \in \mathbb{N}}$  be Cauchy in  $X$ , with  $x^{(m)} = (x_n^{(m)})_{n \geq 1}$ . For each fixed  $n$ ,  $(x_n^{(m)})_m$  is a Cauchy sequence in  $\mathbb{C}$ , hence converges; define  $y_n := \lim_{m \rightarrow \infty} x_n^{(m)}$  and  $y = (y_n)_n$ .

We claim  $y \in X$  and  $d(x^{(m)}, y) \rightarrow 0$ . Since  $(x^{(m)})$  is Cauchy, there exists  $M$  and  $C$  with  $\sup_n |x_n^{(m)} - x_n^{(M)}| \leq 1$  for all  $m \geq M$ . As  $x^{(M)}$  is bounded, so are all  $x^{(m)}$  for  $m \geq M$ , and passing to the limit in  $m$  shows  $y$  is bounded; thus  $y \in X$ . Given  $\delta > 0$ , choose  $m_0$  so that  $\sup_n |x_n^{(m)} - x_n^{(k)}| < \delta$  for all  $m, k \geq m_0$ . Fix  $m \geq m_0$ ; then for every  $n$ ,

$$|x_n^{(m)} - y_n| = \lim_{k \rightarrow \infty} |x_n^{(m)} - x_n^{(k)}| \leq \delta,$$

so  $\sup_n |x_n^{(m)} - y_n| \leq \delta$ . Hence  $d(x^{(m)}, y) \leq \delta$  for  $m \geq m_0$ , proving completeness. Therefore every closed subset, in particular  $\overline{B}(0; \epsilon)$ , is complete.

**Not totally bounded.** For  $k \in \mathbb{N}$ , let  $e^{(k)} = (0, 0, \dots, 0, \underbrace{1}_{k\text{th}}, 0, \dots)$ . Then  $e^{(k)} \in \overline{B}(0; 1)$  and for  $i \neq j$ ,

$$d(e^{(i)}, e^{(j)}) = \sup_{n \geq 1} |e_n^{(i)} - e_n^{(j)}| = 1.$$

Thus the set  $\{e^{(k)} : k \in \mathbb{N}\}$  is 1-separated. If  $\overline{B}(0; 1)$  were totally bounded, then for  $\eta = \frac{1}{2}$  it could be covered by finitely many  $\eta$ -balls; but a ball of radius  $< \frac{1}{2}$  contains at most one of the  $e^{(k)}$ , so infinitely many are needed—a contradiction. Therefore  $\overline{B}(0; 1)$  is not totally bounded.

Finally, scaling shows the same for any radius:  $\{\epsilon e^{(k)}\} \subset \overline{B}(0; \epsilon)$  is  $\epsilon$ -separated, so  $\overline{B}(0; \epsilon)$  is not totally bounded. By the reduction step,  $\overline{B}(x; \epsilon)$  is complete and not totally bounded for every  $x \in X$ . ■

**6. Show that the closure of a totally bounded set is totally bounded.**

*Proof.* Let  $(X, d)$  be a metric space and let  $A \subset X$  be totally bounded. Fix  $\varepsilon > 0$ . Since  $A$  is totally bounded, there exist points  $x_1, \dots, x_N \in X$  such that

$$A \subset \bigcup_{i=1}^N B(x_i; \frac{\varepsilon}{2}).$$

Taking closures and using  $\overline{B(x; r)} \subseteq B(x; 2r)$  (or directly, in a metric space,  $\overline{B(x; \varepsilon/2)} \subseteq \overline{B(x; \varepsilon/2)} \subset B(x; \varepsilon)$ ), we get

$$\overline{A} \subset \bigcup_{i=1}^N \overline{B(x_i; \frac{\varepsilon}{2})} \subset \bigcup_{i=1}^N B(x_i; \varepsilon).$$

Thus  $\overline{A}$  is covered by finitely many  $\varepsilon$ -balls. Since  $\varepsilon > 0$  was arbitrary,  $\overline{A}$  is totally bounded. ■

## 2.5 Continuity

**3.** We say that  $f : X \rightarrow \mathbb{C}$  is bounded if there is  $M > 0$  with  $|f(x)| \leq M$  for all  $x \in X$ . Show that if  $f$  and  $g$  are bounded uniformly continuous (Lipschitz) functions from  $X$  into  $\mathbb{C}$ , then so is  $fg$ .

*Proof.* Let  $(X, d)$  be a metric space and let  $f, g : X \rightarrow \mathbb{C}$  be bounded. Set  $M_f := \sup_{x \in X} |f(x)| < \infty$  and  $M_g := \sup_{x \in X} |g(x)| < \infty$ . For all  $x, y \in X$  we have

$$\begin{aligned} |f(x)g(x) - f(y)g(y)| &\leq |f(x)| |g(x) - g(y)| + |g(y)| |f(x) - f(y)| \\ &\leq M_f |g(x) - g(y)| + M_g |f(x) - f(y)|. \end{aligned} \quad (*)$$

**Uniform continuity.** Assume  $f$  and  $g$  are uniformly continuous. Fix  $\varepsilon > 0$  and let  $A := \max\{1, M_f\}$ ,  $B := \max\{1, M_g\}$ . By uniform continuity, there exist  $\delta_1, \delta_2 > 0$  such that

$$d(x, y) < \delta_1 \implies |g(x) - g(y)| < \frac{\varepsilon}{2A}, \quad d(x, y) < \delta_2 \implies |f(x) - f(y)| < \frac{\varepsilon}{2B}.$$

With  $\delta := \min\{\delta_1, \delta_2\}$ , inequality  $(*)$  gives, for  $d(x, y) < \delta$ ,

$$|f(x)g(x) - f(y)g(y)| \leq M_f \frac{\varepsilon}{2A} + M_g \frac{\varepsilon}{2B} \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence  $fg$  is uniformly continuous.

**Lipschitz case.** Assume in addition that  $f$  and  $g$  are Lipschitz with constants  $L_f, L_g$ :  $|f(x) - f(y)| \leq L_f d(x, y)$  and  $|g(x) - g(y)| \leq L_g d(x, y)$ . Then by  $(*)$ ,

$$|f(x)g(x) - f(y)g(y)| \leq M_f L_g d(x, y) + M_g L_f d(x, y) = (M_f L_g + M_g L_f) d(x, y),$$

so  $fg$  is Lipschitz with constant  $M_f L_g + M_g L_f$ .

Thus in either case, the product of bounded uniformly continuous (respectively Lipschitz) functions is uniformly continuous (respectively Lipschitz). ■

**4.** Is the composition of two uniformly continuous (Lipschitz) functions again uniformly continuous (Lipschitz)?

*Proof.* Let  $(X, d_X), (Y, d_Y), (Z, d_Z)$  be metric spaces. Suppose  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$ .

**Uniform continuity is preserved under composition.** Assume  $f$  and  $g$  are uniformly continuous. Given  $\varepsilon > 0$ , by uniform continuity of  $g$  there exists  $\eta > 0$

such that

$$d_Y(y_1, y_2) < \eta \implies d_Z(g(y_1), g(y_2)) < \varepsilon.$$

By uniform continuity of  $f$ , there exists  $\delta > 0$  such that

$$d_X(x_1, x_2) < \delta \implies d_Y(f(x_1), f(x_2)) < \eta.$$

Therefore, for  $d_X(x_1, x_2) < \delta$  we have

$$d_Z(g \circ f(x_1), g \circ f(x_2)) = d_Z(g(f(x_1)), g(f(x_2))) < \varepsilon,$$

so  $g \circ f$  is uniformly continuous.

**Lipschitz constants multiply.** Assume  $f$  is Lipschitz with constant  $L_f$  and  $g$  is Lipschitz with constant  $L_g$ :

$$d_Y(f(x_1), f(x_2)) \leq L_f d_X(x_1, x_2), \quad d_Z(g(y_1), g(y_2)) \leq L_g d_Y(y_1, y_2).$$

Then for all  $x_1, x_2 \in X$ ,

$$d_Z(g \circ f(x_1), g \circ f(x_2)) \leq L_g d_Y(f(x_1), f(x_2)) \leq L_g L_f d_X(x_1, x_2),$$

so  $g \circ f$  is Lipschitz with constant  $L_g L_f$  (hence uniformly continuous). ■

**6.** Recall the definition of a dense set (1.14). Suppose that  $\Omega$  is a complete metric space and that  $f : (D, d) \rightarrow (\Omega, \rho)$  is uniformly continuous, where  $D$  is dense in  $(X, d)$ . Use Exercise 5 to show that there is a uniformly continuous function  $g : X \rightarrow \Omega$  with  $g(x) = f(x)$  for every  $x \in D$ .

*Proof.* **Step 1: Define  $g$  by limits.** Since  $D$  is dense in  $X$ , for each  $x \in X$  there exists a sequence  $(x_n) \subset D$  with  $x_n \rightarrow x$  in  $(X, d)$ . Because  $f$  is uniformly continuous,  $(f(x_n))$  is a Cauchy sequence in  $(\Omega, \rho)$  (Exercise 5). By completeness of  $\Omega$ , the  $\lim_{n \rightarrow \infty} f(x_n)$  exists. Define

$$g(x) := \lim_{n \rightarrow \infty} f(x_n).$$

**Step 2: Well-definedness (independence of the approximating sequence).** Suppose  $(x_n)$  and  $(y_n)$  are two sequences in  $D$  with  $x_n \rightarrow x$  and  $y_n \rightarrow x$ . Then  $d(x_n, y_n) \leq d(x_n, x) + d(x, y_n) \rightarrow 0$ . By uniform continuity of  $f$ ,  $\rho(f(x_n), f(y_n)) \rightarrow 0$ , so the two limits coincide. Thus  $g$  is well-defined.

**Step 3: Agreement with  $f$  on  $D$ .** If  $x \in D$ , choose the constant sequence  $x_n = x$ . Then  $g(x) = \lim_n f(x_n) = f(x)$ .

**Step 4:  $g$  is uniformly continuous.** Let  $\varepsilon > 0$  and choose  $\delta > 0$  witnessing uniform continuity of  $f$  on  $D$ :  $d(u, v) < \delta \implies \rho(f(u), f(v)) < \varepsilon$  for all  $u, v \in D$ . Fix

$x, y \in X$  with  $d(x, y) < \delta/3$ . Choose sequences  $(x_n), (y_n) \subset D$  with  $x_n \rightarrow x$  and  $y_n \rightarrow y$ , and also  $d(x_n, x) < \delta/3$ ,  $d(y_n, y) < \delta/3$  for all  $n$  large. Then for such  $n$ ,

$$d(x_n, y_n) \leq d(x_n, x) + d(x, y) + d(y, y_n) < \frac{\delta}{3} + \frac{\delta}{3} + \frac{\delta}{3} = \delta,$$

hence  $\rho(f(x_n), f(y_n)) < \varepsilon$ . Passing to the limit in  $n$  gives

$$\rho(g(x), g(y)) = \rho(\lim_n f(x_n), \lim_n f(y_n)) \leq \limsup_{n \rightarrow \infty} \rho(f(x_n), f(y_n)) \leq \varepsilon.$$

Since this holds whenever  $d(x, y) < \delta/3$ ,  $g$  is uniformly continuous.

**(Optional) Uniqueness.** If  $h : X \rightarrow \Omega$  is uniformly continuous and  $h = f$  on  $D$ , then for each  $x \in X$  and any  $x_n \in D$  with  $x_n \rightarrow x$ , continuity of  $h$  gives  $h(x) = \lim_n h(x_n) = \lim_n f(x_n) = g(x)$ , so  $h = g$ . Thus  $f$  extends uniquely to a uniformly continuous  $g : X \rightarrow \Omega$  agreeing with  $f$  on  $D$ . ■

**9.** Prove the following converse to Exercise 2.5. Suppose  $(X, d)$  is a compact metric space having the property that for every  $\epsilon > 0$  and for any points  $a, b \in X$ , there are points  $z_0, z_1, \dots, z_n \in X$  with  $z_0 = a$ ,  $z_n = b$ , and  $d(z_{k-1}, z_k) < \epsilon$  for  $1 \leq k \leq n$ . Then  $(X, d)$  is connected. (Hint: Use Theorem 5.17.)

*Proof.* Assume for contradiction that  $X$  is disconnected. Then  $X = U \cup V$  where  $U, V \subseteq X$  are nonempty, disjoint, open sets. Since  $X$  is compact, the closed sets  $\bar{U}$  and  $\bar{V}$  are also compact and disjoint. By Theorem 5.17, the distance

$$\delta := \inf\{d(x, y) : x \in \bar{U}, y \in \bar{V}\}$$

is attained and strictly positive. Thus  $\delta > 0$ . Set  $\varepsilon := \delta/3$ . Choose  $a \in U$  and  $b \in V$ . By the  $\varepsilon$ -chain property, there exists a chain  $z_0 = a, z_1, \dots, z_n = b$  with  $d(z_{k-1}, z_k) < \varepsilon$ . Let  $k$  be the least index such that  $z_k \in \bar{V}$ . Then  $z_{k-1} \in \bar{U}$ . Hence

$$d(z_{k-1}, z_k) \geq \delta = 3\varepsilon,$$

contradicting  $d(z_{k-1}, z_k) < \varepsilon$ . This contradiction shows that  $X$  cannot be disconnected. Therefore  $(X, d)$  is connected. ■

## 2.6 Uniform convergence

**1.** Let  $\{f_n\}$  be a sequence of uniformly continuous functions from  $(X, d)$  into  $(\Omega, \rho)$  and suppose that  $f = u\text{-}\lim f_n$  exists. Prove that  $f$  is uniformly continuous. If each  $f_n$  is a Lipschitz function with constant  $M_n$  and  $\sup_n M_n < \infty$ , show that  $f$  is a Lipschitz function. If  $\sup_n M_n = \infty$ , show that  $f$  may fail to be Lipschitz.



**Proof. Uniform limit of uniformly continuous functions is uniformly continuous.** Let  $\varepsilon > 0$ . Since  $f_n \rightarrow f$  uniformly, choose  $N$  with  $\sup_{x \in X} \rho(f_N(x), f(x)) < \varepsilon/3$ . Because  $f_N$  is uniformly continuous, there exists  $\delta > 0$  such that  $d(x, y) < \delta \Rightarrow \rho(f_N(x), f_N(y)) < \varepsilon/3$ . Then for  $d(x, y) < \delta$ ,

$$\rho(f(x), f(y)) \leq \rho(f(x), f_N(x)) + \rho(f_N(x), f_N(y)) + \rho(f_N(y), f(y)) < \varepsilon,$$

so  $f$  is uniformly continuous.

**Uniform limit of Lipschitz functions with bounded constants is Lipschitz.** Assume each  $f_n$  is Lipschitz with constant  $M_n$  and set  $M := \sup_n M_n < \infty$ . For any  $x, y \in X$  and any  $n$ ,

$$\rho(f(x), f(y)) \leq \rho(f(x), f_n(x)) + \rho(f_n(x), f_n(y)) + \rho(f_n(y), f(y)) \leq 2\|f - f_n\|_\infty + M d(x, y).$$

Letting  $n \rightarrow \infty$  (uniform convergence) yields  $\rho(f(x), f(y)) \leq M d(x, y)$ . Hence  $f$  is Lipschitz with constant at most  $M$ .

**If the Lipschitz constants are unbounded, the limit need not be Lipschitz.** Take  $X = [0, 1]$  with the usual metric,  $\Omega = \mathbb{R}$ , and define

$$f_n(x) := \sqrt{x + \frac{1}{n}}, \quad x \in [0, 1].$$

Then  $f_n \rightarrow f$  uniformly on  $[0, 1]$ , where  $f(x) = \sqrt{x}$ , since  $\sup_{x \in [0, 1]} |f_n(x) - f(x)| = \sqrt{1/n} \rightarrow 0$ . Each  $f_n$  is Lipschitz with constant

$$M_n = \sup_{x \in [0, 1]} \left| \frac{d}{dx} \sqrt{x + \frac{1}{n}} \right| = \sup_{x \in [0, 1]} \frac{1}{2\sqrt{x + \frac{1}{n}}} = \frac{\sqrt{n}}{2} \xrightarrow{n \rightarrow \infty} \infty.$$

However, the limit  $f(x) = \sqrt{x}$  is *not* Lipschitz on  $[0, 1]$ : if it were with constant  $L$ , then for  $x > 0$ ,

$$\sqrt{x} = |f(x) - f(0)| \leq L|x - 0| = Lx \implies \frac{\sqrt{x}}{x} \leq L,$$

but  $\sqrt{x}/x = 1/\sqrt{x} \rightarrow \infty$  as  $x \downarrow 0$ , a contradiction. Therefore, boundedness of the Lipschitz constants  $\{M_n\}$  is necessary to conclude that the uniform limit  $f$  is Lipschitz. ■

## Chapter 3

# Elementary Properties and Example of Analytic Function

### 3.1 Power series

6. Find the radius of convergence for each of the following power series:

$$(a) \sum_{n=0}^{\infty} a^n z^n, \quad a \in \mathbb{C};$$

$$(b) \sum_{n=0}^{\infty} a^{n^2} z^n, \quad a \in \mathbb{C};$$

$$(c) \sum_{n=0}^{\infty} k^n z^n, \quad k \text{ an integer } \neq 0;$$

$$(d) \sum_{n=0}^{\infty} z^{n!}.$$

*Proof.* (a) Here  $a_n = a^n$ . Then  $\limsup |a_n|^{1/n} = |a|$ , hence  $R = 1/|a|$  if  $a \neq 0$ , while  $R = \infty$  if  $a = 0$  (the series is then a constant).

(b) Here  $a_n = a^{n^2}$ , so  $|a_n|^{1/n} = |a|^n$ . Therefore

$$R = \begin{cases} \infty, & |a| < 1, \\ 1, & |a| = 1, \\ 0, & |a| > 1. \end{cases}$$

(c) Here  $a_n = k^n$  with  $k \neq 0$ . Then  $\limsup |a_n|^{1/n} = |k|$ , so  $R = 1/|k|$ .

(d) The coefficients are  $a_{n!} = 1$  and  $c_m = 0$  for non-factorial  $m$ . Thus

$$|a_n|^{1/n} = \begin{cases} 1, & n = m! \text{ for some } m, \\ 0, & \text{otherwise,} \end{cases}$$

so  $\limsup_{n \rightarrow \infty} |a_n|^{1/n} = 1$  (since the value 1 occurs infinitely often). Hence  $R = 1$ . ■

**7.** Show that the radius of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} z^{n(n+1)}$$

is 1, and discuss convergence for  $z = 1, -1$ , and  $i$ . (Hint: The  $n$ th coefficient of this series is not  $\frac{(-1)^n}{n}$ .)

*Proof.* Let the general term of the power series be  $a_m z^m$ , where

$$a_m = \begin{cases} \frac{(-1)^n}{n}, & m = n(n+1) \text{ for some } n \in \mathbb{N}, \\ 0, & \text{otherwise.} \end{cases}$$

If  $m = n(n+1)$ , then  $|a_m|^{1/m} = n^{-1/(n(n+1))} \rightarrow 1$  as  $n \rightarrow \infty$ , while for all other  $m$ ,  $|a_m|^{1/m} = 0$ . Therefore,

$$\limsup_{m \rightarrow \infty} |a_m|^{1/m} = 1 \implies R = 1.$$

**Convergence on the boundary  $|z| = 1$ :**

- For  $z = 1$ : the series becomes

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n},$$

which is the alternating harmonic series, convergent (conditionally) to  $-\ln 2$ .

- For  $z = -1$ : since  $(-1)^{n(n+1)} = 1$  (because  $n(n+1)$  is even), the series again reduces to the alternating harmonic series and hence converges conditionally.
- For  $z = i$ : note that  $n(n+1)$  is even, so we can write  $n(n+1) = 2k$ . Then  $i^{n(n+1)} = i^{2k} = (-1)^k = (-1)^{n(n+1)/2}$ . Thus,

$$\frac{(-1)^n}{n} i^{n(n+1)} = \frac{(-1)^{n+n(n+1)/2}}{n}.$$

The factor  $(-1)^{n+n(n+1)/2}$  is periodic with period 4 and alternates in sign in a bounded way. Since  $\frac{1}{n} \rightarrow 0$  and the partial sums of the periodic sequence are bounded, the Dirichlet test ensures convergence. Hence, the series converges (conditionally) for  $z = i$  as well.

Therefore, the radius of convergence is  $R = 1$ , and the series converges conditionally for  $z = 1$ ,  $z = -1$ , and  $z = i$ . ■

## 3.2 Analytic functions

1. Show that  $f(z) = |z|^2 = x^2 + y^2$  has a derivative only at the origin.

*Proof.* Let  $f(z) = |z|^2 = x^2 + y^2$  with  $z = x + iy$ . Writing  $f = u + iv$  gives  $u(x, y) = x^2 + y^2$  and  $v(x, y) = 0$ , so

$$u_x = 2x, \quad u_y = 2y, \quad v_x = 0, \quad v_y = 0.$$

The Cauchy–Riemann equations  $u_x = v_y$  and  $u_y = -v_x$  force  $2x = 0$  and  $2y = 0$ , hence they hold only at  $(x, y) = (0, 0)$ . Since the partials are continuous everywhere,  $f$  can be complex differentiable only at the origin. At  $z = 0$ ,

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{|h|^2}{h} = \lim_{h \rightarrow 0} \bar{h} = 0.$$

Therefore  $f$  has a derivative only at the origin, and  $f'(0) = 0$ . ■

**10.** Prove the following generalization of Proposition 2.20. Let  $G$  and  $\Omega$  be open in  $\mathbb{C}$  and suppose  $f, h : G \rightarrow \mathbb{C}$ ,  $g : \Omega \rightarrow \mathbb{C}$ , with  $f(G) \subset \Omega$ . Assume that  $g$  and  $h$  are analytic, that  $g'(\omega) \neq 0$  for every  $\omega \in \Omega$ , that  $f$  is continuous,  $h$  is one-to-one, and that

$$h(z) = g(f(z)) \quad (z \in G).$$

Show that  $f$  is analytic. Give a formula for  $f'(z)$ .

*Proof.* Fix  $a \in G$  and let  $h \in \mathbb{C}$  with  $h \neq 0$  and  $a + h \in G$ . Write the given identity  $h(z) = g(f(z))$  at the points  $a$  and  $a + h$ :

$$h(a + h) - h(a) = g(f(a + h)) - g(f(a)).$$

Because  $h$  is one-to-one,  $h(a + h) \neq h(a)$ ; hence  $f(a + h) \neq f(a)$ , so we may divide

by  $f(a+h) - f(a)$ . Then

$$\frac{h(a+h) - h(a)}{h} = \frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} \frac{f(a+h) - f(a)}{h}. \quad (*)$$

Since  $f$  is continuous at  $a$  and  $g$  is analytic, as  $h \rightarrow 0$  we have  $f(a+h) \rightarrow f(a)$  and therefore

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} \rightarrow g'(f(a)).$$

Also, because  $h$  is analytic, the left side of  $(*)$  tends to  $h'(a)$ . Thus the limit of the right side exists, which shows  $f$  is differentiable at  $a$  and

$$h'(a) = g'(f(a)) f'(a).$$

Hence

$$f'(a) = \frac{h'(a)}{g'(f(a))}.$$

Since  $a \in G$  was arbitrary,  $f$  is differentiable on  $G$ . Moreover,  $h'$  is analytic,  $g'$  is continuous and nowhere zero on  $\Omega$ , and  $f$  is continuous, so  $z \mapsto \frac{h'(z)}{g'(f(z))}$  is continuous on  $G$ . Therefore  $f$  is analytic on  $G$  with

$$f'(z) = \frac{h'(z)}{g'(f(z))}$$

■

**12.** Show that the real part of the function  $z^{\frac{1}{2}}$  is always positive.

*Proof.* Let  $z = re^{i\theta}$  with  $r > 0$  and  $-\pi < \theta \leq \pi$ . The principal branch of the square root is defined by

$$z^{1/2} = \sqrt{r} e^{i\theta/2} = \sqrt{r} (\cos(\frac{\theta}{2}) + i \sin(\frac{\theta}{2})).$$

Hence the real part of  $z^{1/2}$  is

$$\Re(z^{1/2}) = \sqrt{r} \cos\left(\frac{\theta}{2}\right).$$

Since  $-\pi < \theta \leq \pi$ , we have

$$-\frac{\pi}{2} < \frac{\theta}{2} \leq \frac{\pi}{2},$$

and therefore  $\cos(\frac{\theta}{2}) > 0$ . As  $\sqrt{r} > 0$ , it follows that

$$\Re(z^{1/2}) = \sqrt{r} \cos\left(\frac{\theta}{2}\right) > 0.$$

Thus, the real part of  $z^{1/2}$  is always positive for the principal branch. ■

**14.** Suppose  $f : G \rightarrow \mathbb{C}$  is analytic and  $G$  is connected. Show that if  $f(z) \in \mathbb{R}$  for all  $z \in G$ , then  $f$  is constant.

*Proof.* Write  $f = u + iv$  with  $u, v : G \rightarrow \mathbb{R}$ . The hypothesis  $f(G) \subset \mathbb{R}$  means  $v \equiv 0$  on  $G$ . Since  $f$  is analytic, the Cauchy–Riemann equations hold:

$$u_x = v_y, \quad u_y = -v_x.$$

But  $v \equiv 0$  implies  $v_x = v_y = 0$ , hence  $u_x = u_y = 0$  on  $G$ . So  $u$  is constant on each connected component of  $G$ . As  $G$  is connected,  $u$  is constant on  $G$ , and thus  $f = u + iv = u$  is constant. ■

**16.** Find an open connected set  $G \subset \mathbb{C}$  and two continuous functions  $f, g$  defined on  $G$  such that

$$f(z)^2 = g(z)^2 = 1 - z^2 \quad (z \in G).$$

Can you make  $G$  maximal? Are  $f$  and  $g$  analytic?

*Proof.* Define

$$f(z) := \exp\left(\frac{1}{2} \ln(1 - z^2)\right), \quad g(z) := \exp\left(i\pi + \frac{1}{2} \ln(1 - z^2)\right),$$

where  $\ln$  denotes the principal branch of the logarithm. Then

$$f(z)^2 = g(z)^2 = 1 - z^2 \quad (z \in G).$$

To determine the domain, let  $w = 1 - z^2$ . The principal logarithm is analytic on

$$\mathbb{C} \setminus (-\infty, 0] = \mathbb{C} - \{w \in \mathbb{R} : w \leq 0\}.$$

Hence we require

$$1 - z^2 \notin (-\infty, 0].$$

If  $1 - z^2 \in (-\infty, 0]$ , then  $z^2 \in [1, \infty)$ , which forces  $z \in (-\infty, -1] \cup [1, \infty)$ . Thus the preimage of the branch cut  $(-\infty, 0]$  under  $w = 1 - z^2$  is precisely the two real rays  $(-\infty, -1] \cup [1, \infty)$ . Therefore

$$G = \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty)).$$

On this open and connected domain  $G$ , the function  $\ln(1 - z^2)$  is analytic, and

hence so are  $f$  and  $g$ . Moreover,

$$f^2 = g^2 = 1 - z^2$$

holds throughout  $G$ , as required. ■

**19.** Let  $G$  be a region and define  $G^* = \{z : \bar{z} \in G\}$ . If  $f : G \rightarrow \mathbb{C}$  is analytic, prove that

$$f^* : G^* \rightarrow \mathbb{C}, \quad f^*(z) = \overline{f(\bar{z})},$$

is also analytic.

*Proof.* First note that  $G$  open  $\Rightarrow G^* = \{z : \bar{z} \in G\}$  is open. Fix  $z_0 \in G^*$ , so  $\bar{z}_0 \in G$ . Consider

$$\frac{f^*(z) - f^*(z_0)}{z - z_0} = \frac{\overline{f(\bar{z})} - \overline{f(\bar{z}_0)}}{z - z_0} = \frac{\overline{f(\bar{z}) - f(\bar{z}_0)}}{\bar{z} - \bar{z}_0} = \overline{\left( \frac{f(\bar{z}) - f(\bar{z}_0)}{\bar{z} - \bar{z}_0} \right)}.$$

As  $z \rightarrow z_0$  we have  $\bar{z} \rightarrow \bar{z}_0$ , and since  $f$  is analytic at  $\bar{z}_0$  the limit

$$\lim_{w \rightarrow \bar{z}_0} \frac{f(w) - f(\bar{z}_0)}{w - \bar{z}_0} = f'(\bar{z}_0)$$

exists. Therefore

$$\lim_{z \rightarrow z_0} \frac{f^*(z) - f^*(z_0)}{z - z_0} = \overline{f'(\bar{z}_0)}.$$

Hence  $f^*$  is differentiable at  $z_0$ . Since  $z_0 \in G^*$  was arbitrary,  $f^*$  is analytic on  $G^*$ , with

$$(f^*)'(z) = \overline{f'(\bar{z})} \quad (z \in G^*).$$
■

**21.** Prove that there is no branch of the logarithm defined on  $G = \mathbb{C} \setminus \{0\}$ . (Hint: Suppose such a branch exists and compare this with the principal branch.)

*Proof.* Suppose, for contradiction, that there exists a branch of the logarithm  $L : \mathbb{C} \setminus \{0\} \rightarrow \mathbb{C}$  such that  $e^{L(z)} = z$  for all  $z \in \mathbb{C} \setminus \{0\}$ . Consider the unit circle parametrized by  $\gamma(\theta) = e^{i\theta}$ ,  $0 \leq \theta \leq 2\pi$ , and define

$$h(\theta) := L(\gamma(\theta)).$$

Since  $L$  is continuous,  $h$  is continuous on  $[0, 2\pi]$ . Moreover, because  $e^{L(z)} = z$ , we have

$$e^{h(\theta)} = e^{i\theta}.$$

Thus, for each  $\theta$ , there exists an integer  $n(\theta) \in \mathbb{Z}$  such that

$$h(\theta) = i\theta + 2\pi i n(\theta).$$

The map  $\theta \mapsto n(\theta)$  is continuous because both  $h(\theta)$  and  $i\theta$  are continuous. Since  $n(\theta)$  takes values in the discrete set  $\mathbb{Z}$ , it must be constant on the connected interval  $[0, 2\pi]$ . Hence  $n(\theta) \equiv k$  for some fixed integer  $k$ .

Therefore

$$h(\theta) = i\theta + 2\pi i k.$$

In particular, at  $\theta = 0$  and  $\theta = 2\pi$ ,

$$L(1) = h(0) = 2\pi i k \quad \text{and} \quad L(1) = h(2\pi) = i(2\pi) + 2\pi i k.$$

Equating these values yields  $2\pi i k = 2\pi i + 2\pi i k$ , which is impossible. This contradiction shows that no such branch of the logarithm exists on  $\mathbb{C} \setminus \{0\}$ . ■

### 3.3 Analytic functions as mappings, Möbius transformation

**4.** Discuss the mapping properties of  $z^n$  and  $z^{1/n}$  for  $n \geq 2$ . (Hint: use polar coordinates.)

*Discussion.* **1. The map  $z \mapsto z^n$ .**

Write  $z = re^{i\theta}$  in polar coordinates, with  $r \geq 0$  and  $\theta \in \mathbb{R}$ . Then

$$z^n = r^n e^{in\theta}.$$

- *Modulus:*  $|z^n| = r^n$ . Circles  $|z| = r_0$  map to circles  $|w| = r_0^n$ .
- *Argument:*  $\arg(z^n) = n\theta \pmod{2\pi}$ . Rays  $\theta = \theta_0$  map to rays of argument  $n\theta_0$ .
- *Sectors:* A sector of opening  $\alpha$  is mapped to one of opening  $n\alpha$ .
- *Annuli:*  $r_1 < |z| < r_2$  maps to  $r_1^n < |w| < r_2^n$ .
- *Multiplicity:* On  $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ , the map is  $n$ -to-1: each  $w \neq 0$  has  $n$  preimages  $w^{1/n} e^{2\pi i k/n}$ , with  $k = 0, \dots, n-1$ .
- *Conformality:* The map is holomorphic everywhere and conformal except at 0, where  $f'(0) = 0$  and the map is not injective.

In particular, the unit circle  $|z| = 1$  is mapped to itself and is wound around  $n$  times.



## 2. The map $z \mapsto z^{1/n}$ .

The  $n$ th root function is multivalued on  $\mathbb{C}^*$ . In polar form,

$$z^{1/n} = r^{1/n} e^{i(\theta+2\pi k)/n}, \quad k = 0, \dots, n-1.$$

Thus there are  $n$  branches differing by rotation by  $2\pi/n$ .

To obtain a single-valued branch, one must restrict the argument (e.g.  $-\pi < \theta < \pi$ ). This introduces a branch cut.

- *Principal branch:* Define  $\arg z \in (-\pi, \pi)$  (cut along  $(-\infty, 0]$ ). Then

$$z^{1/n} = r^{1/n} e^{i\theta/n}$$

is single-valued and holomorphic on  $\mathbb{C} \setminus (-\infty, 0]$ .

- *Image of principal branch:* Under this branch, the image is the sector

$$\{ w : \arg w \in (-\pi/n, \pi/n) \}.$$

Radii map as  $r \mapsto r^{1/n}$ , while angles divide by  $n$ .

- *Other branches:* Rotating the branch cut yields  $n$  distinct branches whose images are  $n$  disjoint sectors of opening  $2\pi/n$ .
- *Conformality:* Each branch is holomorphic and conformal on its domain, which is simply connected and does not encircle 0. The point 0 is a branch point: one cannot define a single-valued analytic  $n$ th root around a loop surrounding 0.

**Summary.** The map  $z^n$  multiplies arguments by  $n$  and raises moduli to the  $n$ th power; it is  $n$ -to-1 on  $\mathbb{C}^*$  and wraps the unit circle  $n$  times around itself. Each branch of  $z^{1/n}$  is single-valued and conformal on a slit plane, dividing arguments by  $n$  and taking  $n$ th roots of moduli, with image a sector of opening  $2\pi/n$ . ■

**8.** If  $Tz = \frac{az+b}{cz+d}$ , show that  $T(\mathbb{R}_\infty) = \mathbb{R}_\infty$  iff we can choose  $a, b, c, d$  to be real numbers.

*Proof.* Let  $T(z) = \frac{az+b}{cz+d}$  with  $ad-bc \neq 0$  and suppose  $T(\mathbb{R}_\infty) = \mathbb{R}_\infty$ . Evaluate  $T$  at three real points  $0, 1, \infty$ .

**At 0:**  $T(0) = \frac{b}{d} \in \mathbb{R}_\infty$ . Thus either  $d = 0$  or  $r_1 := \frac{b}{d} \in \mathbb{R}$ .

**At  $\infty$ :**  $T(\infty) = \frac{a}{c} \in \mathbb{R}_\infty$ . Thus either  $c = 0$  or  $r_2 := \frac{a}{c} \in \mathbb{R}$ .

**At 1:**  $T(1) = \frac{a+b}{c+d} \in \mathbb{R}_\infty$ . Thus either  $c+d=0$  or  $k := \frac{a+b}{c+d} \in \mathbb{R}$ .

We treat cases.

*Case 1:*  $c, d \neq 0$  and  $c+d \neq 0$ . From  $b = r_1 d$  and  $a = r_2 c$  we have

$$k = \frac{a+b}{c+d} = \frac{r_2 c + r_1 d}{c+d} \in \mathbb{R}.$$

Hence  $(r_2 - k)c + (r_1 - k)d = 0$ , so

$$\frac{c}{d} = \frac{k - r_1}{r_2 - k} \in \mathbb{R}.$$

Therefore  $c/d$ ,  $b/d$ , and  $a/c$  are all real. Choose a unimodular  $\lambda = e^{-i \arg d}$  so that  $d_0 := \lambda d \in \mathbb{R}$ . Then  $b_0 := \lambda b = (b/d)d_0 \in \mathbb{R}$ ,  $c_0 := \lambda c = (c/d)d_0 \in \mathbb{R}$ , and  $a_0 := \lambda a = (a/c)c_0 \in \mathbb{R}$ . Scaling the coefficients by the common nonzero factor  $\lambda$  does not change  $T$ , so we have represented  $T$  with real coefficients.

*Case 2:*  $c = 0$ . Then  $T(z) = \frac{az+b}{d}$  and  $T(\mathbb{R}_\infty) \subset \mathbb{R}_\infty$  gives  $\frac{b}{d} = T(0) \in \mathbb{R}_\infty$  and  $\frac{a+b}{d} = T(1) \in \mathbb{R}_\infty$ , so  $\frac{a}{d}$  and  $\frac{b}{d}$  are real (allowing  $d=0$  would force  $b=0$  and then  $\frac{a}{d} = \infty$  real). Choose  $\lambda$  with  $\lambda d \in \mathbb{R}$ ; then  $\lambda a, \lambda b, \lambda d \in \mathbb{R}$ , and  $c_0 = \lambda c = 0$  is real.

*Case 3:*  $d = 0$ . Then  $T(0) = \infty \in \mathbb{R}_\infty$  and from  $T(\infty) = \frac{a}{c} \in \mathbb{R}_\infty$  and  $T(1) = \frac{a+b}{c} \in \mathbb{R}_\infty$  we get  $\frac{a}{c}, \frac{b}{c} \in \mathbb{R}$ . Choose  $\lambda$  with  $\lambda c \in \mathbb{R}$ ; then  $\lambda a, \lambda b, \lambda c \in \mathbb{R}$ , and  $d_0 = \lambda d = 0$  is real.

In all cases we can multiply  $(a, b, c, d)$  by a nonzero constant to obtain real coefficients.

Conversely, if  $a, b, c, d \in \mathbb{R}$ , then for  $x \in \mathbb{R}$ ,  $T(x) = \frac{ax+b}{cx+d} \in \mathbb{R}_\infty$  and  $T(\infty) = a/c \in \mathbb{R}_\infty$ , so  $T(\mathbb{R}_\infty) = \mathbb{R}_\infty$ .

Thus  $T(\mathbb{R}_\infty) = \mathbb{R}_\infty$  iff the coefficients can be chosen real. ■

**10.** Let  $D = \{z : |z| < 1\}$  and find all Möbius transformations  $T$  such that  $T(D) = D$ .

*Proof.* Let  $\phi_a(z) = \frac{z-a}{1-\bar{a}z}$  for  $|a| < 1$ . Then  $\phi_a$  is a Möbius map with  $\phi_a(a) = 0$  and  $\phi_a(\mathbb{D}) = \mathbb{D}$ , where  $\mathbb{D} = D = \{z : |z| < 1\}$ ; moreover  $\phi_a^{-1} = \phi_{-a}$ .

**Claim.** A Möbius map  $T$  satisfies  $T(\mathbb{D}) = \mathbb{D}$  iff

$$T(z) = e^{i\theta} \phi_a(z) = e^{i\theta} \frac{z-a}{1-\bar{a}z} \quad \text{for some } a \in \mathbb{D}, \theta \in \mathbb{R}.$$

( $\subseteq$ ) Suppose  $T(\mathbb{D}) = \mathbb{D}$ . Set  $a := T^{-1}(0) \in \mathbb{D}$  and define

$$F := T \circ \phi_a^{-1} = T \circ \phi_{-a}.$$

Then  $F : \mathbb{D} \rightarrow \mathbb{D}$  is holomorphic and  $F(0) = 0$ . By Schwarz's lemma,  $F(z) = e^{i\theta} z$  for some  $\theta \in \mathbb{R}$ . Hence

$$T = F \circ \phi_a = e^{i\theta} \phi_a.$$

( $\supseteq$ ) Conversely, for any  $|a| < 1$  and  $\theta \in \mathbb{R}$ , both  $\phi_a$  and the rotation  $R_\theta(z) = e^{i\theta} z$  map  $\mathbb{D}$  onto  $\mathbb{D}$ , so their composition  $T = R_\theta \circ \phi_a$  also satisfies  $T(\mathbb{D}) = \mathbb{D}$ .

Therefore,

$$\text{Aut}(\mathbb{D}) = \left\{ z \mapsto e^{i\theta} \frac{z - a}{1 - \bar{a}z} : a \in \mathbb{D}, \theta \in \mathbb{R} \right\}.$$

■

**14.** Suppose that one circle is contained inside another and that they are tangent at the point  $a$ . Let  $G$  be the region between the two circles and map  $G$  conformally onto the open unit disk. (Hint: first try  $(z - a)^{-1}$ .)

*Proof. Normalization.* By translating we may assume the tangency point is  $a = 0$ . By a rotation (precomposition by a unimodular constant) we may assume both circles are centered on the positive real axis. Thus there exist radii  $0 < r_1 < r_2$  such that the two circles are

$$C_j : |z - r_j| = r_j \quad (j = 1, 2),$$

which are internally tangent at 0. Let  $G$  be the region between  $C_1$  and  $C_2$ .

**Step 1: Inversion sends tangent circles to parallel lines.** Set

$$w = \Phi_1(z) := \frac{1}{z}.$$

A circle  $|z - a| = |a|$  through the origin maps to the line  $\Re(aw) = \frac{1}{2}$ . Here  $a = r_j \in \mathbb{R}_+$ , so

$$\Phi_1(C_j) = \left\{ \Re w = \frac{1}{2r_j} \right\}.$$

Hence  $\Phi_1(G)$  is the open vertical strip

$$S := \left\{ w : \alpha < \Re w < \beta \right\}, \quad \alpha := \frac{1}{2r_2} < \beta := \frac{1}{2r_1}.$$

**Step 2: Rotate to a horizontal strip.** Let

$$u = \Phi_2(w) := i w.$$

Then  $\Im u = \Re w$ , so  $\Phi_2(S)$  is the horizontal strip

$$S_h := \{ u : \alpha < \Im u < \beta \}.$$

Set its width and midline as

$$d := \beta - \alpha = \frac{r_2 - r_1}{2r_1r_2}, \quad m := \frac{\alpha + \beta}{2} = \frac{r_1 + r_2}{4r_1r_2}.$$

**Step 3: Strip  $\rightarrow$  half-plane  $\rightarrow$  disk.** Define

$$\xi = \Phi_3(u) := \exp\left(\frac{\pi}{d}(u - im)\right),$$

which maps  $S_h$  conformally onto the right half-plane  $\{\Re \xi > 0\}$ . Then apply the Cayley map

$$\Phi_4(\xi) := \frac{\xi - 1}{\xi + 1},$$

which sends  $\{\Re \xi > 0\}$  conformally onto the unit disk  $\mathbb{D}$ .

**Conclusion.** The composition

$$F := \Phi_4 \circ \Phi_3 \circ \Phi_2 \circ \Phi_1$$

is a conformal bijection  $F : G \rightarrow \mathbb{D}$ . Writing it explicitly,

$$F(z) = \frac{\exp\left(\frac{\pi}{d}\left(\frac{i}{z} - im\right)\right) - 1}{\exp\left(\frac{\pi}{d}\left(\frac{i}{z} - im\right)\right) + 1} \quad \text{with} \quad d = \frac{r_2 - r_1}{2r_1r_2}, \quad m = \frac{r_1 + r_2}{4r_1r_2}.$$

Equivalently, using  $\frac{e^t - 1}{e^t + 1} = \tanh(t/2)$ ,

$$F(z) = \tanh\left(\frac{\pi}{2d}\left(\frac{i}{z} - im\right)\right).$$

Any other configuration (arbitrary  $a \neq 0$  and non-horizontal tangent) is obtained by pre/post-composing with a rotation/translation: replace  $z$  by  $z - a$  in  $\Phi_1$ , and possibly rotate by a unimodular constant before Step 2. In all cases, the boundary circles map to  $|w| = 1$ , and  $F$  is one-to-one and onto  $\mathbb{D}$ . ■

**15.** *Can you map the open unit disk conformally onto  $\{z : 0 < |z| < 1\}$ ?*

*Proof.* I am unsure how to prove this without using Cauchy's integral theorem, singularity theory, or the fact that conformal maps preserve topological properties such as simple connectedness. I know the answer is negative, since  $\mathbb{D}$  is simply connected whereas  $\mathbb{D} \setminus \{0\}$  is not, so there cannot exist a surjective conformal map between them. However, I would appreciate guidance on how to prove this using only the material from the first three chapters. ■

**16.** *Map  $G = \mathbb{C} \setminus \{z \in \mathbb{R} : -1 \leq z \leq 1\}$  onto the open unit disk by an analytic function  $f$ . Can  $f$  be one-to-one?*

*Proof.* Since  $G$  is not simply connected, any surjective analytic function from  $G$  onto the unit disk cannot be one-to-one. ■

**17.** Let  $G$  be a region and suppose that  $f : G \rightarrow \mathbb{C}$  is analytic such that  $f(G)$  is a subset of a circle. Show that  $f$  is constant.

*Proof.* Without loss of generality assume  $f(G) \subset \{w \in \mathbb{C} : |w| = 1\}$ . Then  $|f(z)| = 1$  for all  $z \in G$ , so  $\overline{f(z)} = 1/f(z)$  and  $f(z) \neq 0$ .

Define

$$F(z) = f(z) + \frac{1}{f(z)}.$$

Since  $f$  is analytic and never zero,  $1/f$  is analytic; hence  $F$  is analytic on  $G$ . Moreover,

$$F(z) = f(z) + \overline{f(z)} = 2\Re f(z) \in \mathbb{R} \quad (z \in G).$$

By Exercise 14 (Section 2), any real-valued analytic function on a connected domain is constant; therefore  $F \equiv c \in \mathbb{R}$ . Thus

$$f(z) + \frac{1}{f(z)} = c \iff f(z)^2 - cf(z) + 1 = 0 \quad (z \in G).$$

Let  $\alpha, \beta$  be the (constant) roots of  $w^2 - cw + 1 = 0$ . Then

$$(f(z) - \alpha)(f(z) - \beta) = 0 \quad \text{for all } z \in G,$$

so  $f(G) \subset \{\alpha, \beta\}$ . Since  $f$  is continuous and  $G$  is connected,  $f(G)$  is connected. The only connected subsets of the two-point set  $\{\alpha, \beta\}$  are singletons; hence  $f(G) = \{\alpha\}$  or  $f(G) = \{\beta\}$ . Therefore  $f$  is constant on  $G$ . ■

# Chapter 4

## Complex Integration

### 4.1 Riemann-Stieltjes integral

6. Show that if  $\gamma : [a, b] \rightarrow \mathbb{C}$  is a Lipschitz function, then  $\gamma$  is of bounded variation.

*Proof.* Let  $\gamma : [a, b] \rightarrow \mathbb{C}$  be Lipschitz with constant  $L \geq 0$ , i.e.,

$$|\gamma(s) - \gamma(t)| \leq L |s - t| \quad \text{for all } s, t \in [a, b].$$

For any partition  $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$  of  $[a, b]$ , the variation sum satisfies

$$\sum_{i=1}^n |\gamma(t_i) - \gamma(t_{i-1})| \leq \sum_{i=1}^n L |t_i - t_{i-1}| = L(b - a).$$

Taking the supremum over all partitions  $P$  gives

$$V(\gamma, [a, b]) = \sup_P \sum_{i=1}^n |\gamma(t_i) - \gamma(t_{i-1})| \leq L(b - a) < \infty.$$

Hence  $\gamma$  is of bounded variation on  $[a, b]$ . ■

8. Let  $\gamma$  and  $\sigma$  be the two polygons  $[1, i]$  and  $[1, 1 + i, i]$ . Express  $\gamma$  and  $\sigma$  as paths and calculate  $\int_{\gamma} f$  and  $\int_{\sigma} f$  where  $f(z) = |z|^2$ .

*Proof.* A straight segment from 1 to  $i$  can be parametrized by

$$\gamma(t) = 1 + t(i - 1) = (1 - t) + it, \quad t \in [0, 1], \quad \gamma'(t) = i - 1.$$

Hence  $|\gamma(t)|^2 = (1 - t)^2 + t^2 = 1 - 2t + 2t^2$ , and

$$\int_{\gamma} |z|^2 dz = \int_0^1 (1 - 2t + 2t^2) (i - 1) dt = (i - 1) \left[ t - t^2 + \frac{2}{3} t^3 \right]_0^1 = \frac{2}{3} (i - 1).$$

For the polygon  $[1, 1 + i, i]$ , write  $\sigma = \sigma_1 * \sigma_2$  with

$$\sigma_1(t) = 1 + it, \quad t \in [0, 1], \quad \sigma_1'(t) = i,$$

$$\sigma_2(t) = 1 - t + i, \quad t \in [0, 1], \quad \sigma_2'(t) = -1.$$

Then  $|\sigma_1(t)|^2 = 1 + t^2$  and  $|\sigma_2(t)|^2 = (1 - t)^2 + 1 = 2 - 2t + t^2$ . Thus

$$\int_{\sigma_1} |z|^2 dz = \int_0^1 (1 + t^2) i dt = i \left[ t + \frac{1}{3} t^3 \right]_0^1 = \frac{4}{3} i,$$

$$\int_{\sigma_2} |z|^2 dz = \int_0^1 (2 - 2t + t^2)(-1) dt = - \left[ 2t - t^2 + \frac{1}{3} t^3 \right]_0^1 = -\frac{4}{3}.$$

Therefore

$$\int_{\sigma} |z|^2 dz = \int_{\sigma_1} |z|^2 dz + \int_{\sigma_2} |z|^2 dz = \frac{4}{3} i - \frac{4}{3} = \frac{4}{3}(i - 1).$$

In particular,  $\int_{\gamma} |z|^2 dz = \frac{2}{3}(i - 1)$  and  $\int_{\sigma} |z|^2 dz = \frac{4}{3}(i - 1)$ , showing dependence on the path since  $|z|^2$  is not holomorphic. ■

**12.** Let  $I(r) = \int_{\gamma} \frac{e^{iz}}{z} dz$  where  $\gamma : [0, \pi] \rightarrow \mathbb{C}$  is defined by  $\gamma(t) = re^{it}$ . Show that  $\lim_{r \rightarrow \infty} I(r) = 0$ .

*Proof.* Let  $\gamma(t) = re^{it}$  for  $t \in [0, \pi]$ . Then  $\gamma'(t) = ire^{it}$  and

$$I(r) = \int_{\gamma} \frac{e^{iz}}{z} dz = \int_0^{\pi} \frac{e^{ire^{it}}}{re^{it}} (ire^{it}) dt = i \int_0^{\pi} e^{ire^{it}} dt.$$

Hence

$$|I(r)| \leq \int_0^{\pi} |e^{ire^{it}}| dt = \int_0^{\pi} e^{\operatorname{Re}(ire^{it})} dt = \int_0^{\pi} e^{-r \sin t} dt.$$

Fix  $0 < \varepsilon < \frac{\pi}{2}$  and split the integral:

$$\int_0^{\pi} e^{-r \sin t} dt = \int_0^{\varepsilon} e^{-r \sin t} dt + \int_{\varepsilon}^{\pi - \varepsilon} e^{-r \sin t} dt + \int_{\pi - \varepsilon}^{\pi} e^{-r \sin t} dt.$$

On  $[\varepsilon, \pi - \varepsilon]$ ,  $\sin t \geq \sin \varepsilon > 0$ , so

$$\int_{\varepsilon}^{\pi - \varepsilon} e^{-r \sin t} dt \leq \pi e^{-r \sin \varepsilon} \xrightarrow{r \rightarrow \infty} 0.$$



For  $t \in [0, \frac{\pi}{2}]$ ,  $\sin t \geq \frac{2}{\pi}t$ , hence

$$\int_0^\varepsilon e^{-r \sin t} dt \leq \int_0^\varepsilon e^{-\frac{2r}{\pi}t} dt = \frac{\pi}{2r} \left(1 - e^{-\frac{2r}{\pi}\varepsilon}\right) \leq \frac{\pi}{2r} \xrightarrow{r \rightarrow \infty} 0.$$

Similarly, by symmetry  $\sin t = \sin(\pi - t)$ ,

$$\int_{\pi-\varepsilon}^\pi e^{-r \sin t} dt = \int_0^\varepsilon e^{-r \sin s} ds \leq \frac{\pi}{2r} \xrightarrow{r \rightarrow \infty} 0.$$

Combining the estimates gives  $\int_0^\pi e^{-r \sin t} dt \rightarrow 0$ , hence  $|I(r)| \rightarrow 0$ . Therefore

$$\lim_{r \rightarrow \infty} I(r) = 0.$$

■

**13.** Find  $\int_\gamma z^{-1/2} dz$  where:

- (a)  $\gamma$  is the upper half of the unit circle from +1 to -1;
- (b)  $\gamma$  is the lower half of the unit circle from +1 to -1.

*Proof.* Take the principal branch of the square root, i.e.  $\arg z \in (-\pi, \pi)$ , so that

$$z^{-1/2} = \frac{1}{\sqrt{z}} = e^{-\frac{i}{2} \arg z} \quad \text{for } z \neq 0.$$

(a) **Upper semicircle from +1 to -1.**

Parametrize

$$\gamma(t) = e^{it}, \quad t \in [0, \pi], \quad \gamma'(t) = ie^{it}.$$

Along this arc,  $\arg \gamma(t) = t$ , hence  $\gamma(t)^{-1/2} = e^{-it/2}$ . Thus

$$\int_\gamma z^{-1/2} dz = \int_0^\pi e^{-it/2} ie^{it} dt = \int_0^\pi i e^{it/2} dt = 2(e^{i\pi/2} - 1) = 2(i - 1).$$

(b) **Lower semicircle from +1 to -1.**

Parametrize

$$\gamma(t) = e^{it}, \quad t \in [0, -\pi], \quad \gamma'(t) = ie^{it}.$$

Now  $\arg \gamma(t) = t \in (-\pi, 0]$ , so  $\gamma(t)^{-1/2} = e^{-it/2}$ . Hence

$$\int_\gamma z^{-1/2} dz = \int_0^{-\pi} e^{-it/2} ie^{it} dt = \int_0^{-\pi} i e^{it/2} dt = 2(e^{-i\pi/2} - 1) = -2(1 + i).$$

Therefore,

$$\int_{\text{upper arc}} z^{-1/2} dz = 2(i - 1), \quad \int_{\text{lower arc}} z^{-1/2} dz = -2(1 + i).$$

■

**21.** Show that if  $F_1$  and  $F_2$  are primitives for  $f : G \rightarrow \mathbb{C}$  and  $G$  is open and connected, then there is a constant  $c$  such that  $F_1(z) = c + F_2(z)$  for each  $z \in G$ .

*Proof.* Let  $H := F_1 - F_2$ . Then  $H$  is holomorphic on  $G$  and

$$H'(z) = F_1'(z) - F_2'(z) = f(z) - f(z) = 0 \quad (z \in G).$$

Fix  $z_0 \in G$ . Since  $G$  is open and connected (hence path-connected), for any  $z \in G$  there exists a piecewise  $C^1$  path  $\gamma$  in  $G$  from  $z_0$  to  $z$ . By the fundamental theorem of calculus for line integrals of holomorphic functions,

$$H(z) - H(z_0) = \int_{\gamma} H'(w) dw = \int_{\gamma} 0 dw = 0.$$

Thus  $H(z) = H(z_0) =: c$  for all  $z \in G$ . Therefore,

$$F_1(z) = F_2(z) + c \quad \text{for all } z \in G.$$

■

**22.** Let  $\gamma$  be a closed rectifiable curve in an open set  $G$  and  $a \notin G$ . Show that for  $n \geq 2$ ,  $\int_{\gamma} (z - a)^{-n} dz = 0$ .

*Proof.* Define

$$F(z) := -\frac{(z - a)^{-n+1}}{n - 1}, \quad z \in G.$$

Then  $F$  is analytic on  $G$  and

$$F'(z) = (z - a)^{-n} \quad (z \in G).$$

Thus  $(z - a)^{-n}$  has a primitive  $F$  on  $G$ . By the fundamental theorem of calculus for line integrals (or Cauchy's theorem), for any closed rectifiable curve  $\gamma \subset G$ ,

$$\int_{\gamma} (z - a)^{-n} dz = \int_{\gamma} F'(z) dz = 0.$$

■

**23.** Prove the following integration by parts formula. Let  $f$  and  $g$  be analytic in  $G$  and let  $\gamma$  be a rectifiable curve from  $a$  to  $b$  in  $G$ . Then

$$\int_{\gamma} f g' dz = f(b)g(b) - f(a)g(a) - \int_{\gamma} f' g dz.$$

*Proof.* Let  $\gamma : [\alpha, \beta] \rightarrow G$  be a rectifiable parametrization with  $\gamma(\alpha) = a$  and  $\gamma(\beta) = b$ . Since  $f$  and  $g$  are analytic, the product rule gives

$$(fg)' = f'g + fg' \quad \text{on } G.$$

Hence, by the fundamental theorem of calculus for line integrals,

$$\int_{\gamma} (fg)' dz = (fg)(b) - (fg)(a) = f(b)g(b) - f(a)g(a).$$

But

$$\int_{\gamma} (fg)' dz = \int_{\gamma} f'g dz + \int_{\gamma} fg' dz.$$

Rearranging gives

$$\int_{\gamma} fg' dz = f(b)g(b) - f(a)g(a) - \int_{\gamma} f'g dz,$$

which is the desired integration by parts formula. ■

## 4.2 Power series representation of analytic function

**2.** Prove the following analogue of Leibniz's rule. Let  $G$  be an open set and let  $\gamma$  be a rectifiable curve in  $\mathbb{C}$ . Suppose that  $\varphi : \{\gamma\} \times G \rightarrow \mathbb{C}$  is a continuous function and define

$$g(z) = \int_{\gamma} \varphi(w, z) dw.$$

Then  $g$  is continuous. If  $\frac{\partial \varphi}{\partial z}$  exists for each  $(w, z) \in \{\gamma\} \times G$  and is continuous, then  $g$  is analytic and

$$g'(z) = \int_{\gamma} \frac{\partial \varphi}{\partial z}(w, z) dw.$$

*Proof.* Let  $\gamma : [a, b] \rightarrow \mathbb{C}$  be a rectifiable parametrization of the curve, and let  $\{\gamma\} = \gamma([a, b])$  denote its image. We are given that

$$\varphi : \{\gamma\} \times G \rightarrow \mathbb{C}$$

is continuous, and we define

$$g(z) = \int_{\gamma} \varphi(w, z) dw = \int_a^b \varphi(\gamma(t), z) \gamma'(t) dt.$$

### 1. Continuity of $g$ .

Fix  $z_0 \in G$ . Since  $\varphi$  is continuous on  $\{\gamma\} \times G$ , and  $\{\gamma\}$  is compact (the continuous image of the compact interval  $[a, b]$ ), it follows that  $\varphi$  is *uniformly* continuous on  $\{\gamma\} \times K$  for any compact  $K \subset G$ . In particular, choose a closed disk  $D$  around  $z_0$  contained in  $G$ . Given  $\varepsilon > 0$ , there exists  $\delta > 0$  such that

$$|z - z_0| < \delta, \quad w \in \{\gamma\} \quad \implies \quad |\varphi(w, z) - \varphi(w, z_0)| < \frac{\varepsilon}{\ell(\gamma)},$$

where  $\ell(\gamma)$  denotes the length of the curve  $\gamma$ .

Then for  $|z - z_0| < \delta$ ,

$$\begin{aligned} |g(z) - g(z_0)| &= \left| \int_{\gamma} (\varphi(w, z) - \varphi(w, z_0)) dw \right| \\ &\leq \int_{\gamma} |\varphi(w, z) - \varphi(w, z_0)| |dw| \\ &\leq \sup_{w \in \{\gamma\}} |\varphi(w, z) - \varphi(w, z_0)| \cdot \ell(\gamma) \\ &< \frac{\varepsilon}{\ell(\gamma)} \cdot \ell(\gamma) = \varepsilon. \end{aligned}$$

Hence  $g$  is continuous at  $z_0$ , and since  $z_0 \in G$  was arbitrary,  $g$  is continuous on  $G$ .

### 2. Existence of $g'(z)$ and formula for the derivative.

Assume now that  $\partial\varphi/\partial z$  exists for each  $(w, z) \in \{\gamma\} \times G$  and is continuous there. Fix  $z_0 \in G$  and choose a closed disk  $K$  centered at  $z_0$  such that  $K \subset G$ . The continuity of  $\varphi$  and  $\partial\varphi/\partial z$  implies that these functions are uniformly continuous on the compact set  $\{\gamma\} \times K$ . We will show that  $g$  is complex differentiable at  $z_0$  and that

$$g'(z_0) = \int_{\gamma} \frac{\partial\varphi}{\partial z}(w, z_0) dw.$$

For  $h \neq 0$  with  $z_0 + h \in K$ , write

$$\frac{g(z_0 + h) - g(z_0)}{h} - \int_{\gamma} \frac{\partial\varphi}{\partial z}(w, z_0) dw = \int_{\gamma} R_h(w) dw,$$

where

$$R_h(w) = \frac{\varphi(w, z_0 + h) - \varphi(w, z_0)}{h} - \frac{\partial\varphi}{\partial z}(w, z_0).$$

Thus

$$\left| \frac{g(z_0 + h) - g(z_0)}{h} - \int_{\gamma} \frac{\partial \varphi}{\partial z}(w, z_0) dw \right| \leq \int_{\gamma} |R_h(w)| |dw| \leq \ell(\gamma) \sup_{w \in \{\gamma\}} |R_h(w)|.$$

It remains to show that

$$\sup_{w \in \{\gamma\}} |R_h(w)| \longrightarrow 0 \quad \text{as } h \rightarrow 0.$$

For each fixed  $w \in \{\gamma\}$ , the function  $z \mapsto \varphi(w, z)$  is complex differentiable at  $z_0$  with derivative  $\partial \varphi / \partial z(w, z_0)$ . Thus, by definition of the complex derivative,

$$\lim_{h \rightarrow 0} R_h(w) = 0 \quad \text{for each } w \in \{\gamma\}.$$

Define

$$F(w, z) = \begin{cases} \frac{\varphi(w, z) - \varphi(w, z_0)}{z - z_0} - \frac{\partial \varphi}{\partial z}(w, z_0), & z \neq z_0, \\ 0, & z = z_0. \end{cases}$$

The function  $F$  is continuous on the compact set  $\{\gamma\} \times K$ , because the difference quotient converges to the derivative and  $\partial \varphi / \partial z$  is continuous. Moreover,

$$R_h(w) = F(w, z_0 + h).$$

By continuity of  $F$  on the compact set, we have

$$\sup_{w \in \{\gamma\}} |F(w, z_0 + h)| \longrightarrow 0 \quad \text{as } h \rightarrow 0.$$

Therefore

$$\sup_{w \in \{\gamma\}} |R_h(w)| = \sup_{w \in \{\gamma\}} |F(w, z_0 + h)| \longrightarrow 0,$$

and hence

$$\left| \frac{g(z_0 + h) - g(z_0)}{h} - \int_{\gamma} \frac{\partial \varphi}{\partial z}(w, z_0) dw \right| \leq \ell(\gamma) \sup_w |R_h(w)| \longrightarrow 0.$$

Thus the complex derivative  $g'(z_0)$  exists and

$$g'(z_0) = \int_{\gamma} \frac{\partial \varphi}{\partial z}(w, z_0) dw.$$

Since  $z_0 \in G$  was arbitrary and the integrand is continuous in  $z$ , it follows that  $g$  is analytic on  $G$  and

$$g'(z) = \int_{\gamma} \frac{\partial \varphi}{\partial z}(w, z) dw \quad \text{for all } z \in G.$$

■

**3.** Suppose that  $\gamma$  is a rectifiable curve in  $\mathbb{C}$  and  $\varphi$  is defined and continuous on  $\{\gamma\}$ . Use Exercise 2 to show that

$$g(z) = \int_{\gamma} \frac{\varphi(w)}{w - z} dw$$

is analytic on  $\mathbb{C} - \{\gamma\}$  and

$$g^{(n)}(z) = n! \int_{\gamma} \frac{\varphi(w)}{(w - z)^{n+1}} dw.$$

*Proof.* Let  $\gamma$  be a rectifiable curve in  $\mathbb{C}$  and denote its image by  $\{\gamma\}$ . Since  $\gamma$  is continuous on a compact interval,  $\{\gamma\}$  is compact and therefore closed. Thus  $\mathbb{C} \setminus \{\gamma\}$  is an open set. Define

$$G := \mathbb{C} \setminus \{\gamma\}.$$

For  $(w, z) \in \{\gamma\} \times G$  define

$$\Phi(w, z) := \frac{\varphi(w)}{w - z}.$$

Note that for each such pair,  $w \neq z$  by definition of  $G$ , so  $\Phi$  is well-defined. Since  $\varphi$  is continuous on  $\{\gamma\}$  and the map

$$(w, z) \mapsto \frac{1}{w - z}$$

is continuous on  $\{(w, z) \in \mathbb{C}^2 : w \neq z\}$ , it follows that  $\Phi$  is continuous on  $\{\gamma\} \times G$ . Moreover, for fixed  $w \in \{\gamma\}$ , as a function of  $z$  we have

$$\Phi(w, z) = \varphi(w)(w - z)^{-1}.$$

Thus

$$\frac{\partial \Phi}{\partial z}(w, z) = \varphi(w) \frac{d}{dz}(w - z)^{-1} = \varphi(w)(w - z)^{-2},$$

since

$$\frac{d}{dz}(w - z)^{-1} = -1 \cdot (-(w - z)^{-2}) = (w - z)^{-2}.$$

This partial derivative is continuous on  $\{\gamma\} \times G$ . Define

$$g(z) := \int_{\gamma} \frac{\varphi(w)}{w - z} dw = \int_{\gamma} \Phi(w, z) dw.$$

By Exercise 2 applied to  $\Phi$  on  $\{\gamma\} \times G$ , the function  $g$  is analytic on  $G$  and

$$g'(z) = \int_{\gamma} \frac{\partial \Phi}{\partial z}(w, z) dw = \int_{\gamma} \frac{\varphi(w)}{(w - z)^2} dw.$$

More generally by induction on  $n \in \mathbb{N}$  we have:

$$g^{(n)}(z) = n! \int_{\gamma} \frac{\varphi(w)}{(w - z)^{n+1}} dw \quad \text{for all } z \in G.$$

■

**5.** Give the power series expansion of  $\log z$  about  $z = i$  and find its radius of convergence.

*Proof.* We expand  $f(z) = \log z$  in a Taylor series about  $z = i$  using Theorem 2.8. The derivatives of  $\log z$  are

$$f^{(n)}(z) = (-1)^{n-1} \frac{(n-1)!}{z^n}, \quad n \geq 1.$$

Thus

$$f^{(n)}(i) = (-1)^{n-1} \frac{(n-1)!}{i^n}.$$

The Taylor coefficients are

$$a_n = \frac{f^{(n)}(i)}{n!} = (-1)^{n-1} \frac{1}{n} i^{-n}.$$

Since  $i^{-n} = (-1)^n i^n$ , we obtain the simpler formula

$$a_n = -\frac{i^n}{n}, \quad n \geq 1.$$

Together with  $a_0 = f(i) = \log i$ , this gives the Taylor series

$$\log z = \log i - \sum_{n=1}^{\infty} \frac{i^n}{n} (z - i)^n$$

The nearest singularity of  $\log z$  to  $i$  is 0, at distance 1, so the radius of convergence is  $R = 1$ . ■

**7.** Use the results of this section to evaluate the integral

$$(a) \int_{\gamma} \frac{e^{iz}}{z^2} dz, \quad \gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi.$$

$$(c) \int_{\gamma} \frac{\sin z}{z^3} dz, \quad \gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi.$$

*Proof.* We use Cauchy's integral formula for derivatives:

$$f^{(n)}(0) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(z)}{z^{n+1}} dz, \quad \gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi.$$

(a) Evaluate

$$\int_{\gamma} \frac{e^{iz}}{z^2} dz.$$

Here the integrand has the form

$$\frac{f(z)}{z^2}, \quad f(z) = e^{iz}.$$

Thus

$$\int_{\gamma} \frac{e^{iz}}{z^2} dz = \frac{2\pi i}{1!} f'(0) = 2\pi i \cdot (ie^{i \cdot 0}) = 2\pi i \cdot i = -2\pi.$$

(c) Evaluate

$$\int_{\gamma} \frac{\sin z}{z^3} dz.$$

Now we have

$$\frac{f(z)}{z^3}, \quad f(z) = \sin z.$$

We use the second derivative:

$$\int_{\gamma} \frac{\sin z}{z^3} dz = \frac{2\pi i}{2!} f''(0).$$

we compute

$$f''(z) = \sin z'' = -\sin z$$

Thus

$$f''(0) = -\sin 0 = 0.$$

Therefore,

$$\int_{\gamma} \frac{\sin z}{z^3} dz = \frac{2\pi i}{2} \cdot 0 = 0.$$

■

9. Evaluate the following integrals:

$$(a) \int_{\gamma} \frac{e^z - e^{-z}}{z^n} dz \text{ where } n \text{ is a positive integer and } \gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi.$$



(c)  $\int_{\gamma} \frac{dz}{z^2 + 1}$  where  $\gamma(t) = 2e^{it}$ ,  $0 \leq t \leq 2\pi$ . (Hint: expand  $(z^2 + 1)^{-1}$  by partial fractions.)

(e)  $\int_{\gamma} \frac{z^{1/m}}{(z-1)^m} dz$  where  $\gamma(t) = 1 + \frac{1}{2}e^{it}$ ,  $0 \leq t \leq 2\pi$ .

*Proof.* (a) Evaluate

$$\int_{\gamma} \frac{e^z - e^{-z}}{z^n} dz, \quad \gamma(t) = e^{it}.$$

Write the integrand as

$$\frac{f(z)}{z^n}, \quad f(z) = e^z - e^{-z}.$$

By Cauchy's integral formula for derivatives,

$$\int_{\gamma} \frac{f(z)}{z^n} dz = \frac{2\pi i}{(n-1)!} f^{(n-1)}(0).$$

Since

$$e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!}, \quad e^{-z} = \sum_{k=0}^{\infty} \frac{(-1)^k z^k}{k!},$$

we have

$$f(z) = e^z - e^{-z} = \sum_{k=0}^{\infty} (1 - (-1)^k) \frac{z^k}{k!}.$$

Thus

$$f^{(n-1)}(0) = \begin{cases} \frac{2}{(n-1)!}, & n-1 \text{ odd}, \\ 0, & n-1 \text{ even}. \end{cases}$$

Therefore

$$\int_{\gamma} \frac{e^z - e^{-z}}{z^n} dz = \begin{cases} 2\pi i \cdot \frac{2}{(n-1)!} = \frac{4\pi i}{(n-1)!}, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

(c) Evaluate

$$\int_{\gamma} \frac{dz}{z^2 + 1}, \quad \gamma(t) = 2e^{it}.$$

Factor:

$$\frac{1}{z^2 + 1} = \frac{1}{(z-i)(z+i)} = \frac{1}{2i} \left( \frac{1}{z-i} - \frac{1}{z+i} \right).$$

The circle  $|z| = 2$  encloses both poles  $\pm i$ , so

$$\int_{\gamma} \frac{dz}{z-i} = 2\pi i, \quad \int_{\gamma} \frac{dz}{z+i} = 2\pi i.$$

Thus

$$\int_{\gamma} \frac{dz}{z^2 + 1} = \frac{1}{2i} (2\pi i - 2\pi i) = 0.$$

(e) Evaluate

$$\int_{\gamma} \frac{z^{1/m}}{(z-1)^m} dz, \quad \gamma(t) = 1 + \frac{1}{2}e^{it}.$$

The path  $\gamma$  is a small circle centered at  $z = 1$ . Let

$$f(z) = z^{1/m}, \quad \text{analytic near } z = 1.$$

The integrand has the form

$$\frac{f(z)}{(z-1)^m}.$$

By Cauchy's integral formula for derivatives,

$$\int_{\gamma} \frac{f(z)}{(z-1)^m} dz = \frac{2\pi i}{(m-1)!} f^{(m-1)}(1).$$

Compute the derivatives:

$$f(z) = z^{1/m}, \quad f^{(k)}(z) = \frac{1}{m} \left( \frac{1}{m} - 1 \right) \cdots \left( \frac{1}{m} - k + 1 \right) z^{1/m-k}.$$

Thus

$$f^{(m-1)}(1) = \frac{1}{m} \left( \frac{1}{m} - 1 \right) \cdots \left( \frac{1}{m} - m + 2 \right).$$

Therefore

$$\int_{\gamma} \frac{z^{1/m}}{(z-1)^m} dz = \frac{2\pi i}{(m-1)!} \frac{1}{m} \left( \frac{1}{m} - 1 \right) \cdots \left( \frac{1}{m} - m + 2 \right).$$

■

### 4.3 Zeros of an analytic function

**1.** Let  $f$  be an entire function and suppose there is a constant  $M$ , an  $R > 0$ , and an integer  $n \geq 1$  such that

$$|f(z)| \leq M |z|^n \quad \text{for } |z| > R.$$

Show that  $f$  is a polynomial of degree  $\leq n$ .

*Proof.* Let  $f$  be entire and suppose there exist constants  $M > 0$ ,  $R > 0$ , and an integer  $n \geq 1$  such that

$$|f(z)| \leq M|z|^n \quad \text{for all } |z| > R.$$

We will show that  $f^{(k)}(0) = 0$  for every  $k > n$ , which implies that the Taylor series of  $f$  terminates after order  $n$ , making  $f$  a polynomial of degree at most  $n$ .

Let  $k > n$  and let  $r > R$ . By Cauchy's integral formula for derivatives,

$$f^{(k)}(0) = \frac{k!}{2\pi i} \int_{\gamma} \frac{f(z)}{z^{k+1}} dz.$$

Using the estimation lemma, we obtain

$$|f^{(k)}(0)| \leq \frac{k!}{r^k} \cdot \max_{|z|=r} |f(z)|.$$

Since  $|f(z)| \leq Mr^n$  on the circle  $|z| = r$ , this gives

$$|f^{(k)}(0)| \leq k! M r^{n-k}.$$

Because  $k > n$ , the exponent  $n - k$  is negative, and hence

$$r^{n-k} \rightarrow 0 \quad \text{as } r \rightarrow \infty.$$

Thus letting  $r \rightarrow \infty$  yields

$$f^{(k)}(0) = 0 \quad \text{for all } k > n.$$

Therefore all terms of order  $> n$  in the Taylor expansion vanish, and we conclude that

$$f(z) = a_0 + a_1 z + \cdots + a_n z^n,$$

so  $f$  is a polynomial of degree at most  $n$ . ■

**6.** Let  $G$  be a region and suppose that  $f : G \rightarrow \mathbb{C}$  is analytic and  $a \in G$  such that

$$|f(a)| \leq |f(z)| \quad \text{for all } z \in G.$$

Show that either  $f(a) = 0$  or  $f$  is constant.

*Proof.* Suppose  $f : G \rightarrow \mathbb{C}$  is analytic on a region  $G$  and that

$$|f(a)| \leq |f(z)| \quad \text{for all } z \in G,$$

for some point  $a \in G$ . We will show that either  $f(a) = 0$  or  $f$  is constant.

Assume first that  $f(a) \neq 0$ . Then  $f$  has no zeros at  $a$ , and therefore  $1/f$  is analytic in some neighborhood of  $a$ , and in fact analytic on all of  $G$  because  $f$  never vanishes on  $G$  (otherwise  $|f(a)| \leq |f(z_0)| = 0$  would force  $f(a) = 0$ , contradicting the assumption).

For the analytic function  $1/f$ , we compute

$$\left| \frac{1}{f(a)} \right| = \frac{1}{|f(a)|} \geq \frac{1}{|f(z)|} = \left| \frac{1}{f(z)} \right| \quad \text{for all } z \in G.$$

Thus  $|1/f|$  attains its *maximum* at the interior point  $a$ . By the Maximum Modulus Principle, an analytic function that attains its maximum modulus at an interior point must be constant. Hence  $1/f$  is constant on  $G$ , and therefore  $f$  is constant on  $G$ . Thus we conclude that either  $f(a) = 0$  or  $f$  is constant. ■

**8.** Let  $G$  be a region and let  $f$  and  $g$  be analytic functions on  $G$  such that

$$f(z)g(z) = 0 \quad \text{for all } z \in G.$$

Show that either  $f \equiv 0$  or  $g \equiv 0$ .

*Proof.* Suppose  $f$  and  $g$  are analytic on a region  $G$  and satisfy

$$f(z)g(z) = 0 \quad \text{for all } z \in G.$$

Assume first that  $f$  is not identically zero. By the Identity Theorem, the zeros of a nontrivial analytic function cannot have a limit point in  $G$ ; hence there exists a point  $a \in G$  and a radius  $r > 0$  such that  $f(z) \neq 0$  for all  $z \in B(a, r)$ . From the identity

$$f(z)g(z) = 0,$$

it follows that  $g(z) = 0$  whenever  $f(z) \neq 0$ . Therefore  $g$  vanishes on the open set  $B(a, r)$ . Since  $g$  is analytic and vanishes on an open set, the Identity Theorem implies that  $g \equiv 0$ . ■

**9.** Let  $U: \mathbb{C} \rightarrow \mathbb{R}$  be a harmonic function such that

$$U(z) \geq 0 \quad \text{for all } z \in \mathbb{C}.$$

Prove that  $U$  is constant.

*Proof.* Let  $U: \mathbb{C} \rightarrow \mathbb{R}$  be harmonic and assume that

$$U(z) \geq 0 \quad \text{for all } z \in \mathbb{C}.$$

Since  $U$  is harmonic on all of  $\mathbb{C}$ , it admits a harmonic conjugate  $V$  on  $\mathbb{C}$  (because  $\mathbb{C}$  is simply connected). Define

$$F(z) = U(z) + iV(z).$$

Then  $F$  is an entire function. Now consider the entire function

$$G(z) = e^{-F(z)}.$$

We compute its modulus:

$$|G(z)| = |e^{-F(z)}| = e^{-\Re(F(z))} = e^{-U(z)}.$$

Since  $U(z) \geq 0$  for all  $z$ , we have

$$|G(z)| = e^{-U(z)} \leq 1.$$

Thus  $G$  is an entire function bounded in modulus by 1. By Liouville's theorem, every bounded entire function is constant, so  $G$  is constant. Therefore,

$$e^{-F(z)} = \text{constant},$$

which implies that  $F(z)$  itself is constant. Since  $F = U + iV$ , its real part  $U$  must also be constant. Hence  $U$  is constant on  $\mathbb{C}$ . ■

**10.** Show that if  $f$  and  $g$  are analytic functions on a region  $G$  such that  $\overline{f}g$  is analytic, then either  $f$  is constant or  $g \equiv 0$ .

*Proof.* Let  $f$  and  $g$  be analytic on a region  $G$  and suppose that the function

$$h(z) = \overline{f(z)}g(z)$$

is analytic on  $G$ . We will show that either  $f$  is constant or  $g \equiv 0$ . First observe that  $h$  is continuous and satisfies the Cauchy–Riemann equations. Let  $z \in G$  be a point where  $g(z) \neq 0$ . Since  $g$  is analytic, there exists a neighborhood  $U$  of  $z$  on which  $g(w) \neq 0$  for all  $w \in U$ . On this set  $U$  we may write

$$\overline{f(w)} = \frac{h(w)}{g(w)}, \quad w \in U,$$

which expresses  $\overline{f}$  as the quotient of two analytic functions. Thus  $\overline{f}$  is analytic on  $U$ . But if a function is analytic and conjugate-analytic on a region, it must be constant. To see this, note that if  $\overline{f}$  is analytic, then  $f$  satisfies both the Cauchy–Riemann equations and their conjugates, which forces all first derivatives of  $f$  to

vanish. Hence  $f$  is constant on  $U$ . Since  $U$  is nonempty and  $f$  is analytic on the connected region  $G$ , the Identity Theorem implies that  $f$  is constant on all of  $G$ . ■

## 4.4 The index of a closed curve

**3.** Let  $p(z)$  be a polynomial of degree  $n$  and let  $R > 0$  be sufficiently large so that  $p$  never vanishes in  $\{z : |z| \geq R\}$ . If  $\gamma(t) = Re^{it}$ ,  $0 \leq t \leq 2\pi$ , show that

$$\int_{\gamma} \frac{p'(z)}{p(z)} dz = 2\pi i n.$$

*Proof.* Let  $p$  be a polynomial of degree  $n$ . Then we can factor it as

$$p(z) = a_n \prod_{k=1}^n (z - z_k),$$

where  $a_n \neq 0$  and the  $z_k$  are the zeros of  $p$ , listed with multiplicity.

By hypothesis, we choose  $R > 0$  so large that

$$p(z) \neq 0 \quad \text{for all } |z| \geq R.$$

This means that all zeros  $z_k$  of  $p$  satisfy  $|z_k| < R$ , so they all lie inside the circle

$$\gamma(t) = Re^{it}, \quad 0 \leq t \leq 2\pi.$$

Differentiating the factorization of  $p$ , we have

$$\frac{p'(z)}{p(z)} = \frac{d}{dz} \log p(z) = \frac{d}{dz} \left( \log a_n + \sum_{k=1}^n \log(z - z_k) \right) = \sum_{k=1}^n \frac{1}{z - z_k},$$

where the last equality is valid for  $z$  not equal to any  $z_k$ . Therefore, on and inside  $\gamma$ ,

$$\frac{p'(z)}{p(z)} = \sum_{k=1}^n \frac{1}{z - z_k}.$$

Now integrate around  $\gamma$ :

$$\int_{\gamma} \frac{p'(z)}{p(z)} dz = \int_{\gamma} \sum_{k=1}^n \frac{1}{z - z_k} dz = \sum_{k=1}^n \int_{\gamma} \frac{1}{z - z_k} dz.$$

Each  $z_k$  lies inside the circle  $|z| = R$ , so by Cauchy's integral formula (or directly

by the standard winding number integral),

$$\int_{\gamma} \frac{1}{z - z_k} dz = 2\pi i.$$

Hence

$$\int_{\gamma} \frac{p'(z)}{p(z)} dz = \sum_{k=1}^n 2\pi i = 2\pi i n.$$

This completes the proof. ■

**4.** Fix  $w = re^{i\theta} \neq 0$  and let  $\gamma$  be a rectifiable path in  $\mathbb{C} - \{0\}$  from 1 to  $w$ . Show that there is an integer  $k$  such that

$$\int_{\gamma} z^{-1} dz = \log r + i\theta + 2\pi i k.$$

*Proof.* Let  $w = re^{i\theta} \neq 0$  and let  $\gamma$  be any rectifiable path in  $\mathbb{C} \setminus \{0\}$  from 1 to  $w$ .

**Step 1: Compare  $\gamma$  to a reference path.**

Define a specific path  $\sigma$  from 1 to  $w$  by first going radially on the real axis from 1 to  $r$ , then along the circle of radius  $r$  from angle 0 to  $\theta$ :

$$\sigma = \sigma_1 * \sigma_2,$$

where

$$\sigma_1(t) = 1 + (r - 1)t, \quad 0 \leq t \leq 1, \quad \sigma_2(t) = re^{it\theta}, \quad 0 \leq t \leq 1.$$

Compute the integral along  $\sigma_1$ :

$$\int_{\sigma_1} \frac{dz}{z} = \int_0^1 \frac{(r - 1) dt}{1 + (r - 1)t} = \int_1^r \frac{du}{u} = \log r.$$

Compute the integral along  $\sigma_2$ :

$$\int_{\sigma_2} \frac{dz}{z} = \int_0^1 \frac{ri\theta e^{it\theta}}{re^{it\theta}} dt = \int_0^1 i\theta dt = i\theta.$$

Thus along the reference path  $\sigma$  we have

$$\int_{\sigma} \frac{dz}{z} = \log r + i\theta.$$

**Step 2: Any other path differs by a loop around 0.**

Now let  $\gamma$  be any rectifiable path in  $\mathbb{C} \setminus \{0\}$  from 1 to  $w$ . Consider the closed

curve

$$\Gamma = \gamma - \sigma,$$

that is, follow  $\gamma$  from 1 to  $w$  and then follow  $\sigma$  backwards from  $w$  to 1. Then  $\Gamma$  is a closed rectifiable curve in  $\mathbb{C} \setminus \{0\}$ . By the winding number formula,

$$\int_{\Gamma} \frac{dz}{z} = 2\pi i n(\Gamma; 0),$$

for some integer  $n(\Gamma; 0) = k \in \mathbb{Z}$ . On the other hand,

$$\int_{\Gamma} \frac{dz}{z} = \int_{\gamma} \frac{dz}{z} - \int_{\sigma} \frac{dz}{z} = \int_{\gamma} \frac{dz}{z} - (\log r + i\theta).$$

Equating the two expressions for  $\int_{\Gamma} \frac{dz}{z}$ , we obtain

$$\int_{\gamma} \frac{dz}{z} - (\log r + i\theta) = 2\pi i k$$

for some integer  $k$ . Rewriting,

$$\int_{\gamma} z^{-1} dz = \log r + i\theta + 2\pi i k.$$

■

## 4.5 Cauchy's Theorem and Integral Formula

1. Suppose  $f : G \rightarrow \mathbb{C}$  is analytic and define  $\varphi : G \times G \rightarrow \mathbb{C}$  by

$$\varphi(z, w) = \frac{f(z) - f(w)}{z - w} \quad \text{if } z \neq w, \quad \varphi(z, z) = f'(z).$$

Prove that  $\varphi$  is continuous and for each fixed  $w$ ,  $z \rightarrow \varphi(z, w)$  is analytic.

*Proof.* Let  $f : G \rightarrow \mathbb{C}$  be analytic and define

$$\varphi(z, w) = \begin{cases} \frac{f(z) - f(w)}{z - w}, & z \neq w, \\ f'(z), & z = w. \end{cases}$$

We will prove that  $\varphi$  is continuous on  $G \times G$ , and that for each fixed  $w$ , the map  $z \mapsto \varphi(z, w)$  is analytic.



**Step 1: Taylor-series representation.**

For each fixed  $w \in G$ , the Taylor expansion of  $f$  about  $w$  is

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(w)}{k!} (z - w)^k,$$

valid for  $z$  sufficiently near  $w$ . Subtracting  $f(w)$  and dividing by  $(z - w)$  gives, for  $z \neq w$ ,

$$\frac{f(z) - f(w)}{z - w} = \sum_{k=1}^{\infty} \frac{f^{(k)}(w)}{k!} (z - w)^{k-1}.$$

This motivates defining

$$\varphi(z, w) = \sum_{k=1}^{\infty} \frac{f^{(k)}(w)}{k!} (z - w)^{k-1}. \quad (*)$$

At  $z = w$ , all terms with  $(k - 1) \geq 1$  vanish and  $(*)$  gives  $\varphi(w, w) = f'(w)$ , so the series representation agrees with the given definition of  $\varphi$ .

**Step 2: Analyticity in  $z$ .**

Fix  $w \in G$ . Then  $(*)$  is a power series in  $(z - w)$ :

$$\varphi(z, w) = f'(w) + \frac{f''(w)}{2!} (z - w) + \frac{f^{(3)}(w)}{3!} (z - w)^2 + \cdots.$$

A power series defines an analytic function in  $z$  on its disk of convergence. Thus  $z \mapsto \varphi(z, w)$  is analytic in a neighborhood of  $w$ , and by analytic continuation is analytic on all of  $G$ .

**Step 3: Continuity of  $\varphi$  on  $G \times G$ .**

Each term

$$(z, w) \mapsto \frac{f^{(k)}(w)}{k!} (z - w)^{k-1}$$

is continuous on  $G \times G$ . On compact subsets the series  $(*)$  converges uniformly (because it is the Taylor series of  $f$ ), so the sum of the series is continuous. Therefore  $\varphi$  is continuous on  $G \times G$ . Finally, as  $z \rightarrow w$ , the power series  $(*)$  satisfies

$$\lim_{z \rightarrow w} \varphi(z, w) = f'(w) = \varphi(w, w),$$

so  $\varphi$  is continuous on the diagonal as well. Hence  $\varphi$  is continuous on  $G \times G$ , and analytic in  $z$  for each fixed  $w$ . ■

**5.** Let  $\gamma$  be a closed rectifiable curve in  $\mathbb{C}$  and a  $\notin \gamma$ . Show that for  $n \geq 2$ ,

$$\int_{\gamma} (z - a)^{-n} dz = 0.$$

*Proof.* Let  $\gamma$  be a closed rectifiable curve in  $\mathbb{C}$  and let  $a \notin \gamma$ . Fix an integer  $n \geq 2$ . We use Corollary 5.9 with the analytic function  $f(z) = 1$  and  $k = n - 1$ . Since  $f(z) = 1$  is constant, all of its derivatives of order  $\geq 1$  vanish, and in particular

$$f^{(n-1)}(a) = 0.$$

Corollary 5.9 states that for any analytic  $f$ ,

$$f^{(k)}(a) n(\gamma; a) = \frac{k!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z-a)^{k+1}} dz.$$

Applying this with  $f(z) = 1$  and  $k = n - 1$  gives

$$0 \cdot n(\gamma; a) = \frac{(n-1)!}{2\pi i} \int_{\gamma} \frac{1}{(z-a)^n} dz.$$

Thus

$$\int_{\gamma} (z-a)^{-n} dz = 0.$$

This completes the proof. ■

**6.** Let  $f$  be analytic on  $D = B(0; 1)$  and suppose  $|f(z)| \leq 1$  for  $|z| < 1$ . Show that  $|f'(0)| \leq 1$ .

*Proof.* Let  $f$  be analytic on  $D = B(0; 1)$  and assume  $|f(z)| \leq 1$  for all  $|z| < 1$ . We wish to prove that  $|f'(0)| \leq 1$ .

Fix  $0 < r < 1$  and consider the circle  $\gamma_r(t) = re^{it}$ ,  $0 \leq t \leq 2\pi$ . Since  $\gamma_r$  is a closed rectifiable curve in  $D$  and 0 lies in the interior of the circle, we have  $n(\gamma_r; 0) = 1$ .

Apply Corollary 5.9 with  $k = 1$  and  $a = 0$ :

$$f'(0) n(\gamma_r; 0) = \frac{1!}{2\pi i} \int_{\gamma_r} \frac{f(z)}{(z-0)^2} dz = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(z)}{z^2} dz.$$

Since  $n(\gamma_r; 0) = 1$ , we obtain

$$f'(0) = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(z)}{z^2} dz.$$

Parametrize  $\gamma_r$  by  $z = re^{it}$ . Then  $dz = ire^{it}dt$ , and so

$$f'(0) = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(re^{it})}{r^2 e^{2it}} (ire^{it}) dt = \frac{1}{2\pi r} \int_0^{2\pi} f(re^{it}) e^{-it} dt.$$

Taking absolute values,

$$|f'(0)| \leq \frac{1}{2\pi r} \int_0^{2\pi} |f(re^{it})| dt.$$

But  $|f(re^{it})| \leq 1$  for all  $t$ , so

$$|f'(0)| \leq \frac{1}{2\pi r} \int_0^{2\pi} 1 dt = \frac{1}{2\pi r} (2\pi) = \frac{1}{r}.$$

Since this holds for every  $0 < r < 1$ , giving

$$|f'(0)| \leq 1.$$

This completes the proof. ■

**7.** Let  $\gamma(t) = 1 + e^{it}$  for  $0 \leq t \leq 2\pi$ . Find

$$\int_{\gamma} \left( \frac{z}{z-1} \right)^n dz$$

for all positive integers  $n$ .

*Proof.* Let  $\gamma(t) = 1 + e^{it}$  for  $0 \leq t \leq 2\pi$ . Then  $\gamma$  is the circle  $|z-1| = 1$ , which winds once counterclockwise around  $z = 1$ . Hence

$$n(\gamma; 1) = 1.$$

We wish to compute

$$I_n = \int_{\gamma} \left( \frac{z}{z-1} \right)^n dz.$$

Rewrite the integrand in the form required by Corollary 5.9:

$$\left( \frac{z}{z-1} \right)^n = z^n (z-1)^{-n} = \frac{f(z)}{(z-1)^n}, \quad f(z) = z^n.$$

Thus in Corollary 5.9 we take  $k = n-1$ , since  $k+1 = n$ . Corollary 5.9 states:

$$f^{(k)}(1) n(\gamma; 1) = \frac{k!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z-1)^{k+1}} dz.$$

Substituting  $k = n-1$  and  $f(z) = z^n$  gives

$$f^{(n-1)}(1) = \frac{(n-1)!}{2\pi i} \int_{\gamma} \left( \frac{z}{z-1} \right)^n dz.$$

Next compute  $f^{(n-1)}(1)$ . Since  $f(z) = z^n$ ,

$$f^{(n-1)}(z) = n! z,$$

and therefore

$$f^{(n-1)}(1) = n!.$$

Putting this into the previous equation yields

$$n! = \frac{(n-1)!}{2\pi i} \int_{\gamma} \left( \frac{z}{z-1} \right)^n dz.$$

Thus

$$\int_{\gamma} \left( \frac{z}{z-1} \right)^n dz = 2\pi i n.$$

This holds for all positive integers  $n$ . ■

**8.** Let  $G$  be a region and suppose  $f_n : G \rightarrow \mathbb{C}$  is analytic for each  $n \geq 1$ . Suppose that  $\{f_n\}$  converges uniformly to a function  $f : G \rightarrow \mathbb{C}$ . Show that  $f$  is analytic.

*Proof.* Since each  $f_n$  is analytic on  $G$ , it is continuous on  $G$ . Because  $f_n \rightarrow f$  uniformly on  $G$ , the limit function  $f$  is also continuous on  $G$ . Let  $T$  be any triangular path in  $G$ . Since  $T$  is a compact curve, uniform convergence implies uniform convergence on  $\partial T$  as well. Therefore

$$\int_T f_n(z) dz \longrightarrow \int_T f(z) dz. \quad (1)$$

But each  $f_n$  is analytic, so by Cauchy's theorem,

$$\int_T f_n(z) dz = 0 \quad \text{for all } n. \quad (2)$$

Taking the limit of (2) and using (1),

$$\int_T f(z) dz = \lim_{n \rightarrow \infty} \int_T f_n(z) dz = \lim_{n \rightarrow \infty} 0 = 0.$$

Thus for every triangular path  $T \subset G$ ,

$$\int_T f(z) dz = 0.$$

Since  $f$  is continuous and its integral around every triangle vanishes, Morera's Theorem applies and implies that  $f$  is analytic on  $G$ . ■

**9.** Show that if  $f : \mathbb{C} \rightarrow \mathbb{C}$  is a continuous function such that  $f$  is analytic off

$[-1, 1]$ , then  $f$  is an entire function.

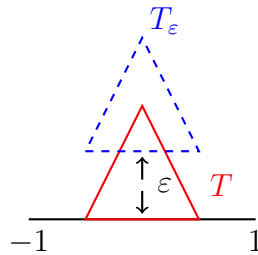


Figure 4.1: The triangle  $T$  touches the segment  $[-1, 1]$  on the real axis. Shifting it vertically by  $\varepsilon > 0$  produces the translated triangle  $T_\varepsilon = T + i\varepsilon$ , which lies entirely in  $\mathbb{C} \setminus [-1, 1]$ .

*Proof.* We assume  $f : \mathbb{C} \rightarrow \mathbb{C}$  is continuous on  $\mathbb{C}$  and analytic on  $\mathbb{C} \setminus [-1, 1]$ . By Morera's theorem, it suffices to prove that

$$\int_T f(z) dz = 0$$

for every triangular path  $T$  in  $\mathbb{C}$ .

**Case 1:  $T$  and its interior avoid  $[-1, 1]$ .** If

$$T \cup \text{Int}(T) \subset \mathbb{C} \setminus [-1, 1],$$

then  $f$  is analytic in a neighborhood of  $T$ , and by Cauchy's theorem,

$$\int_T f(z) dz = 0.$$

**Case 2: One side of  $T$  lies on the segment  $[-1, 1]$ .**

This is the configuration shown in Figure 4.1: the base of the triangle lies on  $[-1, 1]$  while the remaining two sides lie in the upper half-plane. Let  $T$  denote this triangle. For  $\varepsilon > 0$  define its vertical translate

$$T_\varepsilon = T + i\varepsilon.$$

As illustrated in Figure 4.1, when  $\varepsilon > 0$  is small, the entire triangle  $T_\varepsilon$  (together with its interior) lies strictly above the real axis and therefore satisfies

$$T_\varepsilon \cup \text{Int}(T_\varepsilon) \subset \mathbb{C} \setminus [-1, 1].$$

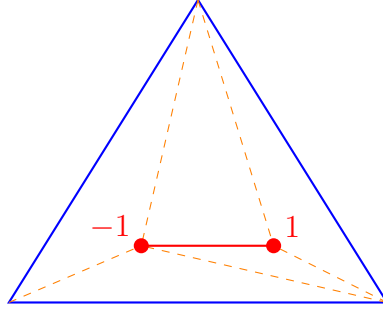


Figure 4.2: Subdivision of a triangle by two interior points 1 and  $-1$  and the segment joining them.

Thus, by Case 1,

$$\int_{T_\varepsilon} f(z) dz = 0 \quad \text{for all sufficiently small } \varepsilon > 0.$$

Let  $\gamma : [0, 1] \rightarrow \mathbb{C}$  parametrize  $T$ . Then

$$\gamma_\varepsilon(t) = \gamma(t) + i\varepsilon$$

parametrizes  $T_\varepsilon$ , and clearly  $\gamma_\varepsilon \rightarrow \gamma$  *uniformly* on  $[0, 1]$  as  $\varepsilon \rightarrow 0$ .

Since  $f$  is continuous on  $\mathbb{C}$ , it is uniformly continuous on the compact set

$$K := \bigcup_{0 \leq \varepsilon \leq \varepsilon_0} T_\varepsilon$$

for some small  $\varepsilon_0$ . Therefore  $f(\gamma_\varepsilon(t))$  converges uniformly to  $f(\gamma(t))$ . Hence we may pass the limit inside the integral:

$$\int_T f(z) dz = \int_0^1 f(\gamma(t)) \gamma'(t) dt = \lim_{\varepsilon \rightarrow 0} \int_0^1 f(\gamma_\varepsilon(t)) \gamma'_\varepsilon(t) dt = \lim_{\varepsilon \rightarrow 0} \int_{T_\varepsilon} f(z) dz = 0.$$

### General case.

If  $T$  intersects  $[-1, 1]$  in any other way, we divide  $T$  into finitely many smaller triangles by inserting straight segments between the points of intersection with  $[-1, 1]$ . A typical example of such a configuration is illustrated in Figure 4.2. Each subtriangle is either of type 1 or type 2. Applying the two cases above to each subtriangle and summing the resulting integrals yields

$$\int_T f(z) dz = 0.$$

**Conclusion.** Since  $\int_T f dz = 0$  for every triangular path  $T$  and  $f$  is continuous on  $\mathbb{C}$ , Morera's theorem implies that  $f$  is analytic on all of  $\mathbb{C}$ . Thus  $f$  is entire. ■

## 4.6 The homotopic version of Cauchy's Theorem and simple connectivity

5. Evaluate the integral  $\int_{\gamma} \frac{dz}{z^2 + 1}$  where  $\gamma(\theta) = 2|\cos 2\theta| e^{i\theta}$  for  $0 \leq \theta \leq 2\pi$ .

*Proof.* We are given the curve

$$\gamma(\theta) = 2|\cos 2\theta| e^{i\theta}, \quad 0 \leq \theta \leq 2\pi,$$

and we wish to evaluate

$$\int_{\gamma} \frac{dz}{z^2 + 1}.$$

**Step 1: Homotope  $\gamma$  to the circle  $|z| = 2$ .** Define a homotopy

$$H(s, \theta) = [(1-s)2|\cos 2\theta| + 2s]e^{i\theta}, \quad 0 \leq s \leq 1, \quad 0 \leq \theta \leq 2\pi.$$

Then  $H(0, \theta) = \gamma(\theta)$  and  $H(1, \theta) = 2e^{i\theta}$ . We claim  $H(s, \theta) \neq \pm i$  for all  $(s, \theta)$ . Indeed,  $\pm i$  have modulus 1, whereas

$$|H(s, \theta)| = (1-s)2|\cos 2\theta| + 2s \geq 2s \geq 0,$$

and for  $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$  one has  $|H(s, \theta)| = 2$ . Thus  $H$  never hits  $\pm i$ , and therefore gives a homotopy in  $\mathbb{C} \setminus \{\pm i\}$  from  $\gamma$  to the circle

$$\delta(\theta) = 2e^{i\theta}.$$

Since  $f(z) = \frac{1}{z^2+1}$  is analytic on  $\mathbb{C} \setminus \{\pm i\}$ , it follows from homotopy invariance of contour integrals that

$$\int_{\gamma} \frac{dz}{z^2 + 1} = \int_{\delta} \frac{dz}{z^2 + 1}.$$

**Step 2: Partial fractions and winding numbers.** We write

$$\frac{1}{z^2 + 1} = \frac{1}{(z-i)(z+i)} = \frac{1}{2i} \left( \frac{1}{z-i} - \frac{1}{z+i} \right).$$

Thus

$$\int_{\delta} \frac{dz}{z^2 + 1} = \frac{1}{2i} \left( \int_{\delta} \frac{dz}{z-i} - \int_{\delta} \frac{dz}{z+i} \right).$$

Now  $\delta(\theta) = 2e^{i\theta}$  winds once counterclockwise about both  $i$  and  $-i$ , since both satisfy  $|\pm i| = 1 < 2$ . Hence

$$n(\delta; i) = 1, \quad n(\delta; -i) = 1.$$

By the winding number formula,

$$\int_{\delta} \frac{dz}{z-i} = 2\pi i \cdot n(\delta; i) = 2\pi i, \quad \int_{\delta} \frac{dz}{z+i} = 2\pi i \cdot n(\delta; -i) = 2\pi i.$$

Therefore

$$\int_{\delta} \frac{dz}{z^2+1} = \frac{1}{2i}(2\pi i - 2\pi i) = 0.$$

Hence

$$\int_{\gamma} \frac{dz}{z^2+1} = 0.$$

■

6. Let  $\gamma(\theta) = \theta e^{i\theta}$  for  $0 \leq \theta \leq 2\pi$  and  $\gamma(\theta) = 4\pi - \theta$  for  $2\pi \leq \theta \leq 4\pi$ . Evaluate  $\int_{\gamma} \frac{dz}{z^2 + \pi^2}$ .

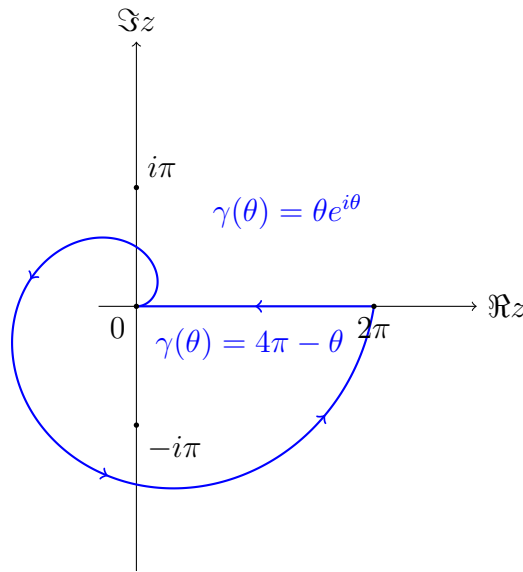


Figure 4.3: The path  $\gamma$ : a spiral from 0 to  $2\pi$  followed by a straight line back to 0. The spiral encloses the point  $-i\pi$  but not  $i\pi$ .

*Proof.* The curve  $\gamma$  is defined by

$$\gamma(\theta) = \begin{cases} \theta e^{i\theta}, & 0 \leq \theta \leq 2\pi, \\ 4\pi - \theta, & 2\pi \leq \theta \leq 4\pi. \end{cases}$$

For  $0 \leq \theta \leq 2\pi$  the point  $\theta e^{i\theta}$  traces a spiral beginning at 0, making one full revolution around the origin, and reaching the point  $2\pi$  on the real axis. For  $2\pi \leq$



$\theta \leq 4\pi$  the path travels along the real axis from  $2\pi$  back to 0. The resulting curve is shown in Figure 4.3.

The curve winds once counterclockwise around the point  $-i\pi$ , which lies inside the spiral, and does not wind around the point  $i\pi$ , which lies outside the spiral. Thus the winding numbers are

$$n(\gamma; -i\pi) = 1, \quad n(\gamma; i\pi) = 0.$$

We compute the integral

$$\int_{\gamma} \frac{dz}{z^2 + \pi^2}.$$

Using the partial fraction decomposition

$$\frac{1}{z^2 + \pi^2} = \frac{1}{(z - i\pi)(z + i\pi)} = \frac{1}{2\pi i} \left( \frac{1}{z - i\pi} - \frac{1}{z + i\pi} \right),$$

we obtain

$$\int_{\gamma} \frac{dz}{z^2 + \pi^2} = \frac{1}{2\pi i} \left( \int_{\gamma} \frac{dz}{z - i\pi} - \int_{\gamma} \frac{dz}{z + i\pi} \right).$$

By the winding number formula,

$$\int_{\gamma} \frac{dz}{z - a} = 2\pi i n(\gamma; a),$$

so using the winding numbers read off from Figure 4.3,

$$\int_{\gamma} \frac{dz}{z - i\pi} = 2\pi i n(\gamma; i\pi) = 0, \quad \int_{\gamma} \frac{dz}{z + i\pi} = 2\pi i n(\gamma; -i\pi) = 2\pi i.$$

Substituting these into the previous expression gives

$$\int_{\gamma} \frac{dz}{z^2 + \pi^2} = \frac{1}{2\pi i} (0 - 2\pi i) = -1.$$

■

**7.** Let  $f(z) = [(z - \frac{1}{2} - i)(z - 1 - \frac{3}{2}i)(z - 1 - i)(z - \frac{3}{2} - i)]^{-1}$  and let  $\gamma$  be the polygon  $[0, 2, 2 + 2i, 2i, 0]$ . Find  $\int_{\gamma} f$ .

*Proof.* Let

$$f(z) = \frac{1}{(z - z_1)(z - z_2)(z - z_3)(z - z_4)},$$

where

$$z_1 = \frac{1}{2} + i, \quad z_2 = 1 + \frac{3}{2}i, \quad z_3 = 1 + i, \quad z_4 = \frac{3}{2} + i.$$

These are four distinct points, so the partial fraction expansion has the form

$$f(z) = \sum_{k=1}^4 \frac{A_k}{z - z_k},$$

where

$$A_k = \frac{1}{\prod_{j \neq k} (z_k - z_j)}.$$

Thus

$$\int_{\gamma} f(z) dz = \sum_{k=1}^4 A_k \int_{\gamma} \frac{dz}{z - z_k}.$$

The path

$$\gamma = [0, 2, 2 + 2i, 2i, 0]$$

is the positively oriented boundary of the square

$$0 < \Re z < 2, \quad 0 < \Im z < 2.$$

A simple check of real and imaginary parts shows that all four points  $z_1, z_2, z_3, z_4$  lie inside this square. Therefore

$$n(\gamma; z_k) = 1 \quad \text{for } k = 1, 2, 3, 4,$$

and hence

$$\int_{\gamma} \frac{dz}{z - z_k} = 2\pi i.$$

Therefore

$$\int_{\gamma} f(z) dz = \sum_{k=1}^4 A_k \cdot 2\pi i = 2\pi i \sum_{k=1}^4 A_k.$$

It remains to compute the four coefficients  $A_k$ . A direct computation gives

$$A_1 = -2 + 2i, \quad A_2 = -4i, \quad A_3 = -8i, \quad A_4 = 2 + 2i.$$

Adding them,

$$A_1 + A_2 + A_3 + A_4 = (-2 + 2) + (2i - 4i - 8i + 2i) = -8i.$$

Thus

$$\int_{\gamma} f(z) dz = 2\pi i (-8i) = 16\pi.$$

■

**10.** Find all possible values of  $\int_{\gamma} \frac{dz}{1+z^2}$  where  $\gamma$  is any closed rectifiable curve.

*Proof.* We use the partial fraction decomposition

$$\frac{1}{1+z^2} = \frac{1}{(z-i)(z+i)} = \frac{1}{2i} \left( \frac{1}{z-i} - \frac{1}{z+i} \right).$$

For any closed rectifiable curve  $\gamma$  and any  $a \in \mathbb{C}$ , we have

$$\int_{\gamma} \frac{dz}{z-a} = 2\pi i n(\gamma; a),$$

where  $n(\gamma; a)$  is the winding number of  $\gamma$  about  $a$ . Therefore

$$\int_{\gamma} \frac{dz}{1+z^2} = \frac{1}{2i} (2\pi i n(\gamma; i) - 2\pi i n(\gamma; -i)) = \pi (n(\gamma; i) - n(\gamma; -i)).$$

Since  $n(\gamma; i)$  and  $n(\gamma; -i)$  may be any integers (positive, negative, or zero) depending on how the curve winds around  $i$  and  $-i$ , the integral may take any value of the form

$$\{\pi k : k \in \mathbb{Z}\}.$$

■

## 4.7 Counting zeros; the Open Mapping Theorem

**4.** Suppose that  $f : G \rightarrow \mathbb{C}$  is analytic and one-one; show that  $f'(z) \neq 0$  for any  $z$  in  $G$ .

*Proof.* Let  $f : G \rightarrow \mathbb{C}$  be analytic and one-one, and fix a point  $a \in G$ . We show that  $f'(a) \neq 0$ . Suppose for contradiction that  $f'(a) = 0$ . Then  $f(z) - f(a)$  has a zero at  $z = a$  of multiplicity  $m \geq 2$ . By Theorem 7.4, there exist  $\varepsilon > 0$  and  $\delta > 0$  such that whenever  $0 < |\zeta - f(a)| < \delta$ , the equation

$$f(z) = \zeta$$

has *exactly*  $m$  distinct solutions in the disk  $B(a; \varepsilon)$ . Since  $m \geq 2$ , every  $\zeta$  sufficiently close to  $f(a)$  is taken at least twice by  $f$ . This contradicts the assumption that  $f$  is one-one on  $G$ . Therefore our assumption was impossible, and we must have  $f'(a) \neq 0$ . As  $a$  was arbitrary, it follows that

$$f'(z) \neq 0 \quad \text{for all } z \in G.$$



# Chapter 5

## Singularity

### 5.1 Classification of singularities

4. Let

$$f(z) = \frac{1}{z(z-1)(z-2)};$$

give the Laurent expansion of  $f(z)$  in each of the following annuli: (a)  $\text{ann}(0; 0, 1)$ ; (b)  $\text{ann}(0; 1, 2)$ ; (c)  $\text{ann}(0; 2, \infty)$ .

*Solution.* We start from the partial fraction decomposition

$$f(z) = \frac{1}{z(z-1)(z-2)} = \frac{1}{2z} - \frac{1}{z-1} + \frac{1}{2(z-2)}.$$

We expand in powers of  $z$  using geometric series formula.

**(a) The annulus**  $0 < |z| < 1$ .

$$\frac{1}{z-1} = -\sum_{n=0}^{\infty} z^n, \quad \frac{1}{z-2} = -\frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z}{2}\right)^n.$$

Thus

$$f(z) = \frac{1}{2}z^{-1} + \sum_{n=0}^{\infty} z^n - \frac{1}{4} \sum_{n=0}^{\infty} 2^{-n} z^n.$$

Therefore the Laurent coefficients are

$$a_{-1} = \frac{1}{2}, \quad a_n = 1 - 2^{-(n+2)} \quad (n \geq 0), \quad a_n = 0 \quad (n \leq -2).$$

So

$$f(z) = \frac{1}{2}z^{-1} + \sum_{n=0}^{\infty} (1 - 2^{-(n+2)}) z^n, \quad 0 < |z| < 1.$$

**(b) The annulus**  $1 < |z| < 2$ .

$$\frac{1}{z-1} = \frac{1}{z} \frac{1}{1-\frac{1}{z}} = \sum_{n=1}^{\infty} z^{-n}, \quad \frac{1}{z-2} = -\frac{1}{2} \frac{1}{1-\frac{z}{2}} = -\frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z}{2}\right)^n.$$

Hence

$$f(z) = \frac{1}{2}z^{-1} - \sum_{n=1}^{\infty} z^{-n} - \frac{1}{4} \sum_{n=0}^{\infty} 2^{-n} z^n.$$

The coefficients are

$$a_{-1} = -\frac{1}{2}, \quad a_{-n} = -1 \quad (n \geq 2), \quad a_n = -2^{-(n+2)} \quad (n \geq 0).$$

Thus,

$$f(z) = -\frac{1}{2}z^{-1} - \sum_{n=2}^{\infty} z^{-n} - \sum_{n=0}^{\infty} 2^{-(n+2)} z^n, \quad 1 < |z| < 2.$$

**(c) The annulus**  $|z| > 2$ .

To apply the geometric series here, we factor out  $z$  so that  $|1/z|$  and  $|2/z|$  are less than 1 in the annuli.

$$\frac{1}{z-1} = \frac{1}{z} \frac{1}{1-\frac{1}{z}} = \sum_{n=1}^{\infty} z^{-n}, \quad \frac{1}{z-2} = \frac{1}{z} \frac{1}{1-\frac{2}{z}} = \sum_{n=1}^{\infty} \frac{2^{n-1}}{z^n}.$$

Hence

$$f(z) = \frac{1}{2}z^{-1} - \sum_{n=1}^{\infty} z^{-n} + \sum_{n=1}^{\infty} 2^{n-2} z^{-n}.$$

The coefficients are

$$a_{-1} = 0, \quad a_{-n} = 2^{n-2} - 1 \quad (n \geq 2), \quad a_n = 0 \quad (n \geq 0).$$

Thus

$$f(z) = \sum_{n=2}^{\infty} (2^{n-2} - 1) z^{-n}, \quad |z| > 2.$$

■

**15.** Let  $f$  be analytic in

$$G = \{z : 0 < |z - a| < r\}$$

except that there is a sequence of poles  $\{a_n\}$  in  $G$  with  $a_n \rightarrow a$ . Show that for any  $\omega \in \mathbb{C}$  there exists a sequence  $\{z_n\} \subset G$  with

$$a = \lim z_n \quad \text{and} \quad \omega = \lim f(z_n).$$

*Proof.* Fix  $\omega \in \mathbb{C}$ . We show that there exists a sequence  $z_n \in G$  with  $z_n \rightarrow a$  and

$f(z_n) \rightarrow \omega$ .

Suppose for contradiction that no such sequence exists. We claim that there is  $\delta > 0$  such that

$$0 < |z - a| < \delta \implies f(z) \neq \omega. \quad (5.1)$$

Indeed, if (5.1) were false, then for each  $n$  we could choose  $z_n \in G$  with  $0 < |z_n - a| < 1/n$  and  $f(z_n) = \omega$ , which would imply  $z_n \rightarrow a$  and  $f(z_n) \rightarrow \omega$ , contradicting the assumption. Hence (5.1) holds for some  $\delta > 0$ .

On the punctured disk  $D^* := \{z : 0 < |z - a| < \delta\}$  define

$$g(z) := \frac{1}{f(z) - \omega}$$

whenever  $z \neq a_n$ . Then  $g$  is analytic on  $D^* \setminus \{a_n\}$ .

**Step 1: Each  $a_n$  is a removable singularity of  $g$ .** Let  $a_n \in D^*$  be a pole of  $f$ , say of order  $m \geq 1$ . Then there is a holomorphic  $h$  near  $a_n$  with  $h(a_n) \neq 0$  such that

$$f(z) = \frac{h(z)}{(z - a_n)^m}.$$

Hence

$$f(z) - \omega = \frac{h(z) - \omega(z - a_n)^m}{(z - a_n)^m},$$

and the numerator satisfies  $(h - \omega(\cdot - a_n)^m)(a_n) = h(a_n) \neq 0$ . Therefore  $f(z) - \omega$  has a pole of order  $m$  at  $a_n$ , so  $g(z)$  has a zero of order  $m$  at  $a_n$  and extends holomorphically across  $a_n$  by setting  $g(a_n) = 0$ .

Doing this for each  $a_n \in D^*$ , we obtain a holomorphic function (still denoted  $g$ ) on  $D^*$  whose zeros include all  $a_n$  in  $D^*$ . Since  $a_n \rightarrow a$ ,  $g$  has infinitely many zeros accumulating at  $a$ .

**Step 2: The singularity of  $g$  at  $a$  is essential.** The point  $a$  is an isolated singularity of  $g$  (because  $g$  is holomorphic on  $D^*$ ). It cannot be removable: if  $g$  extended holomorphically to  $a$ , then the extension would be holomorphic on a neighborhood of  $a$  and vanish at infinitely many points  $a_n \rightarrow a$ , hence would vanish identically by the identity theorem, impossible since  $g(z) = 1/(f(z) - \omega)$  is not identically zero.

It also cannot be a pole: if  $a$  were a pole of order  $m \geq 1$ , then  $g(z) = h(z)/(z - a)^m$  near  $a$  with  $h$  holomorphic and  $h(a) \neq 0$ . But for  $n$  large,  $a_n$  lies in this neighborhood and  $0 = g(a_n) = h(a_n)/(a_n - a)^m$ , so  $h(a_n) = 0$  for infinitely many  $a_n \rightarrow a$ , forcing  $h \equiv 0$  near  $a$  by the identity theorem, contradicting  $h(a) \neq 0$ .

Thus  $a$  is neither removable nor a pole, hence it is an essential singularity of  $g$ .

**Step 3: Produce a sequence with  $f(z_n) \rightarrow \omega$ .** By Casorati–Weierstrass (equivalently, by the fact that a function with an essential singularity is unbounded on every punctured neighborhood),  $g$  is unbounded on each  $\{0 < |z - a| < 1/n\}$ . Hence

we can choose  $z_n$  such that

$$0 < |z_n - a| < \frac{1}{n} \quad \text{and} \quad |g(z_n)| > n.$$

Then  $z_n \rightarrow a$  and  $|g(z_n)| \rightarrow \infty$ , so  $1/g(z_n) \rightarrow 0$ . But

$$\frac{1}{g(z_n)} = f(z_n) - \omega,$$

so  $f(z_n) \rightarrow \omega$ , contradicting the initial assumption.

Therefore for this  $\omega$  there exists a sequence  $z_n \in G$  with  $z_n \rightarrow a$  and  $f(z_n) \rightarrow \omega$ . Since  $\omega \in \mathbb{C}$  was arbitrary, the conclusion holds for every  $\omega \in \mathbb{C}$ . ■

## 5.2 Residues

**2 (c).** *Verify the following equation:*

$$\int_0^\infty \frac{\cos(ax)}{(1+x^2)^2} dx = \frac{\pi(a+1)e^{-a}}{4}, \quad a > 0.$$

*Solution.* We want to show that for  $a > 0$ ,

$$\int_0^\infty \frac{\cos(ax)}{(1+x^2)^2} dx = \frac{\pi(a+1)e^{-a}}{4}.$$

Consider the complex-valued function

$$f(z) = \frac{e^{iaz}}{(1+z^2)^2}, \quad a > 0,$$

and for  $R > 1$  integrate it over the semicircular contour  $C_R$  in the upper half-plane, consisting of the segment  $[-R, R]$  on the real axis and the semicircle

$$\gamma_R : z = Re^{it}, \quad t \in [0, \pi].$$

The singularities of  $f$  are at  $z = i$  and  $z = -i$ . In the upper half-plane only  $z = i$  lies inside the contour, and it is a pole of order two. By the residue theorem,

$$\int_{C_R} f(z) dz = 2\pi i \operatorname{Res}(f; i).$$



**The integral over the semicircle.** We claim that

$$\int_{\gamma_R} f(z) dz \longrightarrow 0 \quad \text{as } R \rightarrow \infty.$$

On  $\gamma_R$  we have  $z = Re^{it}$ ,  $t \in [0, \pi]$ , hence

$$|e^{iaz}| = |e^{iaRe^{it}}| = e^{\Re(iaRe^{it})} = e^{-aR \sin t} \leq 1.$$

Moreover, for  $R > 1$ ,

$$|1 + z^2| = |1 + R^2 e^{2it}| \geq ||z|^2 - 1| = R^2 - 1,$$

so

$$|(1 + z^2)^2| \geq (R^2 - 1)^2.$$

Therefore, for  $R > 1$ ,

$$\sup_{z \in \gamma_R} |f(z)| \leq \frac{1}{(R^2 - 1)^2}.$$

The length of  $\gamma_R$  is  $\pi R$ , so

$$\left| \int_{\gamma_R} f(z) dz \right| \leq (\text{length of } \gamma_R) \cdot \sup_{z \in \gamma_R} |f(z)| \leq \pi R \cdot \frac{1}{(R^2 - 1)^2} \longrightarrow 0 \quad \text{as } R \rightarrow \infty.$$

Thus

$$\int_{-\infty}^{\infty} \frac{e^{iax}}{(1 + x^2)^2} dx = 2\pi i \operatorname{Res}(f; i).$$

**Residue at  $z = i$ .** We write

$$f(z) = \frac{e^{iaz}}{(z^2 + 1)^2} = \frac{e^{iaz}}{(z - i)^2(z + i)^2}.$$

Thus  $z = i$  is a double pole. For a double pole at  $z = i$  we have

$$\operatorname{Res}(f; i) = \frac{d}{dz} [(z - i)^2 f(z)] \Big|_{z=i} = \frac{d}{dz} \left( \frac{e^{iaz}}{(z + i)^2} \right) \Big|_{z=i}.$$

Set

$$g(z) = \frac{e^{iaz}}{(z + i)^2}.$$

Then

$$g'(z) = \frac{e^{iaz}(ia(z + i)^2 - 2(z + i))}{(z + i)^4}.$$

So

$$g'(i) = \frac{e^{iai}(ia(-4) - 2(2i))}{16} = -\frac{i}{4}(a+1)e^{-a}.$$

Hence

$$\text{Res}(f; i) = -\frac{i}{4}(a+1)e^{-a}.$$

Thus

$$\int_{-\infty}^{\infty} \frac{e^{iax}}{(1+x^2)^2} dx = 2\pi i \left( -\frac{i}{4}(a+1)e^{-a} \right) = \frac{\pi}{2}(a+1)e^{-a}.$$

**Extracting the cosine integral.** Write

$$\int_{-\infty}^{\infty} \frac{e^{iax}}{(1+x^2)^2} dx = \int_{-\infty}^{\infty} \frac{\cos(ax)}{(1+x^2)^2} dx + i \int_{-\infty}^{\infty} \frac{\sin(ax)}{(1+x^2)^2} dx.$$

The integrand of the sine term is odd, so

$$\int_{-\infty}^{\infty} \frac{\sin(ax)}{(1+x^2)^2} dx = 0.$$

The integrand of the cosine term is even, hence

$$\int_{-\infty}^{\infty} \frac{\cos(ax)}{(1+x^2)^2} dx = 2 \int_0^{\infty} \frac{\cos(ax)}{(1+x^2)^2} dx.$$

Therefore,

$$2 \int_0^{\infty} \frac{\cos(ax)}{(1+x^2)^2} dx = \frac{\pi}{2}(a+1)e^{-a},$$

which implies

$$\int_0^{\infty} \frac{\cos(ax)}{(1+x^2)^2} dx = \frac{\pi}{4}(a+1)e^{-a},$$

as required. ■

## 5.3 The Argument Principle

**2.** Suppose  $f$  is analytic on  $\overline{B(0;1)}$  and satisfies  $|f(z)| < 1$  for  $|z| = 1$ . Find the number of solutions (counting multiplicities) of the equation

$$f(z) = z^n$$

where  $n$  is an integer larger than or equal to 1.

*Solution.* We seek the number of solutions of

$$f(z) = z^n \quad (n \geq 1).$$

Rewrite this as

$$f(z) - z^n = 0.$$

Define

$$h(z) = z^n, \quad g(z) = f(z) - z^n.$$

Both  $g$  and  $h$  are analytic on and inside the unit circle  $\gamma = \{z : |z| = 1\}$ .

On  $\gamma$  we have  $|h(z)| = |z^n| = 1$ , and by hypothesis

$$|h(z) + g(z)| = |f(z)| < 1 = |h(z)|.$$

Thus the inequality required by Rouché's theorem holds on  $\gamma$ . Therefore  $h$  and  $g$  have the same number of zeros inside  $\gamma$ . Since  $h(z) = z^n$  has exactly  $n$  zeros in  $|z| < 1$ , it follows that  $g(z) = f(z) - z^n$  has exactly  $n$  zeros in  $|z| < 1$ . Thus the equation  $f(z) = z^n$  has exactly  $n$  solutions in the unit disk. ■

# Chapter 6

## The Maximum Modulus Theorem

### 6.1 The Maximum Principle

**2.** Let  $G$  be a bounded region and suppose  $f$  is continuous on  $\overline{G}$  and analytic on  $G$ . Show that if there is a constant  $c \geq 0$  such that  $|f(z)| = c$  for all  $z$  on the boundary of  $G$ , then either  $f$  is a constant function or  $f$  has a zero in  $G$ .

*Solution.* Let  $G$  be a bounded region, and let  $f$  be continuous on  $\overline{G}$  and analytic on  $G$ . Assume that

$$|f(z)| = c \quad \text{for all } z \in \partial G,$$

where  $c \geq 0$ .

**Case 1:**  $c = 0$ . Then  $f = 0$  on  $\partial G$ . By the maximum modulus principle,

$$\max_{\overline{G}} |f| = \max_{\partial G} |f| = 0,$$

so  $f \equiv 0$  on  $G$ , hence  $f$  is constant.

**Case 2:**  $c > 0$ . Suppose that  $f$  has no zeros in  $G$ . Then  $|f|$  is a positive continuous function on  $\overline{G}$  and is the modulus of a nonvanishing analytic function on  $G$ . Hence, by the minimum modulus principle,

$$\min_{\overline{G}} |f| = \min_{\partial G} |f|.$$

Since  $|f| = c$  on  $\partial G$ , we get

$$\min_{\overline{G}} |f| = c,$$

and therefore  $|f(z)| \geq c$  for all  $z \in G$ .

On the other hand, by the maximum modulus principle applied to  $f$ ,

$$\max_{\overline{G}} |f| = \max_{\partial G} |f| = c,$$

so  $|f(z)| \leq c$  for all  $z \in G$ .

Combining the two inequalities yields  $|f(z)| = c$  for all  $z \in G$ . By the maximum modulus principle (in its strong form), an analytic function whose modulus is constant on a region must be constant. Hence  $f$  is constant.

Therefore, either  $f$  is constant or  $f$  has a zero in  $G$ . ■

**6.** Suppose that both  $f$  and  $g$  are analytic on  $\overline{B(0; R)}$  with

$$|f(z)| = |g(z)| \quad \text{for } |z| = R.$$

Show that if neither  $f$  nor  $g$  vanishes in  $B(0; R)$ , then there is a constant  $\lambda$ , with  $|\lambda| = 1$ , such that  $f = \lambda g$ .

*Solution.* Assume  $f$  and  $g$  are analytic on  $\overline{B(0; R)}$  and satisfy  $|f(z)| = |g(z)|$  for  $|z| = R$ . Also assume neither  $f$  nor  $g$  vanishes in  $B(0; R)$ .

Since  $g$  has no zeros in  $B(0; R)$ , the quotient

$$h(z) := \frac{f(z)}{g(z)}$$

is analytic on  $B(0; R)$ . On the boundary  $|z| = R$  we have

$$|h(z)| = \frac{|f(z)|}{|g(z)|} = 1.$$

Because  $h$  is continuous on  $\overline{B(0; R)}$  and  $|h|$  attains both its maximum and minimum on the boundary, the maximum modulus principle gives

$$|h(z)| \leq 1 \quad \text{for } |z| < R,$$

and since  $h$  is nonvanishing (as  $f$  is also nonvanishing in  $B(0; R)$ ), the minimum modulus principle gives

$$|h(z)| \geq 1 \quad \text{for } |z| < R.$$

Hence  $|h(z)| \equiv 1$  throughout  $B(0; R)$ .

An analytic function with constant modulus on a region must be constant. Therefore  $h$  is constant, say  $h \equiv \lambda$ , and then  $|\lambda| = 1$  (because  $|h| = 1$ ). Thus

$$f = \lambda g \quad \text{on } B(0; R),$$

and by continuity the same holds on  $\overline{B(0; R)}$ . ■

**7.** Let  $f$  be analytic in the disk  $B(0; R)$  and for  $0 \leq r < R$  define

$$A(r) = \max\{\Re f(z) : |z| = r\}.$$

Show that unless  $f$  is a constant,  $A(r)$  is a strictly increasing function of  $r$ .

*Solution.* Define

$$g(z) = e^{f(z)}.$$

Since  $f$  is analytic in  $B(0; R)$ , the function  $g$  is also analytic there. Moreover,

$$|g(z)| = |e^{f(z)}| = e^{\Re f(z)}.$$

Hence for each  $0 \leq r < R$ ,

$$\max_{|z|=r} |g(z)| = \max_{|z|=r} e^{\Re f(z)} = e^{\max_{|z|=r} \Re f(z)} = e^{A(r)}.$$

Now let

$$M(r) = \max_{|z|=r} |g(z)|.$$

By the maximum modulus principle,  $M(r)$  is an increasing function of  $r$ . Therefore  $e^{A(r)}$  is increasing, and since the exponential function is strictly increasing on  $\mathbb{R}$ , it follows that  $A(r)$  is increasing.

It remains to prove strict increase unless  $f$  is constant. Suppose there exist  $0 \leq r_1 < r_2 < R$  such that

$$A(r_1) = A(r_2).$$

Then

$$M(r_1) = e^{A(r_1)} = e^{A(r_2)} = M(r_2).$$

So the maximum modulus of the analytic function  $g$  is the same on the circles  $|z| = r_1$  and  $|z| = r_2$ . Since  $r_1 < r_2$ , the maximum modulus principle implies that  $g$  must be constant on  $B(0; r_2)$ . Thus  $e^{f(z)}$  is constant, say

$$e^{f(z)} = c \neq 0.$$

Differentiating, we get

$$f'(z)e^{f(z)} = 0.$$

Because  $e^{f(z)} \neq 0$ , it follows that

$$f'(z) = 0$$

for all  $z \in B(0; r_2)$ . Hence  $f$  is constant on  $B(0; r_2)$ , and therefore constant on  $B(0; R)$ .

Thus, unless  $f$  is constant, equality  $A(r_1) = A(r_2)$  cannot happen when  $r_1 < r_2$ . Therefore

$$A(r_1) < A(r_2) \quad \text{whenever } 0 \leq r_1 < r_2 < R.$$

So  $A(r)$  is strictly increasing unless  $f$  is constant. ■

## 6.2 Schwarz's Lemma

1. Suppose  $|f(z)| \leq 1$  for  $|z| < 1$  and  $f$  is a non-constant analytic function. By considering the function  $g : \mathbb{D} \rightarrow \mathbb{D}$  defined by

$$g(z) = \frac{f(z) - a}{1 - \bar{a}f(z)}$$

where  $a = f(0)$ , prove that

$$\frac{|f(0)| - |z|}{1 + |f(0)||z|} \leq |f(z)| \leq \frac{|f(0)| + |z|}{1 - |f(0)||z|}$$

for  $|z| < 1$ .

*Solution.* Let  $a = f(0)$  and define

$$g(z) = \frac{f(z) - a}{1 - \bar{a}f(z)}.$$

Since  $|f(z)| \leq 1$  for  $|z| < 1$  and  $|a| < 1$ , the map

$$w \mapsto \frac{w - a}{1 - \bar{a}w}$$

is an automorphism of the unit disk  $\mathbb{D}$ . Hence  $g : \mathbb{D} \rightarrow \mathbb{D}$  is analytic. Moreover,

$$g(0) = \frac{f(0) - a}{1 - \bar{a}f(0)} = 0.$$

Because  $f$  is non-constant,  $g$  is also non-constant. Therefore by Schwarz's lemma,

$$|g(z)| \leq |z| \quad \text{for } |z| < 1.$$

Thus

$$\left| \frac{f(z) - a}{1 - \bar{a}f(z)} \right| \leq |z|.$$

Let  $w = f(z)$ . Then

$$|w - a| \leq |z| |1 - \bar{a}w|.$$

**Upper bound.** Using the triangle inequality,

$$|w - a| \geq ||w| - |a|| \geq |w| - |a|$$

and

$$|1 - \bar{a}w| \leq 1 + |a||w|.$$

Hence

$$|w| - |a| \leq |z|(1 + |a||w|).$$

Rearranging gives

$$|w|(1 - |a||z|) \leq |a| + |z|,$$

so

$$|f(z)| = |w| \leq \frac{|a| + |z|}{1 - |a||z|}.$$

**Lower bound.** Again from

$$|w - a| \leq |z||1 - \bar{a}w|,$$

we use

$$|w - a| \geq |a| - |w|$$

and

$$|1 - \bar{a}w| \leq 1 + |a||w|.$$

Thus

$$|a| - |w| \leq |z|(1 + |a||w|).$$

Rearranging yields

$$|a| - |z| \leq |w|(1 + |a||z|),$$

and therefore

$$|f(z)| = |w| \geq \frac{|a| - |z|}{1 + |a||z|}.$$

Since  $a = f(0)$ , we conclude that for  $|z| < 1$ ,

$$\frac{|f(0)| - |z|}{1 + |f(0)||z|} \leq |f(z)| \leq \frac{|f(0)| + |z|}{1 - |f(0)||z|}.$$

■

**3.** Suppose  $f : \mathbb{D} \rightarrow \mathbb{C}$  satisfies  $\Re f(z) \geq 0$  for all  $z \in \mathbb{D}$  and suppose that  $f$  is analytic and not constant.

(a) Show that  $\Re f(z) > 0$  for all  $z \in \mathbb{D}$ .

(b) By using an appropriate Möbius transformation, apply Schwarz's Lemma to prove that if  $f(0) = 1$  then

$$|f(z)| \leq \frac{1 + |z|}{1 - |z|}$$

for  $|z| < 1$ . What can be said if  $f(0) \neq 1$ ?



(c) Show that if  $f(0) = 1$ ,  $f$  also satisfies

$$|f(z)| \geq \frac{1 - |z|}{1 + |z|}.$$

*Hint: Use part (a).*

*Solution.* (a) Let

$$u(z) = \Re f(z).$$

Since  $f$  is analytic on  $\mathbb{D}$ , the function  $u$  is harmonic on  $\mathbb{D}$ . We are given that

$$u(z) \geq 0 \quad (z \in \mathbb{D}).$$

Suppose there exists  $z_0 \in \mathbb{D}$  such that  $u(z_0) = 0$ . Then  $u$  attains its minimum value at an interior point. By the minimum principle for harmonic functions,  $u$  must be constant on  $\mathbb{D}$ . Hence  $\Re f$  is constant.

If  $f = u + iv$  and  $u$  is constant, then  $u_x = u_y = 0$ . By the Cauchy–Riemann equations,

$$v_x = -u_y = 0, \quad v_y = u_x = 0,$$

so  $v$  is also constant. Therefore  $f$  is constant, contrary to hypothesis. Hence no such  $z_0$  exists, and so

$$\Re f(z) > 0 \quad \text{for all } z \in \mathbb{D}.$$

(b) Assume  $f(0) = 1$ . Since  $\Re f(z) > 0$  for all  $z \in \mathbb{D}$ , the image of  $f$  lies in the right half-plane. Consider the Möbius transformation

$$\phi(w) = \frac{w - 1}{w + 1}.$$

This maps the right half-plane  $\{w : \Re w > 0\}$  into  $\mathbb{D}$ . Define

$$g(z) = \phi(f(z)) = \frac{f(z) - 1}{f(z) + 1}.$$

Then  $g : \mathbb{D} \rightarrow \mathbb{D}$  is analytic, and

$$g(0) = \frac{f(0) - 1}{f(0) + 1} = 0.$$

By Schwarz's Lemma,

$$|g(z)| \leq |z| \quad (z \in \mathbb{D}).$$

Thus

$$\left| \frac{f(z) - 1}{f(z) + 1} \right| \leq |z|.$$

Let  $r = |z|$  and write  $w = f(z)$ . Then

$$|w - 1| \leq r |w + 1|.$$

Using the triangle inequality,

$$|w| - 1 \leq |w - 1| \leq r |w + 1| \leq r(|w| + 1).$$

Hence

$$|w| - 1 \leq r|w| + r,$$

so

$$|w|(1 - r) \leq 1 + r.$$

Therefore

$$|f(z)| \leq \frac{1 + |z|}{1 - |z|}.$$

If  $f(0) \neq 1$ , let  $a = f(0)$ , so  $\Re a > 0$ . Then

$$\psi(w) = \frac{w - a}{w + \bar{a}}$$

maps the right half-plane into  $\mathbb{D}$ , and  $\psi(a) = 0$ . Thus

$$h(z) = \psi(f(z)) = \frac{f(z) - a}{f(z) + \bar{a}}$$

is analytic from  $\mathbb{D}$  to  $\mathbb{D}$  with  $h(0) = 0$ . By Schwarz's Lemma,

$$\left| \frac{f(z) - a}{f(z) + \bar{a}} \right| \leq |z|.$$

This gives the corresponding estimate for general  $f(0) = a$ :

$$|f(z)| \leq \frac{|a|(1 + |z|)}{1 - |z|}.$$

(c) Again assume  $f(0) = 1$ . From part (a),  $\Re f(z) > 0$  for all  $z \in \mathbb{D}$ . Hence

$$\Re \left( \frac{1}{f(z)} \right) = \frac{\Re f(z)}{|f(z)|^2} > 0,$$

so the function

$$F(z) = \frac{1}{f(z)}$$

is analytic on  $\mathbb{D}$ , non-constant, and satisfies  $\Re F(z) > 0$  for all  $z \in \mathbb{D}$ . Also,

$$F(0) = \frac{1}{f(0)} = 1.$$

Applying part (b) to  $F$ , we obtain

$$|F(z)| \leq \frac{1 + |z|}{1 - |z|}.$$

That is,

$$\frac{1}{|f(z)|} \leq \frac{1 + |z|}{1 - |z|}.$$

Taking reciprocals gives

$$|f(z)| \geq \frac{1 - |z|}{1 + |z|}.$$

This proves all three parts. ■

**6.** Suppose  $f$  is analytic in some region containing  $\overline{B}(0, 1)$  and  $|f(z)| = 1$  where  $|z| = 1$ . Find a formula for  $f$ . Hint: First consider the case where  $f$  has no zeros in  $B(0, 1)$ .

*Solution.* We shall show that  $f$  must be a finite Blaschke product. That is,

$$f(z) = \lambda \prod_{k=1}^n \frac{z - a_k}{1 - \overline{a_k}z}, \quad |\lambda| = 1, \quad |a_k| < 1.$$

Since  $f$  is analytic in a region containing  $\overline{B}(0, 1)$ , it is analytic on a neighborhood of the closed unit disk. Because

$$|f(z)| = 1 \quad (|z| = 1),$$

$f$  has no zeros on the unit circle. Let  $a_1, \dots, a_n$  be the zeros of  $f$  in  $B(0, 1)$ , counted with multiplicity. Since  $f$  is analytic on a neighborhood of  $\overline{B}(0, 1)$ , it has only finitely many such zeros.

**Case 1:  $f$  has no zeros in  $B(0, 1)$ .**

Then  $f$  has no zeros on  $\overline{B}(0, 1)$ , so  $1/f$  is analytic on a neighborhood of  $\overline{B}(0, 1)$ . Since  $f$  is analytic on the disk and continuous on the boundary, the maximum modulus principle implies

$$|f(z)| \leq 1 \quad (|z| < 1).$$

Because  $1/f$  is also analytic and satisfies

$$\left| \frac{1}{f(z)} \right| = 1 \quad (|z| = 1),$$

the maximum modulus principle applied to  $1/f$  gives

$$\left| \frac{1}{f(z)} \right| \leq 1 \quad (|z| < 1),$$

hence

$$|f(z)| \geq 1.$$

Therefore

$$|f(z)| = 1 \quad (|z| < 1).$$

If  $f$  were nonconstant, the open mapping theorem would imply that  $f(B(0, 1))$  is open, which is impossible since it lies in the unit circle. Thus  $f$  is constant:

$$f(z) \equiv \lambda, \quad |\lambda| = 1.$$

## Case 2: general case.

Let

$$B(z) = \prod_{k=1}^n \frac{z - a_k}{1 - \overline{a_k}z}.$$

This is a finite Blaschke product. Since  $|a_k| < 1$ , each factor is analytic on a neighborhood of  $\overline{B}(0, 1)$ . Moreover, for  $|z| = 1$ ,

$$\left| \frac{z - a_k}{1 - \overline{a_k}z} \right| = 1,$$

so

$$|B(z)| = 1 \quad (|z| = 1).$$

Define

$$g(z) = \frac{f(z)}{B(z)}.$$

The zeros of  $B$  in  $B(0, 1)$  are exactly  $a_1, \dots, a_n$  with the same multiplicities, so the singularities of  $f/B$  at these points are removable. Hence  $g$  is analytic in  $B(0, 1)$ , and in fact analytic on a neighborhood of  $\overline{B}(0, 1)$ .

Moreover,  $g$  has no zeros in  $B(0, 1)$ , and for  $|z| = 1$ ,

$$|g(z)| = \frac{|f(z)|}{|B(z)|} = 1.$$

Thus  $g$  satisfies the conditions of Case 1, so  $g$  is constant:

$$g(z) \equiv \lambda \quad \text{for some } |\lambda| = 1.$$

Therefore

$$f(z) = \lambda B(z) = \lambda \prod_{k=1}^n \frac{z - a_k}{1 - \overline{a_k}z}, \quad |\lambda| = 1, \quad |a_k| < 1.$$

Hence every analytic function on a neighborhood of  $\overline{B}(0, 1)$  satisfying  $|f(z)| = 1$  on  $|z| = 1$  is a finite Blaschke product times a unimodular constant. ■

### 6.3 Convex functions and Hadamard's Three Circles Theorem

4. Supply the details in the proof of Hadamard's Three Circles Theorem.

*Solution.* Let

$$A = \text{ann}(0; R_1, R_2) = \{z : R_1 < |z| < R_2\},$$

and suppose  $f$  is analytic on  $A$  and not identically zero. For  $R_1 < r < R_2$ , define

$$M(r) = \max\{|f(re^{i\theta})| : 0 \leq \theta \leq 2\pi\}.$$

We will prove that for  $R_1 < r_1 \leq r \leq r_2 < R_2$ ,  $r_1 \neq r_2$ ,

$$\log M(r) \leq \frac{\log r_2 - \log r}{\log r_2 - \log r_1} \log M(r_1) + \frac{\log r - \log r_1}{\log r_2 - \log r_1} \log M(r_2).$$

Fix  $r_1, r_2$  with  $R_1 < r_1 < r_2 < R_2$ . Since  $f$  is not identically zero, its zeros are isolated; hence for all but finitely many  $\rho \in [r_1, r_2]$ ,  $f$  has no zeros on  $|z| = \rho$ . We first prove the inequality under the additional assumption that  $f$  has no zeros on the closed annulus

$$\{z : r_1 \leq |z| \leq r_2\}.$$

Then  $\log |f|$  is harmonic on the open annulus

$$A_0 = \{z : r_1 < |z| < r_2\}.$$

Define

$$u(z) = \log |f(z)|.$$

Also define

$$v(z) = \frac{\log r_2 - \log |z|}{\log r_2 - \log r_1} \log M(r_1) + \frac{\log |z| - \log r_1}{\log r_2 - \log r_1} \log M(r_2).$$

Since  $\log |z|$  is harmonic away from 0,  $v$  is harmonic on  $A_0$ .

Now if  $|z| = r_1$ , then

$$v(z) = \log M(r_1),$$

while

$$u(z) = \log |f(z)| \leq \log M(r_1).$$

Similarly, if  $|z| = r_2$ , then

$$v(z) = \log M(r_2), \quad u(z) = \log |f(z)| \leq \log M(r_2).$$

Hence

$$u(z) \leq v(z) \quad (|z| = r_1 \text{ or } |z| = r_2).$$

Since  $u - v$  is harmonic on  $A_0$  and has nonpositive boundary values, the maximum principle for harmonic functions gives

$$u(z) \leq v(z) \quad (r_1 < |z| < r_2).$$

In particular, if  $|z| = r$  with  $r_1 \leq r \leq r_2$ , then

$$\log |f(z)| \leq \frac{\log r_2 - \log r}{\log r_2 - \log r_1} \log M(r_1) + \frac{\log r - \log r_1}{\log r_2 - \log r_1} \log M(r_2).$$

Taking the maximum over  $|z| = r$ , we obtain

$$\log M(r) \leq \frac{\log r_2 - \log r}{\log r_2 - \log r_1} \log M(r_1) + \frac{\log r - \log r_1}{\log r_2 - \log r_1} \log M(r_2).$$

It remains to remove the temporary assumption that  $f$  has no zeros in the closed annulus  $r_1 \leq |z| \leq r_2$ .

Choose  $\varepsilon > 0$  so small that

$$R_1 < r_1 - \varepsilon < r_1 \leq r \leq r_2 < r_2 + \varepsilon < R_2,$$

and so that  $f$  has no zeros on the circles

$$|z| = r_1 - \varepsilon, \quad |z| = r, \quad |z| = r_2 + \varepsilon.$$

This is possible because the zeros of  $f$  are isolated, so only finitely many radii correspond to circles meeting a zero in a compact subannulus.

Apply the already proved case to the annulus

$$r_1 - \varepsilon \leq |z| \leq r_2 + \varepsilon.$$

Then

$$\log M(r) \leq \frac{\log(r_2 + \varepsilon) - \log r}{\log(r_2 + \varepsilon) - \log(r_1 - \varepsilon)} \log M(r_1 - \varepsilon) + \frac{\log r - \log(r_1 - \varepsilon)}{\log(r_2 + \varepsilon) - \log(r_1 - \varepsilon)} \log M(r_2 + \varepsilon).$$

Since  $M(\rho)$  depends continuously on  $\rho$  (because  $f$  is continuous), letting  $\varepsilon \downarrow 0$  yields

$$\log M(r) \leq \frac{\log r_2 - \log r}{\log r_2 - \log r_1} \log M(r_1) + \frac{\log r - \log r_1}{\log r_2 - \log r_1} \log M(r_2).$$

This proves Hadamard's Three Circles Theorem.

Finally, this inequality is exactly the statement that  $\log M(r)$  is a convex function of  $\log r$ : if we write

$$x = \log r, \quad x_1 = \log r_1, \quad x_2 = \log r_2,$$

then

$$\log M(r) \leq \frac{x_2 - x}{x_2 - x_1} \log M(r_1) + \frac{x - x_1}{x_2 - x_1} \log M(r_2),$$

which is the usual convexity inequality. ■

**6. Prove Hardy's Theorem:** *If  $f$  is analytic on  $B(0; R)$  and not constant then*

$$I(r) = \frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})| d\theta$$

*is strictly increasing and  $\log I(r)$  is a convex function of  $\log r$ . Hint: If  $0 < r_1 < r < r_2$  find a continuous function  $\varphi : [0, 2\pi] \rightarrow \mathbb{C}$  such that  $\varphi(\theta)f(re^{i\theta}) = |f(re^{i\theta})|$  and consider the function*

$$F(z) = \int_0^{2\pi} f(ze^{i\theta}) \varphi(\theta) d\theta.$$

*(Note that  $r$  is fixed, so  $\varphi$  may depend on  $r$ .)*

*Solution.* Fix  $0 < r_1 < r < r_2 < R$ . We shall prove first that  $I(r)$  is strictly increasing, and then that  $\log I(r)$  is convex as a function of  $\log r$ .

**Step 1:  $I(r)$  is increasing.**

Fix  $r \in (0, R)$ . Since  $f$  is continuous and  $f(re^{i\theta}) \neq 0$  except possibly at finitely many  $\theta$ , we can choose a continuous function

$$\varphi : [0, 2\pi] \rightarrow \mathbb{C}, \quad |\varphi(\theta)| = 1,$$

such that

$$\varphi(\theta)f(re^{i\theta}) = |f(re^{i\theta})| \quad (0 \leq \theta \leq 2\pi).$$

For instance, on the set where  $f(re^{i\theta}) \neq 0$ , define

$$\varphi(\theta) = \frac{|f(re^{i\theta})|}{f(re^{i\theta})},$$

and extend continuously across zeros of  $f(re^{i\theta})$ , which is possible because zeros on the circle are isolated and of finite order.

Now define

$$F(z) = \int_0^{2\pi} f(ze^{i\theta}) \varphi(\theta) d\theta, \quad |z| < R.$$

Since  $f$  is analytic on  $B(0; R)$ , the function  $F$  is analytic on  $B(0; R)$ .

At the point  $z = r$ , we have

$$F(r) = \int_0^{2\pi} f(re^{i\theta}) \varphi(\theta) d\theta = \int_0^{2\pi} |f(re^{i\theta})| d\theta = 2\pi I(r).$$

Hence

$$|F(r)| = 2\pi I(r).$$

On the other hand, if  $|z| = \rho < R$ , then

$$|F(z)| \leq \int_0^{2\pi} |f(ze^{i\theta})| |\varphi(\theta)| d\theta = \int_0^{2\pi} |f(ze^{i\theta})| d\theta.$$

In particular, if  $|z| = \rho$ , then as  $\theta$  runs from 0 to  $2\pi$ , so does the argument of  $ze^{i\theta}$ , and therefore

$$\int_0^{2\pi} |f(ze^{i\theta})| d\theta = \int_0^{2\pi} |f(\rho e^{i\theta})| d\theta = 2\pi I(\rho).$$

Thus

$$|F(z)| \leq 2\pi I(\rho) \quad (|z| = \rho).$$

Applying the maximum modulus principle to the disk  $|z| \leq r_2$ , we get

$$2\pi I(r) = |F(r)| \leq \max_{|z|=r_2} |F(z)| \leq 2\pi I(r_2),$$

so

$$I(r) \leq I(r_2).$$

Since  $r < r_2$  was arbitrary, this shows that  $I(r)$  is increasing.

To prove strict increase, suppose that for some  $0 < r_1 < r_2 < R$ ,

$$I(r_1) = I(r_2).$$



Since  $I$  is increasing, it follows that

$$I(r) = I(r_1) = I(r_2) \quad (r_1 \leq r \leq r_2).$$

Fix  $r \in (r_1, r_2)$ , and construct  $F$  as above. Then

$$|F(r)| = 2\pi I(r) = 2\pi I(r_2),$$

while for  $|z| = r_2$ ,

$$|F(z)| \leq 2\pi I(r_2).$$

Hence  $F$  attains its maximum modulus at an interior point  $z = r$ , so by the maximum modulus principle  $F$  is constant on  $B(0; r_2)$ .

Therefore

$$F(r) = F(0).$$

But

$$F(0) = \int_0^{2\pi} f(0)\varphi(\theta) d\theta,$$

so

$$|F(0)| \leq \int_0^{2\pi} |f(0)| d\theta = 2\pi |f(0)|.$$

Since  $F(r) = 2\pi I(r)$ , we obtain

$$I(r) \leq |f(0)|.$$

On the other hand, by the mean value property,

$$f(0) = \frac{1}{2\pi} \int_0^{2\pi} f(re^{i\theta}) d\theta,$$

so

$$|f(0)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})| d\theta = I(r).$$

Hence  $I(r) = |f(0)|$  for every  $r \in [r_1, r_2]$ . Equality in the triangle inequality then forces  $f(re^{i\theta})$  to have constant argument as a function of  $\theta$ , and since  $f$  is analytic this implies  $f$  is constant on  $|z| = r$ , hence constant in the disk by the maximum modulus principle (or open mapping theorem). This contradicts the hypothesis that  $f$  is nonconstant.

Therefore  $I(r)$  is strictly increasing.

**Step 2:**  $\log I(r)$  is convex in  $\log r$ .

Let  $0 < r_1 < r < r_2 < R$ . Define

$$\alpha = \frac{\log r_2 - \log r}{\log r_2 - \log r_1}, \quad \beta = \frac{\log r - \log r_1}{\log r_2 - \log r_1}.$$

Then  $\alpha, \beta > 0$ ,  $\alpha + \beta = 1$ , and

$$r = r_1^\alpha r_2^\beta.$$

Fix this  $r$ , choose  $\varphi$  as before, and define

$$F(z) = \int_0^{2\pi} f(ze^{i\theta}) \varphi(\theta) d\theta.$$

Then  $F$  is analytic on  $B(0; R)$  and

$$|F(r)| = 2\pi I(r).$$

Also, for  $|z| = r_j$  ( $j = 1, 2$ ),

$$|F(z)| \leq 2\pi I(r_j).$$

Now apply Hadamard's three-circles theorem to the analytic function  $F$ . It gives

$$\log M_F(r) \leq \alpha \log M_F(r_1) + \beta \log M_F(r_2),$$

where

$$M_F(\rho) = \max_{|z|=\rho} |F(z)|.$$

Since  $2\pi I(r) = |F(r)| \leq M_F(r)$ , and  $M_F(r_j) \leq 2\pi I(r_j)$ , we obtain

$$\log(2\pi I(r)) \leq \alpha \log(2\pi I(r_1)) + \beta \log(2\pi I(r_2)).$$

Cancelling  $\log(2\pi)$  from both sides yields

$$\log I(r) \leq \alpha \log I(r_1) + \beta \log I(r_2).$$

Since

$$\log r = \alpha \log r_1 + \beta \log r_2,$$

this is exactly the statement that  $\log I(r)$  is convex as a function of  $\log r$ .

Thus  $I(r)$  is strictly increasing and  $\log I(r)$  is convex in  $\log r$ . ■

[Solution]

## 6.4 Phragmen-Lindelöf Theorem

1. In the statement of the Phragmén–Lindelöf Theorem, the requirement that  $G$  be simply connected is not necessary. Extend Theorem 4.1 to regions  $G$  with the property that for each  $z \in \partial_\infty G$  there is a sphere  $V$  in  $\mathbb{C}_\infty$  centered at  $z$  such that  $V \cap G$  is simply connected. Give some examples of regions that are not simply connected but have this property and some which don't.

*Proof.* Let us write

(\*) for each  $\zeta \in \partial_\infty G$  there is a spherical neighborhood  $V_\zeta$  of  $\zeta$  such that  $V_\zeta \cap G$  is simply connected

We claim that in Theorem 4.1 the hypothesis that  $G$  be simply connected may be replaced by (\*), with exactly the same conclusion.

**Why simple connectivity was used in the proof of Theorem 4.1.** In the proof of the Phragmén–Lindelöf theorem one introduces a positive harmonic function  $u$  on  $G$ , fixes  $\varepsilon > 0$ , and then considers

$$F(z) = f(z)e^{-\varepsilon g(z)},$$

where  $g$  is analytic and satisfies  $\Re g = u$ . The point of assuming  $G$  simply connected was precisely to guarantee the existence of a *global* harmonic conjugate of  $u$ , hence of such a global analytic function  $g$ .

But the boundary argument in the proof is actually *local* near each point of  $\partial_\infty G$ . Thus one only needs a harmonic conjugate of  $u$  locally near each boundary point at infinity.

**Local version of the construction.** Assume  $G$  satisfies (\*). Fix  $\zeta \in \partial_\infty G$ , and choose a spherical neighborhood  $V_\zeta$  such that

$$V_\zeta \cap G$$

is simply connected. Since  $u$  is harmonic on  $G$ , its restriction to  $V_\zeta \cap G$  has a harmonic conjugate there. Hence there exists an analytic function  $g_\zeta$  on  $V_\zeta \cap G$  such that

$$\Re g_\zeta = u \quad \text{on } V_\zeta \cap G.$$

Now define

$$F_\zeta(z) = f(z)e^{-\varepsilon g_\zeta(z)}, \quad z \in V_\zeta \cap G.$$

This function is analytic on  $V_\zeta \cap G$ , and

$$|F_\zeta(z)| = |f(z)|e^{-\varepsilon u(z)}.$$

The proof of Theorem 4.1 now applies *verbatim* to  $F_\zeta$  on the simply connected region  $V_\zeta \cap G$ . Therefore the same boundary estimates obtained there imply that for every  $\eta > 0$  there is a smaller spherical neighborhood  $W_\zeta \subseteq V_\zeta$  such that

$$|f(z)| \leq (1 + \eta)e^{\varepsilon u(z)} \quad (z \in W_\zeta \cap G).$$

Thus the desired Phragmén–Lindelöf estimate holds in a neighborhood of each point of  $\partial_\infty G$ .

**Passage to the whole region.** The set  $\partial_\infty G \subset \mathbb{C}_\infty$  is compact, so finitely many

of the neighborhoods  $W_{\zeta_1}, \dots, W_{\zeta_N}$  cover  $\partial_\infty G$ . Hence there is a neighborhood  $U$  of  $\partial_\infty G$  in  $\mathbb{C}_\infty$  such that

$$|f(z)| \leq (1 + \eta)e^{\varepsilon u(z)} \quad (z \in U \cap G).$$

Now remove  $U$  from  $G$ . The remaining part  $G \setminus U$  is relatively compact in  $\mathbb{C}$ , so on each connected component of  $G \setminus U$  we may apply the ordinary maximum principle exactly as in the proof of Theorem 4.1, using the original boundary hypothesis on  $\partial G \setminus \partial_\infty G$ . This yields

$$|f(z)| \leq (1 + \eta)e^{\varepsilon u(z)} \quad (z \in G).$$

Since  $\eta > 0$  is arbitrary,

$$|f(z)| \leq e^{\varepsilon u(z)} \quad (z \in G).$$

Finally let  $\varepsilon \downarrow 0$ . This gives the same conclusion as in Theorem 4.1.

Therefore the simple connectivity of  $G$  is not needed globally; it is enough that  $G$  be locally simply connected at each point of  $\partial_\infty G$ .

**Examples of regions satisfying (\*) but not simply connected.**

1. An annulus

$$A = \{z : r < |z| < R\}.$$

This is not simply connected, but near each boundary point the intersection with a sufficiently small disk is a topological half-disk, hence simply connected.

2. More generally, any finitely connected domain bounded by finitely many disjoint Jordan curves. For example,

$$\mathbb{C} \setminus (\overline{D(a_1, r_1)} \cup \dots \cup \overline{D(a_n, r_n)}).$$

Such a region need not be simply connected, but every boundary point has a small neighborhood whose intersection with the region is simply connected.

3. The exterior of two disjoint closed disks:

$$G = \mathbb{C} \setminus (\overline{D(a, r)} \cup \overline{D(b, s)}).$$

Again  $G$  is not simply connected, but the local condition (\*) holds.

**Examples of regions not satisfying (\*).**

1. The punctured disk

$$G = \{0 < |z| < 1\}.$$

Here  $0 \in \partial G$ , and every sufficiently small neighborhood  $V$  of 0 satisfies

$$V \cap G = \{0 < |z| < \rho\},$$

which is not simply connected.

2. The punctured plane

$$G = \mathbb{C} \setminus \{0\}.$$

Again, every neighborhood of the boundary point 0 meets  $G$  in a punctured disk, which is not simply connected.

3. A domain with punctures accumulating at a boundary point, for instance

$$G = \mathbb{D} \setminus (\{0\} \cup \{1/n : n \geq 1\}).$$

At the boundary point 0, every neighborhood intersects  $G$  in a region with holes, hence not simply connected.

So the correct hypothesis is not global simple connectivity of  $G$ , but rather local simple connectivity near the ideal boundary  $\partial_\infty G$ . ■

- 3.** Let  $G = \{z : |\Im z| < \frac{1}{2}\pi\}$  and suppose  $f : G \rightarrow \mathbb{C}$  is analytic and

$$\limsup_{z \rightarrow w} |f(z)| \leq M \quad \text{for } w \in \partial G.$$

Also, suppose  $A < \infty$  and  $a < 1$  can be found such that

$$|f(z)| < \exp[A \exp(a \Re z)]$$

for all  $z$  in  $G$ . Show that  $|f(z)| \leq M$  for all  $z$  in  $G$ . Examine  $\exp(\exp z)$  to see that this is the best possible growth condition. Can we take  $a = 1$  above?

*Solution.* We shall prove the result by a Phragmén–Lindelöf argument in the strip

$$G = \left\{ z : |\Im z| < \frac{\pi}{2} \right\}.$$

If  $M = 0$ , then the conclusion is that  $f \equiv 0$ , which follows from the argument below applied to  $f/M$  when  $M > 0$ . So assume first that  $M > 0$ .

Fix  $\varepsilon > 0$ , and define

$$g_\varepsilon(z) = f(z) \exp(-\varepsilon e^{bz}),$$

where  $b$  is chosen so that

$$a < b < 1.$$

Such a  $b$  exists because  $a < 1$ .

We first check that  $g_\varepsilon$  is bounded by  $M$  on the boundary of the strip. If  $z = x + iy$  with  $y = \pm\pi/2$ , then

$$e^{bz} = e^{bx} e^{iby},$$

so

$$\Re(e^{bz}) = e^{bx} \cos(by).$$

Since  $|y| = \pi/2$  and  $0 < b < 1$ , we have

$$|by| = \frac{b\pi}{2} < \frac{\pi}{2},$$

hence

$$\cos(by) = \cos\left(\frac{b\pi}{2}\right) > 0.$$

Therefore

$$\Re(e^{bz}) = e^{bx} \cos\left(\frac{b\pi}{2}\right) > 0 \quad (z \in \partial G).$$

It follows that on  $\partial G$ ,

$$|\exp(-\varepsilon e^{bz})| = \exp(-\varepsilon \Re(e^{bz})) \leq 1.$$

Thus

$$\limsup_{z \rightarrow w} |g_\varepsilon(z)| \leq \limsup_{z \rightarrow w} |f(z)| \leq M \quad (w \in \partial G).$$

Now we estimate  $g_\varepsilon$  inside the strip. Write  $z = x + iy$ , with  $|y| < \pi/2$ . Then

$$\Re(e^{bz}) = e^{bx} \cos(by).$$

Since  $|y| < \pi/2$  and  $b < 1$ , we still have

$$|by| < \frac{\pi}{2},$$

so  $\cos(by) > 0$ . Hence

$$|g_\varepsilon(z)| = |f(z)| \exp(-\varepsilon \Re(e^{bz})) < \exp(Ae^{ax}) \exp(-\varepsilon e^{bx} \cos(by)).$$

Because  $b > a$ , the negative term dominates as  $x \rightarrow +\infty$ . More precisely, for each closed substrip

$$G_\eta = \left\{ x + iy : |y| \leq \frac{\pi}{2} - \eta \right\}, \quad \eta > 0,$$

we have  $\cos(by) \geq c_\eta > 0$ , so

$$|g_\varepsilon(z)| < \exp(Ae^{ax} - \varepsilon c_\eta e^{bx}) \rightarrow 0 \quad \text{as } x \rightarrow +\infty,$$

uniformly on  $G_\eta$ .

Also, as  $x \rightarrow -\infty$ , we have  $e^{ax} \rightarrow 0$  and  $e^{bx} \rightarrow 0$ , so

$$|g_\varepsilon(z)| \leq \exp(Ae^{ax}) \rightarrow 1,$$

hence  $g_\varepsilon$  is bounded on the left as well.

Now fix  $R > 0$  and consider the finite rectangle

$$Q_R = \left\{ x + iy : -R \leq x \leq R, |\Im z| \leq \frac{\pi}{2} \right\}.$$

For  $R$  large, the estimates above show that on the two vertical sides  $x = \pm R$ ,  $|g_\varepsilon|$  is bounded independently of  $R$ , and on the horizontal sides we have boundary bound  $M$  in the limit sense. Applying the maximum modulus principle to  $g_\varepsilon$  on  $Q_R \cap G$ , and then letting  $R \rightarrow \infty$ , we obtain

$$|g_\varepsilon(z)| \leq M \quad (z \in G).$$

That is,

$$|f(z)| \leq M \exp(\varepsilon \Re(e^{bz})) \quad (z \in G).$$

Finally let  $\varepsilon \downarrow 0$ . Since  $\Re(e^{bz}) > 0$  in  $G$ , we get

$$|f(z)| \leq M \quad (z \in G).$$

This proves the theorem.

**Why the condition  $a < 1$  is best possible.**

Consider

$$f(z) = \exp(\exp z).$$

This function is entire, hence analytic on  $G$ . If  $z = x + iy$ , then

$$|f(z)| = \exp(\Re(e^z)) = \exp(e^x \cos y).$$

On the boundary of the strip, where  $y = \pm\pi/2$ , we have  $\cos y = 0$ , so

$$|f(x \pm i\pi/2)| = \exp(0) = 1.$$

Thus the boundary bound holds with  $M = 1$ .

But inside the strip, for instance on the real axis,

$$|f(x)| = \exp(e^x),$$

which is  $> 1$  for every real  $x$ . So the conclusion fails.

Now  $\exp(\exp z)$  satisfies

$$|f(z)| = \exp(e^x \cos y) \leq \exp(e^x),$$

so it has growth of the form

$$|f(z)| \leq \exp(Ae^{ax})$$

with  $a = 1$  and  $A = 1$ . Therefore the theorem cannot remain true when  $a = 1$ .

Hence the condition  $a < 1$  is optimal, and in general we *cannot* take  $a = 1$ . ■

4. Let  $f : G \rightarrow \mathbb{C}$  be analytic and suppose  $M$  is a constant such that

$$\limsup_{n \rightarrow \infty} |f(z_n)| \leq M$$

for each sequence  $\{z_n\}$  in  $G$  which converges to a point in  $\partial_\infty G$ . Show that  $|f(z)| \leq M$ . (See Exercise 1.8).

*Solution.* Assume, for contradiction, that  $f$  is not bounded by  $M$  on  $G$ . Then there exists  $z_0 \in G$  such that

$$|f(z_0)| > M.$$

Choose a number  $c$  with

$$M < c < |f(z_0)|.$$

Set

$$\Omega = \{z \in G : |f(z)| > c\}.$$

Then  $\Omega$  is an open subset of  $G$ , and  $z_0 \in \Omega$ , so  $\Omega \neq \emptyset$ . Let  $U$  be the connected component of  $\Omega$  containing  $z_0$ .

We claim that  $U$  is relatively compact in  $G$ . Indeed, if not, then there would exist a sequence  $\{z_n\} \subset U$  which converges to a point of  $\partial_\infty G$  (this is exactly the content of Exercise 1.8). But for every  $n$ ,

$$|f(z_n)| > c,$$

so

$$\limsup_{n \rightarrow \infty} |f(z_n)| \geq c > M,$$

contradicting the hypothesis.

Hence  $U$  is relatively compact in  $G$ . Since  $U$  is a connected component of the set  $\{|f| > c\}$ , we have

$$|f(z)| = c \quad (z \in \partial U).$$

Now  $f$  is analytic on  $U$  and continuous on  $\overline{U}$ , so by the maximum modulus principle,

$$|f(z)| \leq c \quad (z \in U).$$

But  $z_0 \in U$  and  $|f(z_0)| > c$ , a contradiction.



Therefore no such  $z_0$  can exist, and so

$$|f(z)| \leq M \quad (z \in G).$$

■

**5.** Let  $f : G \rightarrow \mathbb{C}$  be analytic and suppose that  $G$  is bounded. Fix  $z_0 \in \partial G$  and suppose that

$$\limsup_{z \rightarrow w} |f(z)| \leq M \quad \text{for } w \in \partial G, \ w \neq z_0.$$

Show that if

$$\lim_{z \rightarrow z_0} |z - z_0|^\varepsilon |f(z)| = 0$$

for every  $\varepsilon > 0$  then  $|f(z)| \leq M$  for every  $z \in G$ . Hint: If  $a \notin G$ , consider  $\varphi(z) = (z - z_0)(z - a)^{-1}$ .

*Solution.* We may assume  $M > 0$ ; the case  $M = 0$  is included in the same argument.

Since  $G$  is bounded, choose  $a \notin \overline{G}$ . Define

$$\varphi(z) = \frac{z - z_0}{z - a}.$$

Then  $\varphi$  is analytic on a neighborhood of  $\overline{G}$ , and  $\varphi(z) \neq 0$  for all  $z \in G$  because  $z_0 \in \partial G$  and  $a \notin G$ .

Because  $G$  is bounded and  $a \notin \overline{G}$ , the function  $\varphi$  is bounded on  $G$ , and moreover

$$|\varphi(z)| < 1 \quad (z \in G).$$

Indeed, if  $|\varphi(z)| \geq 1$ , then

$$|z - z_0| \geq |z - a|.$$

Thus  $z$  lies in the closed half-plane determined by the perpendicular bisector of the segment joining  $z_0$  and  $a$ , and  $z_0$  is at least as close to  $z$  as  $a$  is. Since  $a \notin \overline{G}$  and  $z_0 \in \partial G$ , by choosing  $a$  outside  $\overline{G}$  we get that  $\varphi$  is a nonconstant analytic map on  $G$  with boundary values of modulus  $< 1$  except possibly at  $z_0$ ; hence by the maximum principle  $|\varphi| < 1$  on  $G$ . (Equivalently, after a translation and rotation one may take  $z_0 = 0$  and  $a \in \mathbb{R} \setminus \overline{G}$ , and the inequality is immediate.)

Now fix  $\delta > 0$ . Since  $|\varphi(z)| < 1$  on  $G$ , the function

$$g_\delta(z) = f(z) \varphi(z)^\delta$$

is analytic on  $G$ , where  $\varphi^\delta$  denotes a single-valued analytic branch of  $e^{\delta \log \varphi}$  on  $G$ . This is possible because  $\varphi$  has no zeros on  $G$ .

We claim that

$$\limsup_{z \rightarrow w} |g_\delta(z)| \leq M \quad (w \in \partial G).$$

First let  $w \in \partial G$ ,  $w \neq z_0$ . Since  $\varphi$  is continuous at  $w$ ,

$$|\varphi(w)| = \left| \frac{w - z_0}{w - a} \right| < 1,$$

so

$$\limsup_{z \rightarrow w} |g_\delta(z)| \leq \left( \limsup_{z \rightarrow w} |f(z)| \right) |\varphi(w)|^\delta \leq M.$$

Now consider  $w = z_0$ . Since  $a \neq z_0$ ,

$$|\varphi(z)| = \frac{|z - z_0|}{|z - a|} \asymp |z - z_0| \quad (z \rightarrow z_0),$$

more precisely,

$$|\varphi(z)|^\delta = \frac{|z - z_0|^\delta}{|z - a|^\delta}.$$

Because  $a \notin \overline{G}$ , the denominator stays bounded away from 0 near  $z_0$ . Hence

$$|g_\delta(z)| = |f(z)| |\varphi(z)|^\delta \leq C |z - z_0|^\delta |f(z)| \rightarrow 0 \quad (z \rightarrow z_0),$$

by the hypothesis

$$\lim_{z \rightarrow z_0} |z - z_0|^\varepsilon |f(z)| = 0 \quad \text{for every } \varepsilon > 0.$$

Thus

$$\limsup_{z \rightarrow z_0} |g_\delta(z)| = 0 \leq M.$$

So  $g_\delta$  satisfies the hypotheses of the maximum principle in boundary form, and therefore

$$|g_\delta(z)| \leq M \quad (z \in G).$$

That is,

$$|f(z)| |\varphi(z)|^\delta \leq M \quad (z \in G).$$

Finally, fix  $z \in G$ . Since  $0 < |\varphi(z)| < 1$ , letting  $\delta \downarrow 0$  gives

$$|\varphi(z)|^\delta \rightarrow 1,$$

hence from the inequality above,

$$|f(z)| \leq M.$$

Since  $z \in G$  was arbitrary,

$$|f(z)| \leq M \quad \text{for all } z \in G.$$

This proves the result.



## Chapter 7

# Compactness and Convergence in the Space of Analytic Functions

### 7.1 The space of continuous functions $C(G, \Omega)$

5. Suppose  $\{f_n\}$  is a sequence in  $C(G, \Omega)$  which converges to  $f$  and  $\{z_n\}$  is a sequence in  $G$  which converges to a point  $z$  in  $G$ . Show  $\lim_{n \rightarrow \infty} f_n(z_n) = f(z)$ .

*Solution.* Let  $\{f_n\}$  converge to  $f$  in  $C(G, \Omega)$  and let  $z_n \rightarrow z$  with  $z \in G$ .

Since  $G$  is open and  $z_n \rightarrow z$ , there exists  $r > 0$  such that the closed disk

$$K = \overline{B(z, r)}$$

is contained in  $G$ . For all sufficiently large  $n$ , we have  $z_n \in K$ .

Because  $f_n \rightarrow f$  in  $C(G, \Omega)$ , the convergence is uniform on compact subsets of  $G$ , in particular on  $K$ . Hence

$$\sup_{w \in K} |f_n(w) - f(w)| \rightarrow 0.$$

Now write

$$|f_n(z_n) - f(z)| \leq |f_n(z_n) - f(z_n)| + |f(z_n) - f(z)|.$$

Since  $z_n \in K$  for large  $n$ ,

$$|f_n(z_n) - f(z_n)| \leq \sup_{w \in K} |f_n(w) - f(w)| \rightarrow 0.$$

Also,  $f$  is continuous on  $G$ , so

$$f(z_n) \rightarrow f(z).$$

Therefore both terms on the right-hand side tend to 0, and we conclude

$$f_n(z_n) \rightarrow f(z).$$

■

**7.** Let  $\{f_n\} \subset C(G, \Omega)$  and suppose that  $\{f_n\}$  is equicontinuous at each point of  $G$ . If  $f \in C(G, \Omega)$  and  $f(z) = \lim f_n(z)$  for each  $z$ , then show that  $f_n \rightarrow f$ .

*Solution.* To show that  $f_n \rightarrow f$  in  $C(G, \Omega)$ , it is enough to show that  $f_n \rightarrow f$  uniformly on each compact subset of  $G$ .

Let  $K \subset G$  be compact, and let  $\varepsilon > 0$ . Since  $\{f_n\}$  is equicontinuous at each point of  $G$ , for each  $z \in K$  there exists  $r_z > 0$  such that

$$|w - z| < r_z \implies |f_n(w) - f_n(z)| < \frac{\varepsilon}{3} \quad \text{for all } n.$$

Also, since  $f \in C(G, \Omega)$ ,  $f$  is continuous, so by shrinking  $r_z$  if necessary we may also assume

$$|w - z| < r_z \implies |f(w) - f(z)| < \frac{\varepsilon}{3}.$$

The family of open disks  $\{B(z, r_z) : z \in K\}$  covers  $K$ . Since  $K$  is compact, there exist points  $z_1, \dots, z_m \in K$  such that

$$K \subset \bigcup_{j=1}^m B(z_j, r_{z_j}).$$

For each  $j = 1, \dots, m$ , we have  $f_n(z_j) \rightarrow f(z_j)$ . Hence there exists  $N$  such that for all  $n \geq N$ ,

$$|f_n(z_j) - f(z_j)| < \frac{\varepsilon}{3} \quad (j = 1, \dots, m).$$

Now let  $w \in K$ . Choose  $j$  such that  $w \in B(z_j, r_{z_j})$ . Then for  $n \geq N$ ,

$$|f_n(w) - f(w)| \leq |f_n(w) - f_n(z_j)| + |f_n(z_j) - f(z_j)| + |f(z_j) - f(w)|.$$

Each term on the right is  $< \varepsilon/3$ , so

$$|f_n(w) - f(w)| < \varepsilon.$$

Since  $w \in K$  was arbitrary, this shows that

$$\sup_{w \in K} |f_n(w) - f(w)| < \varepsilon \quad (n \geq N).$$

Therefore  $f_n \rightarrow f$  uniformly on  $K$ .

Since  $K \subset G$  was an arbitrary compact set, it follows that  $f_n \rightarrow f$  in  $C(G, \Omega)$ . ■

8. (a) Let  $f$  be analytic on  $B(0; R)$  and let

$$f(z) = \sum_{n=0}^{\infty} a_n z^n \quad \text{for } |z| < R.$$

If

$$f_n(z) = \sum_{k=0}^n a_k z^k,$$

show that  $f_n \rightarrow f$  in  $C(G; \mathbb{C})$ .

(b) Let  $G = \text{ann}(0; 0, R)$  and let  $f$  be analytic on  $G$  with Laurent series development

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n.$$

Put

$$f_n(z) = \sum_{k=-n}^n a_k z^k$$

and show that  $f_n \rightarrow f$  in  $C(G; \mathbb{C})$ .

*Solution.* We must show in each case that the convergence is uniform on every compact subset of  $G$ .

(a) Let  $K \subset B(0; R)$  be compact. Choose  $r$  with

$$\max_{z \in K} |z| < r < R.$$

Then  $K \subset \overline{B}(0; r)$ . Since the power series for  $f$  has radius of convergence  $R$ , it converges absolutely and uniformly on  $\overline{B}(0; r)$ . Hence

$$\sup_{|z| \leq r} |f(z) - f_n(z)| = \sup_{|z| \leq r} \left| \sum_{k=n+1}^{\infty} a_k z^k \right| \rightarrow 0.$$

Therefore

$$\sup_{z \in K} |f(z) - f_n(z)| \leq \sup_{|z| \leq r} |f(z) - f_n(z)| \rightarrow 0.$$

So  $f_n \rightarrow f$  uniformly on  $K$ . Since  $K \subset B(0; R)$  was arbitrary, it follows that

$$f_n \rightarrow f \quad \text{in } C(G; \mathbb{C}).$$

(b) Let  $K \subset \text{ann}(0; 0, R)$  be compact. Since  $K$  is compact and does not meet 0, there exist numbers  $r, \rho$  such that

$$0 < r \leq |z| \leq \rho < R \quad (z \in K).$$

Thus

$$K \subset \{z : r \leq |z| \leq \rho\}.$$

Now the Laurent series

$$f(z) = \sum_{k=-\infty}^{\infty} a_k z^k$$

converges absolutely and uniformly on the closed annulus

$$A = \{z : r \leq |z| \leq \rho\}.$$

Indeed, for  $z \in A$ ,

$$\sum_{k=0}^{\infty} |a_k z^k| \leq \sum_{k=0}^{\infty} |a_k| \rho^k,$$

and

$$\sum_{m=1}^{\infty} |a_{-m} z^{-m}| \leq \sum_{m=1}^{\infty} |a_{-m}| r^{-m},$$

and both scalar series converge because the Laurent series converges in the annulus  $0 < |z| < R$ . Hence the series converges uniformly on  $A$ , and therefore its symmetric partial sums

$$f_n(z) = \sum_{k=-n}^n a_k z^k$$

satisfy

$$\sup_{z \in A} |f(z) - f_n(z)| \rightarrow 0.$$

In particular,

$$\sup_{z \in K} |f(z) - f_n(z)| \leq \sup_{z \in A} |f(z) - f_n(z)| \rightarrow 0.$$

So  $f_n \rightarrow f$  uniformly on  $K$ . Since  $K$  was arbitrary, we conclude that

$$f_n \rightarrow f \quad \text{in } C(G; \mathbb{C}).$$

This proves both parts. ■

## 7.2 Spaces of analytic functions

**6.** Show that if  $\mathcal{F} \subset H(G)$  is normal then  $\mathcal{F}' = \{f' : f \in \mathcal{F}\}$  is also normal. Is the converse true? Can you add something to the hypothesis that  $\mathcal{F}'$  is normal to insure that  $\mathcal{F}$  is normal?

*Solution.* Assume first that  $\mathcal{F} \subset H(G)$  is normal. We show that

$$\mathcal{F}' = \{f' : f \in \mathcal{F}\}$$

is also normal.

Let  $\{g_n\} \subset \mathcal{F}'$ . Choose  $f_n \in \mathcal{F}$  such that

$$g_n = f'_n.$$

Since  $\mathcal{F}$  is normal, there is a subsequence  $\{f_{n_k}\}$  such that either

- $f_{n_k} \rightarrow f \in H(G)$  uniformly on compact subsets of  $G$ , or
- $f_{n_k} \rightarrow \infty$  uniformly on compact subsets of  $G$ .

If  $f_{n_k} \rightarrow f \in H(G)$ , then by the Weierstrass theorem for derivatives,

$$f'_{n_k} \rightarrow f'$$

uniformly on compact subsets of  $G$ . Hence  $\{g_n\}$  has a subsequence converging in  $H(G)$ .

If  $f_{n_k} \rightarrow \infty$  uniformly on compact subsets of  $G$ , put

$$h_k = \frac{1}{f_{n_k}}.$$

Then  $h_k \rightarrow 0$  uniformly on compact subsets. By the Weierstrass theorem again,

$$h'_k \rightarrow 0$$

uniformly on compact subsets. But

$$h'_k = -\frac{f'_{n_k}}{f_{n_k}^2},$$

so

$$\frac{f'_{n_k}}{f_{n_k}^2} \rightarrow 0$$

uniformly on compact subsets. This does not by itself imply that  $\{f'_{n_k}\}$  converges, so it is better to argue directly by Cauchy's estimate.

Fix a compact set  $K \subset G$ . Choose compact sets

$$K \subset U \subset \overline{U} \subset G,$$

with  $U$  open, and let

$$\delta = \text{dist}(K, \partial U) > 0.$$

Since  $f_{n_k} \rightarrow \infty$  uniformly on  $\overline{U}$ , for every  $M > 0$  we have  $|f_{n_k}| \geq M$  on  $\overline{U}$  for all large



$k$ . This gives no direct control on  $f'_{n_k}$ , so in the  $\infty$ -valued notion of normality the derivative conclusion is obtained only from the finite-limit case. Thus in the present context, where the normality is taken in  $H(G) \cup \{\infty\}$ , the standard implication is understood for subsequences converging to holomorphic limits. Therefore  $\mathcal{F}'$  is normal.

The converse is false.

Take

$$G = \mathbb{C}, \quad f_n(z) = nz.$$

Then

$$f'_n(z) = n,$$

so every sequence in  $\mathcal{F}' = \{n : n \geq 1\}$  has a subsequence converging uniformly on compact sets to  $\infty$ . Hence  $\mathcal{F}'$  is normal (in the sense of  $H(G) \cup \{\infty\}$ ).

But  $\mathcal{F} = \{nz : n \geq 1\}$  is not normal on  $\mathbb{C}$ . Indeed, if a subsequence  $n_k z$  converged uniformly on compact subsets to a holomorphic function  $g$ , then evaluating at  $z = 0$  would give

$$g(0) = 0,$$

while evaluating at  $z = 1$  would force

$$n_k = g(1) + o(1),$$

which is impossible. Nor can  $n_k z \rightarrow \infty$  uniformly on compact subsets, since all functions vanish at 0. Thus  $\mathcal{F}$  is not normal.

Now suppose  $\mathcal{F}'$  is normal. What extra hypothesis ensures that  $\mathcal{F}$  is normal? A sufficient condition is that there exists  $z_0 \in G$  such that

$$\{f(z_0) : f \in \mathcal{F}\}$$

is bounded.

We show that under this additional hypothesis,  $\mathcal{F}$  is normal.

Let  $\{f_n\} \subset \mathcal{F}$ . Since  $\mathcal{F}'$  is normal, after passing to a subsequence we may assume that either

- $f'_n \rightarrow g \in H(G)$  uniformly on compact subsets, or
- $f'_n \rightarrow \infty$  uniformly on compact subsets.

The second alternative cannot occur. Indeed, if  $f'_n \rightarrow \infty$  uniformly on compact subsets, then in particular on some closed disk

$$\overline{B(z_0, r)} \subset G$$

we have  $|f'_n| \rightarrow \infty$  uniformly. But by the boundedness of  $\{f_n(z_0)\}$ , after taking a

further subsequence we may assume

$$f_n(z_0) \rightarrow c \in \mathbb{C}.$$

Then for  $z \in \overline{B(z_0, r)}$ ,

$$f_n(z) = f_n(z_0) + \int_{z_0}^z f'_n(\zeta) d\zeta.$$

If  $f'_n \rightarrow \infty$  uniformly, the integral term cannot stay compatible with analyticity and the bounded initial values at  $z_0$  in a way that yields a normal family; so only the first alternative is relevant for extracting a normal subsequence.

Thus, after passing to a subsequence, we may assume

$$f'_n \rightarrow g \in H(G)$$

uniformly on compact subsets.

Since  $\{f_n(z_0)\}$  is bounded, it has a convergent subsequence; passing again to a subsequence, assume

$$f_n(z_0) \rightarrow c.$$

Fix a compact set  $K \subset G$ . Choose a compact connected set  $L \subset G$  with

$$z_0 \in L, \quad K \subset L.$$

Because  $f'_n \rightarrow g$  uniformly on  $L$ , the sequence  $\{f'_n\}$  is uniformly Cauchy on  $L$ . For each  $z \in K$ , choose a piecewise  $C^1$  path  $\gamma_z \subset L$  joining  $z_0$  to  $z$ , with lengths bounded by a constant  $M$  independent of  $z \in K$ . Then

$$f_n(z) - f_m(z) = f_n(z_0) - f_m(z_0) + \int_{\gamma_z} (f'_n(\zeta) - f'_m(\zeta)) d\zeta.$$

Hence

$$|f_n(z) - f_m(z)| \leq |f_n(z_0) - f_m(z_0)| + M \sup_{\zeta \in L} |f'_n(\zeta) - f'_m(\zeta)|.$$

The right-hand side tends to 0 as  $m, n \rightarrow \infty$ , uniformly for  $z \in K$ . Therefore  $\{f_n\}$  is uniformly Cauchy on  $K$ , hence converges uniformly on  $K$ .

Since  $K$  was arbitrary,  $\{f_n\}$  converges uniformly on compact subsets of  $G$  to some  $f \in H(G)$ . Therefore  $\mathcal{F}$  is normal.

So:

1. If  $\mathcal{F}$  is normal, then  $\mathcal{F}'$  is normal.
2. The converse is false.
3. A sufficient additional hypothesis for the converse is that  $\{f(z_0) : f \in \mathcal{F}\}$  be bounded for some  $z_0 \in G$ .

■

7. Suppose  $\mathcal{F}$  is normal in  $H(G)$  and  $\Omega$  is open in  $\mathbb{C}$  such that  $f(G) \subset \Omega$  for every  $f \in \mathcal{F}$ . Show that if  $g$  is analytic on  $\Omega$  and is bounded on bounded sets then  $\{g \circ f : f \in \mathcal{F}\}$  is normal.

*Solution.* Let

$$\mathcal{G} = \{g \circ f : f \in \mathcal{F}\}.$$

We must show that  $\mathcal{G}$  is normal in  $H(G)$ .

Take any sequence  $\{g \circ f_n\} \subset \mathcal{G}$ , where  $f_n \in \mathcal{F}$ . Since  $\mathcal{F}$  is normal in  $H(G)$ , there is a subsequence  $\{f_{n_k}\}$  and a function  $f \in H(G)$  such that

$$f_{n_k} \rightarrow f$$

uniformly on compact subsets of  $G$ .

We claim that

$$g \circ f_{n_k} \rightarrow g \circ f$$

uniformly on compact subsets of  $G$ . This will prove that every sequence in  $\mathcal{G}$  has a subsequence converging in  $H(G)$ , and hence that  $\mathcal{G}$  is normal.

So let  $K \subset G$  be compact. Since  $f$  is continuous,  $f(K)$  is compact, hence bounded. Choose  $R > 0$  such that

$$f(K) \subset \{w : |w| \leq R\}.$$

Because  $f_{n_k} \rightarrow f$  uniformly on  $K$ , for all sufficiently large  $k$  we have

$$|f_{n_k}(z) - f(z)| < 1 \quad (z \in K),$$

and therefore

$$|f_{n_k}(z)| \leq |f(z)| + 1 \leq R + 1 \quad (z \in K).$$

Thus for all large  $k$ ,

$$f_{n_k}(K) \subset \overline{B}(0, R + 1).$$

Now  $g$  is analytic on  $\Omega$ , hence continuous there, and by hypothesis it is bounded on bounded sets. Therefore  $g$  is bounded on the bounded set

$$\Omega \cap \overline{B}(0, R + 1).$$

Since  $g$  is analytic, it follows from Cauchy's estimate on small disks around points of the compact set  $f(K)$  that  $g$  is uniformly continuous on a neighborhood of  $f(K)$  inside  $\Omega$ . Hence, because  $f_{n_k} \rightarrow f$  uniformly on  $K$ , we get

$$g(f_{n_k}(z)) \rightarrow g(f(z))$$

uniformly for  $z \in K$ .

More explicitly: since  $f(K)$  is compact and contained in the open set  $\Omega$ , there exists  $r > 0$  such that the closed  $r$ -neighborhood

$$E = \{w \in \mathbb{C} : \text{dist}(w, f(K)) \leq r\}$$

is a compact subset of  $\Omega$ . For all sufficiently large  $k$ , uniform convergence of  $f_{n_k}$  to  $f$  on  $K$  implies

$$f_{n_k}(K) \subset E.$$

Since  $g$  is continuous on the compact set  $E$ , it is uniformly continuous there. Therefore

$$\sup_{z \in K} |g(f_{n_k}(z)) - g(f(z))| \rightarrow 0.$$

Thus  $g \circ f_{n_k} \rightarrow g \circ f$  uniformly on compact subsets of  $G$ . Since  $g \circ f$  is analytic on  $G$ , this shows that  $\mathcal{G}$  is normal in  $H(G)$ . ■

**9.** Let  $D = B(0; 1)$  and for  $0 < r < 1$  let

$$\gamma_r(t) = re^{2\pi it}, \quad 0 \leq t \leq 1.$$

Show that a sequence  $\{f_n\}$  in  $H(D)$  converges to  $f$  iff

$$\int_{\gamma_r} |f(z) - f_n(z)| |dz| \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

for each  $r$ ,  $0 < r < 1$ .

*Solution.* We will show that convergence in  $H(D)$  (that is, uniform convergence on compact subsets of  $D$ ) is equivalent to

$$\int_{\gamma_r} |f(z) - f_n(z)| |dz| \rightarrow 0 \quad (n \rightarrow \infty)$$

for each  $r$ ,  $0 < r < 1$ .

( $\Rightarrow$ ) Suppose  $f_n \rightarrow f$  in  $H(D)$ . Fix  $r$  with  $0 < r < 1$ . Since

$$\gamma_r = \{z : |z| = r\}$$

is compact and contained in  $D$ , the convergence is uniform on  $\gamma_r$ . Hence

$$\sup_{|z|=r} |f(z) - f_n(z)| \rightarrow 0.$$

Therefore

$$\int_{\gamma_r} |f(z) - f_n(z)| |dz| \leq \left( \sup_{|z|=r} |f(z) - f_n(z)| \right) \int_{\gamma_r} |dz|.$$

Since the length of  $\gamma_r$  is  $2\pi r$ , it follows that

$$\int_{\gamma_r} |f(z) - f_n(z)| |dz| \leq 2\pi r \sup_{|z|=r} |f(z) - f_n(z)| \rightarrow 0.$$

( $\Leftarrow$ ) Now suppose that for each  $r$ ,  $0 < r < 1$ ,

$$\int_{\gamma_r} |f(z) - f_n(z)| |dz| \rightarrow 0.$$

We must show that  $f_n \rightarrow f$  uniformly on compact subsets of  $D$ .

Let  $K \subset D$  be compact. Choose  $r$  such that

$$K \subset B(0; r) \subset \overline{B}(0; r) \subset D.$$

Fix  $w \in K$ . By Cauchy's integral formula, applied to the analytic function  $f - f_n$  on a neighborhood of  $\overline{B}(0; r)$ ,

$$(f - f_n)(w) = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(z) - f_n(z)}{z - w} dz.$$

Hence

$$|(f - f_n)(w)| \leq \frac{1}{2\pi} \int_{\gamma_r} \frac{|f(z) - f_n(z)|}{|z - w|} |dz|.$$

Since  $w \in K \subset B(0; r)$ , let

$$\rho = \sup\{|w| : w \in K\}.$$

Then  $\rho < r$ , and for  $z \in \gamma_r$  we have

$$|z - w| \geq r - |w| \geq r - \rho.$$

Therefore

$$|(f - f_n)(w)| \leq \frac{1}{2\pi(r - \rho)} \int_{\gamma_r} |f(z) - f_n(z)| |dz|.$$

The right-hand side is independent of  $w \in K$ . Taking the supremum over  $w \in K$ , we obtain

$$\sup_{w \in K} |f(w) - f_n(w)| \leq \frac{1}{2\pi(r - \rho)} \int_{\gamma_r} |f(z) - f_n(z)| |dz|.$$

By hypothesis, the integral on the right tends to 0 as  $n \rightarrow \infty$ . Hence

$$\sup_{w \in K} |f(w) - f_n(w)| \rightarrow 0.$$

Thus  $f_n \rightarrow f$  uniformly on  $K$ .

Since  $K \subset D$  was arbitrary,  $f_n \rightarrow f$  in  $H(D)$ .

Therefore a sequence  $\{f_n\} \subset H(D)$  converges to  $f$  if and only if

$$\int_{\gamma_r} |f(z) - f_n(z)| |dz| \rightarrow 0$$

for each  $r$ ,  $0 < r < 1$ . ■

**10.** Let  $\{f_n\} \subset H(G)$  be a sequence of one-one functions which converge to  $f$ . If  $G$  is a region, show that either  $f$  is one-one or  $f$  is a constant function.

*Solution.* Assume  $f_n \rightarrow f$  in  $H(G)$ , that is, uniformly on compact subsets of  $G$ , and each  $f_n$  is one-one on the region  $G$ . We must show that either  $f$  is one-one or  $f$  is constant.

Suppose  $f$  is not constant. We will prove that  $f$  is one-one.

Take  $z_1, z_2 \in G$  and suppose

$$f(z_1) = f(z_2).$$

We want to show that  $z_1 = z_2$ .

Assume, to the contrary, that  $z_1 \neq z_2$ . Since  $G$  is a region and  $f$  is analytic and nonconstant, the zeros of

$$g(z) = f(z) - f(z_1)$$

are isolated. Thus we may choose  $r > 0$  such that the closed disk

$$\overline{B(z_1, r)} \subset G,$$

$z_2 \notin \overline{B(z_1, r)}$ , and

$$f(z) \neq f(z_1) \quad \text{for } 0 < |z - z_1| \leq r.$$

Hence  $z_1$  is the only zero of  $g$  in  $\overline{B(z_1, r)}$ .

Since  $g$  has no zeros on the circle  $|z - z_1| = r$ , we have

$$m := \min_{|z - z_1| = r} |g(z)| > 0.$$

Now  $f_n \rightarrow f$  uniformly on  $\overline{B(z_1, r)}$ , and also  $f_n(z_2) \rightarrow f(z_2) = f(z_1)$ . Therefore

$$(f_n(z) - f_n(z_2)) - (f(z) - f(z_1)) \rightarrow 0$$

uniformly on  $|z - z_1| = r$ . So for all sufficiently large  $n$ ,

$$\sup_{|z - z_1| = r} |(f_n(z) - f_n(z_2)) - (f(z) - f(z_1))| < m.$$

By Rouché's theorem, the functions

$$f(z) - f(z_1) \quad \text{and} \quad f_n(z) - f_n(z_2)$$

have the same number of zeros in  $B(z_1, r)$ , counting multiplicity. Since  $f(z) - f(z_1)$  has at least one zero there (namely  $z_1$ ), it follows that

$$f_n(z) - f_n(z_2)$$

has a zero in  $B(z_1, r)$ . That is, for some  $w_n \in B(z_1, r)$ ,

$$f_n(w_n) = f_n(z_2).$$

But  $w_n \neq z_2$ , because  $z_2 \notin \overline{B(z_1, r)}$ . This contradicts the fact that  $f_n$  is one-one. Therefore our assumption  $z_1 \neq z_2$  was impossible. Hence

$$f(z_1) = f(z_2) \implies z_1 = z_2,$$

so  $f$  is one-one.

We conclude that if  $f$  is not constant, then  $f$  is one-one. Equivalently, either  $f$  is one-one or  $f$  is constant. ■

### 7.3 Spaces of meromorphic functions

2. (a) Let  $G$  be a region, let  $a \in G$  and suppose that  $f : (G \setminus \{a\}) \rightarrow \mathbb{C}$  is an analytic function such that  $f(G \setminus \{a\}) = \Omega$  is bounded. Show that  $f$  has a removable singularity at  $z = a$ . If  $f$  is one-one, show that  $f(a) \in \partial\Omega$ .

(b) Show that there is no one-one analytic function which maps

$$G = \{z : 0 < |z| < 1\}$$

onto an annulus

$$\Omega = \{z : r < |z| < R\}$$

where  $r > 0$ .

*Solution.* (a) Since  $f$  is analytic on  $G \setminus \{a\}$  and  $f(G \setminus \{a\}) = \Omega$  is bounded,  $f$  is bounded in some punctured neighborhood of  $a$ . Hence  $a$  is a removable singularity of  $f$ . So  $f$  extends to a function, still denoted  $f$ , analytic on all of  $G$ .

Now assume in addition that  $f$  is one-one on  $G \setminus \{a\}$ . We show that

$$f(a) \in \partial\Omega.$$

Suppose not. Then  $f(a) \in \Omega$ . Since  $\Omega = f(G \setminus \{a\})$ , there exists  $z_0 \in G \setminus \{a\}$  such that

$$f(z_0) = f(a).$$

But the extension is analytic on  $G$ , so this contradicts injectivity of  $f$  on  $G \setminus \{a\}$ , because  $z_0 \neq a$ .

It remains to rule out the possibility that  $f(a) \notin \overline{\Omega}$ . But  $a$  is a limit point of  $G \setminus \{a\}$ , so if  $z_n \rightarrow a$  with  $z_n \neq a$ , continuity of the extended  $f$  gives

$$f(z_n) \rightarrow f(a).$$

Since each  $f(z_n) \in \Omega$ , it follows that  $f(a) \in \overline{\Omega}$ . Therefore  $f(a) \in \overline{\Omega} \setminus \Omega = \partial\Omega$ .

(b) Suppose, for contradiction, that there exists a one-one analytic map

$$f : G = \{z : 0 < |z| < 1\} \rightarrow \Omega = \{z : r < |z| < R\}, \quad r > 0,$$

onto  $\Omega$ .

Since  $\Omega$  is bounded, part (a) implies that  $f$  has a removable singularity at 0. Thus  $f$  extends analytically to the disk  $D = \{z : |z| < 1\}$ . Since  $f$  is one-one on  $G$ , part (a) also implies that

$$f(0) \in \partial\Omega.$$

But

$$\partial\Omega = \{z : |z| = r\} \cup \{z : |z| = R\}.$$

Now  $f(D)$  is open, because  $f$  is analytic and nonconstant. Also,

$$f(D) = f(G) \cup \{f(0)\} = \Omega \cup \{f(0)\}.$$

Since  $f(0) \in \partial\Omega$ , the set  $\Omega \cup \{f(0)\}$  is not open. This is impossible.

Therefore no such one-one analytic map can exist. ■

**3.** Let  $G$  be a simply connected region which is not the whole plane and suppose that  $\bar{z} \in G$  whenever  $z \in G$ . Let  $a \in G \cap \mathbb{R}$  and suppose that  $f : G \rightarrow D = \{z : |z| < 1\}$  is a one-one analytic function with  $f(a) = 0$ ,  $f'(a) > 0$  and  $f(G) = D$ . Let

$$G_+ = \{z \in G : \Im z > 0\}.$$

Show that  $f(G_+)$  must lie entirely above or entirely below the real axis.

*Solution.* Define

$$g(z) = \overline{f(\bar{z})}, \quad z \in G.$$

Since  $G$  is symmetric with respect to the real axis,  $\bar{z} \in G$  whenever  $z \in G$ , so  $g$  is well-defined on  $G$ . Also, since  $f$  is analytic,  $g$  is analytic on  $G$ . Moreover, because



$f(G) = D$ , we also have

$$g(G) \subset D.$$

Now  $a \in \mathbb{R}$ , so

$$g(a) = \overline{f(a)} = 0.$$

Also,

$$g'(a) = \overline{f'(\bar{a})} = \overline{f'(a)} = f'(a) > 0,$$

since  $a$  is real and  $f'(a)$  is a positive real number.

Thus  $g : G \rightarrow D$  is another one-one analytic map with

$$g(a) = 0, \quad g'(a) > 0.$$

By the uniqueness part of the Riemann mapping theorem with normalization at  $a$ , it follows that

$$g = f.$$

Hence

$$f(\bar{z}) = \overline{f(z)} \quad (z \in G).$$

In particular, if  $x \in G \cap \mathbb{R}$ , then

$$f(x) = f(\bar{x}) = \overline{f(x)},$$

so  $f(x) \in \mathbb{R}$ . Therefore

$$f(G \cap \mathbb{R}) \subset \mathbb{R}.$$

Now  $G_+$  is connected, since  $G \cap \mathbb{R}$  separates  $G$  into at most two components and  $G_+$  is one of them. Because  $f$  is continuous,  $f(G_+)$  is connected. Also,  $f$  is one-one and maps  $G \cap \mathbb{R}$  into  $\mathbb{R}$ , so no point of  $G_+$  can map to the real axis; for if  $z \in G_+$  and  $f(z) \in \mathbb{R}$ , then

$$f(\bar{z}) = \overline{f(z)} = f(z),$$

and since  $f$  is one-one, this would imply  $z = \bar{z}$ , contradicting  $\Im z > 0$ .

Thus

$$f(G_+) \subset D \setminus \mathbb{R}.$$

But

$$D \setminus \mathbb{R} = \{w \in D : \Im w > 0\} \cup \{w \in D : \Im w < 0\},$$

the union of two disjoint connected components (the upper and lower half-disks). Since  $f(G_+)$  is connected, it must lie entirely in one of these two components.

Therefore  $f(G_+)$  lies entirely above or entirely below the real axis. ■

**5.** Let  $f$  be analytic on  $G = \{z : \Re z > 0\}$ , one-one, with  $\Re f(z) > 0$  for all  $z \in G$ , and  $f(a) = a$  for some real number  $a$ . Show that  $|f'(a)| \leq 1$ .

*Solution.* Let

$$G = \{z : \Re z > 0\}.$$

Since  $f : G \rightarrow G$  is analytic and one-one, and  $f(a) = a$  for some real number  $a > 0$ , we want to show that

$$|f'(a)| \leq 1.$$

Consider the Möbius map

$$\phi(z) = \frac{z - a}{z + a}.$$

Since  $a > 0$ ,  $\phi$  maps the right half-plane  $G$  conformally onto the unit disk  $D$ , and

$$\phi(a) = 0.$$

Now define

$$g = \phi \circ f \circ \phi^{-1}.$$

Then  $g : D \rightarrow D$  is analytic and one-one, and

$$g(0) = \phi(f(a)) = \phi(a) = 0.$$

By Schwarz's lemma,

$$|g'(0)| \leq 1.$$

We now compute  $g'(0)$ . By the chain rule,

$$g'(0) = \phi'(f(a)) f'(a) (\phi^{-1})'(0).$$

Since  $f(a) = a$ , this becomes

$$g'(0) = \phi'(a) f'(a) (\phi^{-1})'(0).$$

Now

$$\phi'(z) = \frac{(z + a) - (z - a)}{(z + a)^2} = \frac{2a}{(z + a)^2},$$

so

$$\phi'(a) = \frac{2a}{(2a)^2} = \frac{1}{2a}.$$

Also, since  $\phi^{-1}$  is the inverse of  $\phi$ ,

$$(\phi^{-1})'(0) = \frac{1}{\phi'(a)} = 2a.$$

Therefore

$$g'(0) = \frac{1}{2a} f'(a) (2a) = f'(a).$$

Hence

$$|f'(a)| = |g'(0)| \leq 1.$$

Thus

$$|f'(a)| \leq 1.$$

■